ỌBÁFẸ́MI AWÓLỌ́WỌ̀ UNIVERSITY

ILÉ-IFẸ̀, NIGERIA

FACULTY OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING DEPARTMENT

THIS DOCUMENT IS NOT FOR SALE !

CPE 316: Introduction to Artificial Intelligence

Rain Semester, 2024-2025 Session

LECTURE AND LABORATORY ASSIGNMENTS

This document is meant to support Lectures and your Laboratory work in CPE 316. Attempts has been made to present the content of this document as accurately as possible. However, cases of typographical, grammatical or other errors are likely. This is unintentional and I will be happy if you could please draw my attention to such errors. Send email to: *oodejobi@yahoo.com* as soon as you find error or mistake.

ỌDẸ́JỌBÍ Ọdẹ́túnjí Àjàdí

JUNE, 2026

CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1

"ARTIFICIAL INTELLIGENCE": AN INTRODUCTION

We have explored several issues surrounding the conventional idea of "Artificial Intelligence" (AI). A standard or consensual definitions for the terms "Artificial Intelligence" are yet to be presented in the literature. Fundamentally, a set criteria by which to distinguish an "Artificial" from "Non-Artificial" instance, entity or agency is yet to be presented. Also, a set of criteria or prescription by which to distinguish an "Intelligent" from "Unintelligent" (or stupidity) agency or entity is yet to be presented. Indeed, an individual to which intelligence (or genius) is ascribed in one context or environment may, in fact, be considered to be stupid in another.

1.1 Conventional perspective of "Artificial Intelligence"

The conventional perspective of "Artificial Intelligence" is grounded in the European Intellectual Tradition (EIT). Indeed, the title "Artificial Intelligence" emerged at a Summer Workshop called in 1956 by John McCarthy and attended by Marvin Minsky, Claude Shannon, Solomonoff Ray, Nathaniel Rochester, Allen Newell, Herbert Simon, amongst others. Two (2) approaches to "Artificial Intelligence" were propounded: (i) Stochastic/Statistical/Probabilistic and (ii) Theory based formulation of logic with symbols. These approaches situated the substances of "intelligence" in the numerical rendering of human mental abstraction.

In the literature on the conventional approach to "Artificial Intelligence", the idea of "Intelligence" and "Artificial" are treated as though they are self-evidence enough. Therefore, the terms "Intelligence" and "Artificial" are often treated as though they do NOT require or deserve explicit definitions. As a result, a variety of connotations have been ascribed to "Artificial Intelligence". Answers to questions, such as those listed as (i) through (iv) below, are often treated superficially, ignored, jettisoned and/or considered as inconsequential in conventional discussion on "Artificial Intelligence".

(i.) What is the substance of (or the competence for) Intelligence ?

(ii.) In what way or form is (i) expressed (or performed) ?

(iii.) To what criteria does (ii) conform and how can it be assessed ?

(iv.) What is the most effective language for rendering, formulating and/or expressing: (i), (ii) and (iii).

1.1.1 Schools of thought in Conventional AI

The questions listed as (i) through (iv) above are addressed through Two (2) schools of thought in conventional AI, namely: (a) Hard-AI and (b) Soft-AI.

(a.) In the "Hard-AI" or "Human-level Intelligence" school of thought, the ultimate aim is to create an **autonomous mechanism** that could imitate and even, perhaps, surpass human capacity for intelligence.

(b.) The ultimate aim in "Soft-AI" or "Utilitarian Intelligence" school of thought is to create an **automate mechanism** that could simulate, and perhaps be used to support, efficient expression of human mental activity.

To the question "Could there ever be a machine that will think like human beings and be more intelligent than human beings?" Richard Feynman responded that: "Machine that thinks like human beings is impossible" but "machines that are intelligent depends on the definition of intelligence". Though his explanation aligns with the Soft-AI school of thought the procedure and/or admissible criteria to which "Intelligence" is expected to conform is not provided. Indeed, a consensual definition for (i) Thinking and (ii) intelligence is yet to be provided in the conventional literature. Therefore, a criteria to which a thinking and/or intelligence agency must conform is yet to be advanced in the European Intellectual Tradition (EIT).

Within *Àròkún*, the capacity to ask and answer questions such as those listed as (i) through (iv) above is the foundation of meaningful enquiry. Whereas a meaningful "enquiry" seeks to conform to the criteria of *Kíkún* ("Coherence"), an informed enquiry seeks to construct *Àlàyé* (an"Explanation") that conforms to the criteria of *Títọ́* ("Consistency"). Competency is self-recursive. Therefore, competency is effected through an agency. Infinity is inherent in a self-recursion. Every Previous state has a unique Previous state in an infinite process. Also, every Next state has a new and unique Next in an infinite process. Hence, the instances of term with which to express an infinite process is inexhaustible. Therefore, an instance in which infinity is inherent, such as competency, is inexpressible. Performance is recursive. Performance is used to simplify and/or render Competency into action and/or expression through a process. Therefore, the process used to render performance is expressible to the extent that instrument of language permits. The action and/or expression used to render a performance is assessed through the instrument of language used to prescribe its process.

An informed explanation comprises finite instances of expression. An explanation is therefore, formulated and prescribed through an instrument of context-neutral language. An informed explanation is a simplification of aspects of a meaningful experience. The aim here, therefore, is an enquiry into the formulation and prescription of a process through which "Intelligence" had been ascribed to the performance of the activity of biological agencies. This is with the aim to automate the process to the extent that computing techniques and technology permits.

Answers to questions (i) to (iv) listed above require that the key terms in the Topic or Title of an enquiry are rendered and simplified through *Asọfín* ("Definition").

1.1.2 The definition of *Àṣọfín* ("Definition")

An instance to be defined is *Odù* ("An Axiom"). An axiom is simplified through an expression. An expression comprises finite sequence of *Ohùn* ("Term"). Attempt to render *Àṣọfín* (The "Definition") for the term *Àṣọfín* ("Definition") is recursive. *Ìrújú* ("Ambiguity") is inherent in a recursive expression. The ambiguity arises because the axiom to be defined is also a term in the expression of its own definition. Therefore, "Correct/Yes" and "Incorrect/No" are equally admissible to an ambiguous expression.

For example, the expression: "This is a sentence." is recursive. This is because the Axiom "Sentence" is also a term in its own expression. Therefore, "Yes" and "No" are equally admissible to the expression: "This is a sentence".

What is *Àṣọfín* (a "Definition")?

The *Yorùbá* definition of the term *Àṣọfín* ("Definition") is rendered into expression as:

A succinct *ìwí* ("narrative") for expressing *àìdámọ̀* ("an unfamiliar") experience or instance using *dídámọ̀* ("familiar") *Ohùn* ("Terms").

Fundamentally, the "familiarity" and "unfamiliarity" ascribed to an instance or experience is grounded in instrument of language. An expression is used to render an Axiom into its Terms. *Ìdámọ̀* ("The identity") rendered and/or reckoned with an expression is grounded in its instances of *Ohùn* ("Term"). Axioms and term are formulated and prescribed through an instrument of language. The information content in an expression is accessible to individual *Ènìyàn* ("Human agency") to the extent of his/her familiarity with the terms and axioms of its instrument of language. An individual's competence in an instrument of language is situated in the ability to access the information in the expressions constructed with it. An individual's performance with an instrument of language is situated in the ability to used its terms to construct admissible and precise expressions.

Why must we "Define"?

A clear definition for a field of study is used to inform and assess:

(a.) **The character of admissible evidence:** {Source, medium, presentation, representation, precision, checkers, collection}

(b.) **Admissible prescription and representation of a process** {Admissible content and instances in its Symbols, Alphabet, Dictionary, Vocabulary, Axioms, Terms, Format, Grammar}.

(c.) **Admissible formulation of agency for a process** {What constitutes State/Location, Transition/Displacement, Interval/Gap as well as their Relation/Connection and manipulation: (Metaphor for rendering Iteration/Induction, Algorithm/Deduction, Heuristics/Default)}

1.1.3 Criteria for assessing the narrative of an informed definition

Ìwí ("A narrative") grounded in "familiar metaphor" and "precise terms" puts better clarity to *Ìdámọ̀* ("the identity") of the instance being defined. The following are the features off "an informative definition" in *Àròkún*:

(i.) It is NOT "Cyclic". A "cyclic definition" is that in which the term being defined is in the narrative of its definition. A cyclic definition is not explicit because it is self-recursive. For example, defining Artificial intelligence: "A machine with capacity to demonstrate **intelligent** behaviour" is cycle or recursive.

(ii.) It is does NOT "beg the question". A definition begs the question when it avoids the term being defined by defining a similar or different term. For example, Evolution as a definition for Creation. Such definition uses terms such as "Can be", "Should be", "Could be", etc.

(iii.) It is NOT "absurd". An absurd definition is neither grounded in language tools of logic and/or polarity nor metaphor. For example, Big-bang as a definition for Creation.

(iv.) It is NOT "compounding". A definition is compounding when a term that is more complex than the term being defined is in the narrative of its definition. For example, the definition of Knowledge as: "Epistemological state of justified true belief".

(v.) It is "Precise". A definition is precise when terms in its expression are used to ascribe identity to unique instances or familiar range. For example, the definition "Fishing is when we fish for fish in the fish river" is imprecise.

(vi.) It is "Consistent". A definition is inconsistent when at least two (2) correct but mutually exclusive and contradictory information can be ascribed to its expression. An inconsistent definition is self-contradictory. For example, The definition a aquatic organism as: "A living things that lives on water".

NOTE THAT:

(a.) A "name" (*Orúkọ*) is used to ascribe nominal identity to an Instance, Entity, Agency or Experience. Therefore, **a name is NOT a definition**.

(b.) A "number" (*Oǹkà*) is used to ascribe numerical identity to an Instance, Entity, Agency or Experience. Therefore, **a number is NOT a definition**.

(c.) Mutually exclusion is inherent in an informed definition. Therefore, the definition of **X** excludes **NOT X**.

Fundamentally, a definition is expressed in the habitual language in which an enquiry is conducted. An informed definition is expressed using an instrument of context-neutral language. A context-neutral definition is grounded in mental abstraction such as Theory, Model, These, Hypothesis, Hierarchy or Method. *Títọ́* ("Consistency") is the fundamental criteria for assessing the logical content of an informed definition. A meaningful definition is expressed with an instrument of context-sensitive language. A context-sensitive definition is situated in a mental conception that is grounded in *Àfijúwe* ("Metaphor"). Metaphor grounded in collective human sensory experiences is the most effective rendering of a meaningful "Definition". *Kíkún* ("Coherence") is the fundamental criteria for assessing the application or action grounded in a meaningful definition.

1.1.4 The narrative for the definition of an area of study

The narrative for the definition of an area or field of study should respond to three (3) important enquiries:

(i.) **The object of study:** This responds to the "**What?**", "**Where?**", "**When?**" and "**Which?**" of an enquiry. What is being studied in "Artificial Intelligence?": "Observation of the performance of the activity of biological agency to which "intelligence" has been ascribed". The aspect of the observation that had been expressed in human language is the object of study.

(ii.) **The kind of study:** This responds to the "**How?**" of an enquiry. How will the study be conducted? The kind of study in "Artificial Intelligence" is computational. The aspect of the human narrative for "intelligence" that had been rendered into the Pseudo Language of symbols.

(iii.) **The purpose of study:** This responds to the "**Why?**" of an enquiry. Why is the study being conducted?: The purpose of study in "Artificial Intelligence" is to create an autonomous mechanism that imitates the intelligence ascribed to an organism to the extent that technology permits.

1.1.5 Some examples of definition of AI in the literature

Artificial Intelligence is the science and engineering of making intelligent machines. John McCarthy(1955)

Intelligence is the computational part of the ability to achieve goals in the world. John McCarthy(1997)

Artificial Intelligence is the science of making machines that can do the kinds of things that humans can do. Blackburn, S. (2005) Oxford dictionary of Philosophy, 2^{nd} Edition, Oxford University Press

Artificial Intelligence is a discipline concerned with the building of computer programs that performs tasks requiring intelligence when done by humans Oxford dictionary of Computing,2008, 6^{th} Edition, Oxford University Press

Artificial Intelligence is an area of study concerned with making computers copy intelligence human behaviours Oxford Advanced Learner's Dictionary, International Students Edition, Oxford University Press

1.2 TASK 1.1

(i.) Discuss and provide your view in respect of the "criteria for an informed definition" as presented in Subsection 1.1.3. Which of the criteria do you consider to be unnecessary? What criteria do you consider should be included?

(ii.) Select any Six (6) definitions from standard "Artificial Intelligence" Textbooks or Journal articles. Assess the information contents in the selected definition by discussing their strength and weakness based on the criteria presented in Subsection 1.1.3.

(iii.) Using the criteria presented in Section 1.1.4, provide a definition for what you understand by "Artificial Intelligence".

1.3 Conventional theories of "Intelligence"

The modern, and dominant, perspective influencing the subject-matter of "Artificial Intelligence" is grounded in the European Intellectual Tradition (EIT). The Philosophical contributions of authors such as Epicurus, Socrates, Plato, Aristotle, Isaac Newton, René Descartes, Gottfried Leibniz, John Locke, Emmanuel Kant, Bertrand Russell, Alfred Whitehead, John McCarthy, Marvin Minsky, Gorge Boole, Alan Turing, and so forth, influences admissible explanations in modern "Artificial Intelligence". These explanation are grounded in "the process perspective and attitude to the phenomenon of nature". In a process perspective, the phenomenon of nature is treated as though it were a process with exact with inputs and outputs. The foundation of enquiry in a process perspective is to seek for the theory by which the output for all possible combination of inputs can be predicted and/or determined.

The process perspective motivated the idea that the substance of an organism, such as human, can be "seconded" and/or "outsourced" to another "organism". With the advent of computation machines, the attitude has be extended to the idea that "the substance of the activity of an organism, such as human, can be "seconded to", "outsourced to" and/or "replicated in" a mechanism".

Within the *Yorùbá* Intellectual Tradition of *Àròkún*, treating *Ìṣẹ̀rọ* ("a mechanism") as though it were *Ìṣẹ̀mí* ("an organism") is a symptom of *Ìdàndan Àfihun* ("Illusion of Pseudo-completeness"). *Ìdàndan* (An "Illusion") is the act of treating *àìpé* ("an incomplete") instance as though it were *Pé* ("complete"). Completeness is inherent in *wíwà* ("beingness") of *Ìṣẹ̀dá* ("the phenomenon of nature"). However, incompleteness is inherent in the identity (nominal or numerical) as well as binary opposite such as creation and destruction, evolution and devolution, Previous and Next, and so forth, ascribed to the sensory accounts of the phenomenon of nature through instruments of language.

Six (6) of the popular "Theories" influencing the idea of "Intelligence" as used in the field of "Artificial Intelligence" are listed and briefly discussed in the following Subsections.

1.3.1 Brian Theory of Intelligence

The Brian Theory posits that

(i) The substance of intelligence is situated in "the human brain".

(ii) A mechanism in which the human brain is replicated is, or had the potential for, intelligence.

In the Brain theory of Intelligence "Organism" is often misconstrue as, or confused for, "Mechanism". For example, the Alan Turing and Alonzo Church's theory of the universal problem-solving machine is based on the assumption that "human brain is like a mechanism with parts and configuration". This informs the attitude that "At its simplest level, the brain can be treated as a collection of units that exchanges signals".

The Brain Theory of Intelligence underlies the activities in conventional "Artificial Neural Networks" and "Machine Learning".

Ọpọlọ ("The brain") is an important organ in the organism of a biological agency. Fundamentally, within *Àròkún*, the brain is NOT A PART of the human body. The brain is *Ẹ̀yà* ("an ORGAN") in the material manifestation of *Ìṣẹ̀mí* ("Biological beingness"). An organ is such by virtue of the organism in which it is situated. An organ, by itself alone, irrespective of its sophistication, is not sufficient to accounts for the activity of its organism.

There are case of humans, who have suffered damage or impairment to their organs of brain, but are able to perform creative activity consider to be intelligent. Therefore, intelligence is NOT situated in human body organs, such as the Brain.

Within *Àròkún*, ascribing intelligence to an agency on the grounds of its organs, is a symptom of *Ìdàndan Àfinhun* ("The illusion of pseudo-completeness").

1.3.2 Mind Theory of Intelligence

The Mind Theory theory posits that:

(i.) The mind is the substance of the capacity for intelligence.

(ii.) An understanding of the "human mind" is essential and sufficient for the development of an intelligent mechanism.

However, the question of "What is mind?" is yet to be answered in the European Intellectual Tradition. In conventional Philosophy, for example, various arguments and counter-arguments on "The body-mind problem" is yet to provide a definitive solution. Neither the **Monist** ("Organism is totality situated in Material" or "Organism is wholly situated in Mental") nor **Dualist** (Mental and Material are binary opposite instances of an Organism) theory of the "Body-mind problem" has culminated in a consistent explanation.

Material manifestation is necessary for the observation and/or study of the activity or process effected by an agency. The question of "The location of the mind in the beingness of a biological agency" is yet to be answered. Within *Àròkún*, an enquiry into the "What?", "Where?", "When" and "Which" of an activity and/or process requires the determination of its spatial location. Indeed, ascribing intelligence to an agency in the absence of evidences grounded in the location and character of its activity, in this case the mind, is a symptom of *Ìdàndan Àtinúndá* ("Abstraction illusion of completeness").

Several references to the use of the "Theory of mind", in conventional Artificial Intelligence, is yet to culminate a practically demonstrated system. Therefore, the "Mind theory" is yet to have a foothold in conventional Artificial Intelligence.

1.3.3 Language Theory of Intelligence

The Language Theory posits that:

(i.) The capacity for language is the substance of human intelligence.

(ii.) A mechanism with the capacity for language can be refined and/or optimise in such a way that it will eventually manifest intelligence,

Ludwig Wittgenstein and Noam Avram Chomsky have influenced, substantially, the theory of language used in conventional "Artificial Intelligence" and "Computing". However, the conventional perspective is limited to instruments of "Context-neutral" and "Regular" language.

Language theory underlies most of the important developments in modern "Artificial Intelligence". For example, the Alan Turing's idea of "Universal Problem Solve" and "Imitation Game" are grounded in Language theory. The imitation game is the foundation of conventional "Test of Intelligence" that is used to assess AI systems.

Subfields of AI such as "Automated Dialogue System", "Speech Recognition Systems", "Speech Synthesis Systems", "Automatic Grammar Checker", and so forth, are grounded in Language Theory of Intelligence.

Fundamentally, *Èdè-orí* ("Faculty of language") is inherent in *wíwà* ("the beingness") of *Abẹ̀mí* ("Biological agency" or "Organism"). Therefore, "faculty of language" is inextricable from its *Abẹ̀mí* ("Biological agency" or "Organism"). The expressions articulated by an organism are effected through instrument of language. Instruments of language are created through Faculty of language. Self-expression, rather than, expression confers agency on an organism. Self-expression is inextricable from its biological agency.

Indeed, Conventional computational mechanism are created through the Pseudo language of symbols. A Pseudo language of symbols is a mirror image of its instrument of language. Symbols, Alphabet, String, Syntax and Grammar are used to formulated and prescribe Pseudo language of symbols. Pseudo agency is effected through a Pseudo language of symbols. Therefore, the agency of a computation process is a mechanism, NOT an organism. Faculty of language is outside the ambit of a mechanism.

Within *Àròkún*, ascribing intelligence to an agency on the grounds of a Pseudo language of symbols, is a symptom of *Ìdàndan Àfinhun* ("The illusion of pseudo-completeness").

1.3.4 Pattern Theory of Intelligence

The Pattern Theory posits that:

(i.) The substance of the human capacity for intelligence is situated in the ability to create, recognise, manipulate and express "Patterns".

(ii.) A mechanism with the ability to "Generate", "Manipulate", "Store" and "Recognise" Patterns is, or has the potential for, intelligence.

Fundamentally, Patterns are ascribed to expressions based on conformity with prescribed arrangement of symbols and/or signs. The Data ascribed to the pattern in an expression is by virtue of human sensory accounts. Therefore, the capacity to ascribe pattern to an expression is grounded in "Sensory organs". This implies that, an agency in which "Sensory organs" is not inherent can neither reckon nor apprehend nor ascribe pattern to an expression. Therefore, though a mechanism can be used to generate, manipulate and store huge varieties of data for digitally encoded patterns, a mechanism can neither reckon nor apprehend not ascribe pattern to an expression.

Indeed, organisms such as Plants, Insects (e.g. bees and ants) and Birds are capable of creating, manipulating and expression wonderful varieties of patterns. Indeed, the creation of precise and perfectly symmetrical patterns does NOT confer intelligence on an agency. Indeed, every pattern that can be precisely prescribed and described using a Pseudo language of Symbols can be automatically generated using a mechanism.

The Pattern theory of intelligence influences the activities in "Machine vision" and "Pattern Recognition".

Within *Àròkún*, ascribing intelligence to an agency on the ground of Patterns generated through a Pseudo language of symbols, is a symptom of *Ìdàndan Àfinhun* ("The illusion of pseudo-completeness").

1.3.5 Cause-Effect Theory of Intelligence

The "Cause-effect theory" posits that:

(i.) The substance of intelligence is situated in the accounts of the Cause and Effect observed during the performance of *Ìṣe* ("Action") and *Àṣedí* ("Reaction").

(ii.) A mechanism with the ability to imitate the Cause-Effect observed during a performance to which intelligence is ascribed is, or has the potential for, intelligent.

The action and/or reaction ascribed to the sensory accounts of the performance effected by an agency is grounded in instrument of language. Therefore, "Start and Stop(Halt)", "Cause and Effect", "Begin and End" as well as "Precedence and Consequence" are situated in instrument of language. The use of these binary opposite to ascribe intelligence to the activity of an organism, is therefore, falsifiable, on the grounds of incoherence, inconsistency and/or imprecision.

The Isaac Newton's "Laws of motion" used in the Physics of Mechanics is grounded in the "Cause-effect theory" of action and reaction. The "Cause-effect theory" influences, and has been fruitfully used, in the field of Robotics.

Within *Àròkún*, however, *Ìṣe* ("the Action") and *Àṣedí* ("Reaction") are ascribed to the sensory accounts of *Ìṣẹ̀lẹ̀* ("Happening") through instruments of language. Every instance in *Ìṣẹ̀lẹ̀* (a "Happening") is in *Ọjọ́* ("Continuum"). *L'ọ́wọ́* ("Now") is the only accessible instance in the beingness of continuum. The beingness of *L'ọ́wọ́* ("Now") is outside the ambit of *Ìdámọ̀* ("Identification"). Identification is necessary for the rendering and formulation of *Abèjì* ("Binary opposite").

Therefore, the binary opposite of *Ìṣe* ("Action") and *Àṣedí* ("Reaction") are ascribed to, and situated in, a prescribed process. A process is prescribed and/or formulated using an instrument of language. An instrument of context-sensitive language is used to treat *Ìṣe* ("an Action") and its *Àṣedí*

(“Reaction”) as the mutually inclusive and complementary aspects of its agency’s *Àìṣe* (“Inaction”). An instrument of context-neutral language is used to treat *Ìṣe* (“an Action”) and its *Àṣedí* (“Reaction”) as mutually exclusive and contradictory instances.

An instrument of regular language is binary opposite neutral. Therefore, *Ìṣe* (An “Action”) and its *Àṣedí* (“Reaction”) are treated as unrelated instances.

Within *Àròkún*, ascribing intelligence to an agency on the ground of the “Cause and Effect” reckoned during the activities of a biological agency, is a symptom of *Ìdàndan Ohùn* (“The Term illusion of completeness”).

1.3.6 Problem-Solving Theory of Intelligence

This theory posits that:

(i) The capacity for “Problem-Solving” is the substance of “Human intelligence”

(ii) The creation of “Intelligent mechanisms” in the ability to replicate “Problem-solving”.

Several earlier computation mechanisms used to demonstrate “Artificial Intelligence” are grounded in “The problem-solving theory of intelligence” (Newell, A., & Simon, H. A. (1972). Human problem solving. Prentice-Hall.)

However, Problem-Solving is an Algorithmic and Deductive process. Therefore, a problem-solving process is formulated and totally explicable through an instrument of context-neutral language. The transition in a context-neutral process is finite and its binary opposite states are mutually exclusive and contradictory. Therefore, a context-neutral process is non-deterministic and monotonic. A monotonic process will eventual become stale and insipid after several repetitions. Indeed, only one of the binary opposite aspects of a state is admissible, and accounted for, in a consistent context-neutral explanation. Therefore, a context-neutral explanation for the state of “Intelligence” ascribed an agency is falsifiable on the grounds of inconsistency. The inconsistency is situated in the aspects of a binary opposite state that is ignored or jettisoned during the formulation of a context-neutral process.

Moreover, a context-neutral process is expressed to the extent that its instrument of regular language permits. The expressed aspects of a context-neutral process is a partial account of its explanation. Indeed, the expressed aspects of a context-neutral process is falsifiable on the grounds of imprecision.

1.3.7 Summary

Fundamentally, a computational mechanism is used to effect a prescribed process to the extent that Pseudo language of symbols and modern technology permits. A digital computation mechanism is formulated, prescribed and realised through a Pseudo language of symbols based on binary alphabet, that is $\{0, 1\}$. However, **a digital computation mechanism does not process binary string**.

NOTE that a computational mechanism does NOT manipulate the terms of instrument of language. A computation mechanism, or machine, manipulates the material rendering of the symbols used to cue the strings ascribed to the terms of an instrument of language. The capacity to use and/or apprehend the terms and expression of an instrument of language is inherent, and inextricable from, its biological agencies.

For example, modern digital computation mechanisms are created to manipulate the binary rendering of electrical or electronic signals. There is neither "1" nor "0" in the stream of electrical and/or electronic signals. "1" and "0" are ascribed to electrical or electronic signals through a prescription and/or standard. For example, in modern digital computation machines, electrical signal in the range 5.0 V to 3.5 V is assigned to digit '1' while the range 2.5 to 0.0 V is assigned to '0'. The signals in the range 3.4 to 2.6. V is used to demarcate the Two (2) admissible ranges into mutually exclusive instances. Strings of binary digits are then represented and manipulated through electrical circuits that modulates and/or switches between the binary opposite signal ranges used to cue '1' and '0'.

Indeed, the mental activity, such as *Ìṣirò* ("Computing") and *Ìrònún* ("Thinking"), of an individual can neither be outsourced nor seconded nor replicated in another human. Therefore, the idea of be outsourcing or seconding nor replicating human mental activity in a mechanism is situated in *Ìdàndan* ("Illusion").

Human mental activities are effected through instruments of language. Instrument of language are not used to manipulate Symbols and/or String. String and symbols are created and manipulated using Pseudo language of symbols. Treating Symbols and/or Strings as though they are terms of an instrument language is a symptom of *Ìdàndan Àfihun* ("Illusion of Pseudo completeness").

1.3.8 Distinction between Organism and Mechanism

The *Àròkún* distinction between *Ìṣẹ̀mí* ("Organism") and *Ìṣẹ̀rọ* ("Mechanism") is presented in Table 1.1.

Based on Items 1 and 2 in Table 1.1, "Faculty of language" and "Sensory organs" distinguished *Ìṣẹ̀mí* ("Organism") from *Ìṣẹ̀rọ* ("Mechanism"). Therefore, an agency in which Faculty of language and Sensory organs are NOT inherent is NOT an organism. Based on Item 6 in Table 1.1, the capacity to ascribe "identity" and "binary opposite" to an instance is inherent in an organism. Therefore, a mechanism can neither render nor apprehend the identity and/or binary opposite ascribed to an instance.

Please NOTE that the following *Yorùbá* terms are used to reckon the activities of organism (Biological agency) alone: (i) *Ọgbọ́n* ("Intelligent"), (ii) *Àìgbọ́n* ("Unintelligent"), (iii) *Agọ̀* ("Stupidity") and (iv) *Àgunlá* ("Ignorant"). An overview of the terms is discussed as follows.

The binary opposite of *Ọgbọ́n* ("Intelligent") is *Àìgbọ́n* ("Unintelligent"). Therefore, either context-neutral or context-sensitive language can be used to ascribe these binary opposite to the performance of a biological agency. If the context-neutral language is used, then an agency is either Intelligent and Unintelligent, an never both. Therefore, the Intelligent and Unintelligent ascribed to the performance of an agency, through an instrument of context-neutral language, are mutually exclusive and contradictory.

If the context-sensitive language is used, then an agency is Intelligent on the grounds of the content of its Unintelligent. Therefore, the Intelligent and Unintelligent ascribed to the performance of an agency, through an instrument of context-sensitive language, is mutually inclusive and complementary.

Agọ̀ ("Stupidity") is ascribed to the activity of a biological agency through an instrument of regular language. *Agọ̀* (The "Stupidity") ascribed to an individual is on the grounds of NOT knowing what he/she is expected to know.

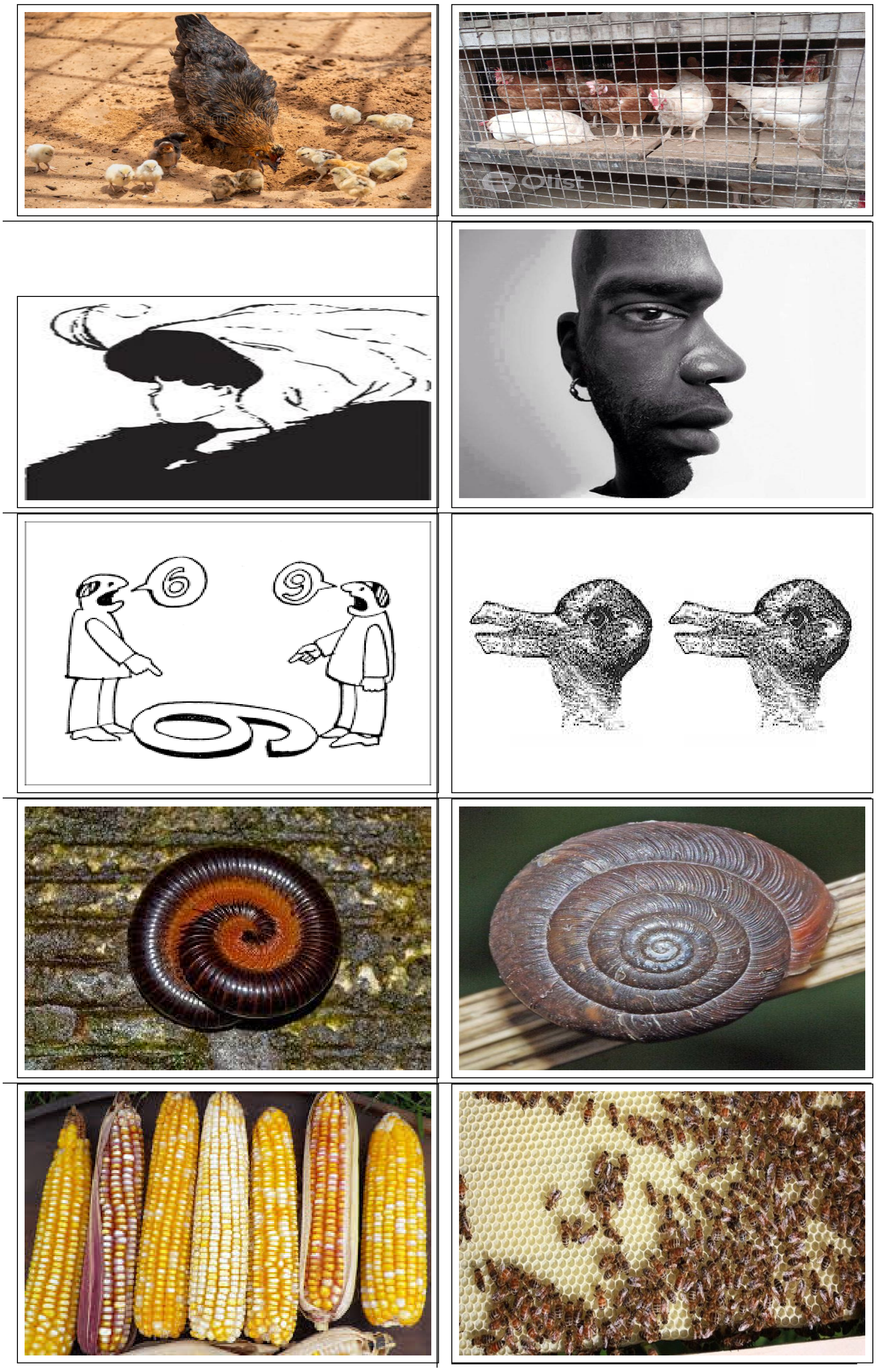
Àgunlá ("Ignorant") is ascribed to the activity of a biological agency through an instrument of regular language. *Àgunlá* (The "Ignorance") ascribed to an individual is on the grounds of Knowing and deciding to ignore what he/she is expected to take action on.

Table 1.1: Distinction *Ìṣẹ̀mí* ("Organism") and *Ìṣẹ̀rọ* ("Mechanism")

Ser. No.	*Ìṣẹ̀mí* ("Organism")	*Ìṣẹ̀rọ* ("Mechanism")
1.	*Èdè-orí* ("Faculty of language") is inherent in his/her/its mental beingness. *Ìpèdé* ("Instruments of language") are created and refined through faculty of language.	Its process and agency are prescribed and formulated through instrument of language. Its agency is created and realised through Pseudo-language.
2.	Sensory and articulatory organs are inherent in his/her/its material manifestation.	Material devices are used to realise the sensors and actuators for the input and output, respectively, of the process it is used to effect.
5.	*Abínibí* ("Offspring-rebirth") is inherent in his/her/its beingness. Offspring-rebirth manifests through *Ìbí* ("Birth") and *Ikú* ("Death").	Offspring is outside the ambit of its beingness. Its manifestation (creation) and cessation (destruction) is effected by a biological agency.
4.	*Atúnrabí* ("Self-rebirth") is inherent in his/her/its beingness. Self-rebirth is effected through *Hù* ("Growth") and *Rà* ("Decay").	Self is outside the ambit of its beingness. Its agency is crafted and refined by a biological agency. Wear and Tear culminates in its damage, depreciation, deterioration and dysfunction.
5.	Breath (Inhale and Exhale) sustains the material manifestation of biological beingness.	Breath is neither required nor inherent in the sustenance of its material manifestation.
6.	Capacity to ascribe *Ìdámọ̀* ("Identity") and *Abèjì* ("Binary opposite") to an instance is inherent in his/her/its beingness. The ability to ascribe *Èmi* ("I" or "Me" or "Self") to his/her/its own agency is grounded in identity and binary opposite.	Its beingness is identity and binary opposite neutral. It can neither ascribe identity nor binary opposite to its and/or another agency.

1.4 TASK 1.2

Study carefully, the adjacent pair of images on Page 13. Try to generate a narrative that describes any Two (2) pair as precisely as you can. Review your narrative in the context of “the Theories of Intelligence” discussed in Section 1.3.



1.5 Conventional assessment and evaluation of AI Systems

Though a number of theories have been propounded, in the conventional literature, a consensual or standard definition for "Intelligence" is yet to emerge. The absence of "a consensual definition" makes it difficult to set standard assessment criteria for the performance to which "Intelligence" conforms. The "Intelligence" ascribed to the performance of an organism is often grounded in unfamiliarity with the competence required to create it. The "intelligence" ascribed to a mechanism seems to be appreciated more by those who are unfamiliar with the methods and means of its creation.

Often time, in the conventional literature, *Ìṣẹ̀rọ* ("Mechanism") is confused for *Ìṣẹ̀mí* ("Biological beingness"). Therefore, the logic of the language used to describe *Abèmí* ("Organism") is also often used to describe *Ìṣẹ̀rọ* ("Mechanism"). For example, the term "memory" is often ascribed a computing mechanism as though it were an organism. However, whereas forgetfulness is inherent in the human capacity to recall the content reckoned into his/her memory, a mechanism can never forget instances recorded into its memory. Human forgetfulness is grounded in fatigue, oversight distraction and other physiological limitations inherent in biological beingness. The inability of a mechanism to recall the instances in its memory is grounded in mechanical faults and/or the withdrawal of energy.

Despite not having a definition for "Artificial Intelligence", a number of early researchers in the field of AI have prescribed condition or criteria to which an "Intelligent machine" should conform. The extent to which a system conforms to the prescribed criteria is used to rate its "intelligence".

1.5.1 Turing Test of Intelligence

Alan Matheson Turing addressed the question: "Can machine think?". He considered the question too meaningless to deserve discussion. He then proceeded to reduce the question to **an imitation game**. Alan Turing prescribed the imitation from the perspective that "If an agency A can imitate another agency B, to the extent that it is difficult to distinguish the performance of A from B then B can be said to have the capacity of A". There are several narratives for the "Turing experience", an example is as presented in Table 1.2. A configuration for the Turing test of intelligence is shown in Figure 1.1.

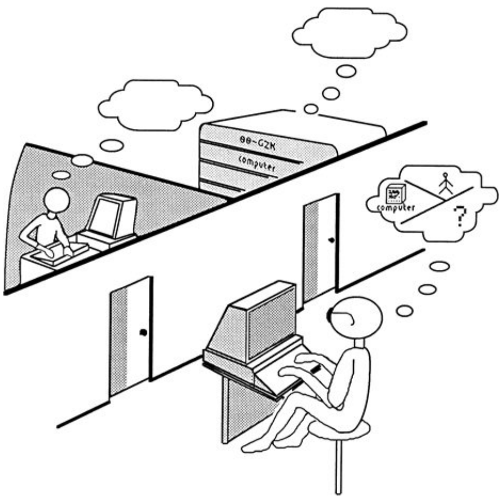


Figure 1.1: A configuration for Turing test of intelligence (Sourced from the web)

Table 1.2: The Turing Test

Given three (3) agencies: (i) a Judge (Ag_J), (ii) Subject (Ag_S) and (iii) an Impostor (Ag_I). These agencies are located in different rooms. The agencies can only communicate through teletype writer or teleprinter. The Judge interrogates the Subject and Impostor. The communication of the Subject and Impostor with the Judge is normalised by the teletype writer. This implies that features inherent the character of the Subject and Impostor, which can influence the decision of the Judge, have been removed. The Judge (Ag_J) interrogates the Subject (Ag_S) and Impostor (Ag_I) through questioning. After "a considerable" question-answering session, and on the basis of the response given by the Subject and the Impostor, the decision of the Judge can be one of the following: (i) Identify and distinguish the Subject from the Impostor. (ii) Confuses the Impostor for the Subject or *vice versa*. (iii) Unable to distinguish the Subject from the Impostor. If the Judge makes decisions (ii) and (iii) then the Impostor (Ag_I) has PASSED the imitation game. If the Judge makes decision (i) then the Impostor (Ag_I) has FAILED the imitation game. In extending the "Turing imitation game" to "Artificial Intelligence", the Impostor Ag_I is replaced with a "Computing machine". The Turing test is a weaker version of the Leibniz's law of "Indiscernible the identity". The law states that "if there is no indiscernible significant differences between the performances of two (2) agencies W and C, then W and C are interchangeable.
The implication of the "Turing test of Intelligence" is that "Given a set of Input or String S_I, if *a Human* (H) and *a Machine* (M) generate outputs strings S_{Ho} and S_{Mo}, respectively, that are indistinguishable, then the machine is, at least, as *intelligent* as the human."

Another version of Turing test involves a dialogue between a human and a machine, over a communication channel such as the Internet (see Figure 1.2). If after a reasonable length of time, the human cannot determine whether or not he/she has been communicating with another human, then the machine is considered to have passed the *Test for intelligent*. The program **ELIZA** (available on the web) is an example of a system developed to achieve such performances.

The beingness of an organism can be imitated to the extent of the self-awareness of an imitator. However, *Ìṣẹ̀mí* ("biological beingness") can neither be outsource nor seconded to another biological agency. Fundamentally, the capacity to imitate does not confer originality on the imitator. The idea of using a mechanism to imitate an organism is falsified on the grounds of *àìpé* ("incompleteness"). Within *Àròkún*, anything falsified on the grounds of *àìpé* ("incompleteness"), is completely and *àbálá* ("irrevocable") falsified.

Figure 1.2: Chat-box configuration for Turing test of intelligence

1.5.2 Challenges to the conventional "test of Intelligence in AI"

There are a number of challenges to the specified aim and objectives of the activities in AI. Some of these objections questioned the procedure adopted in AI. Others questioned the fundamental principles of the purpose of the field of AI.

A school of thought posit a procedural to AI. This school challenges the AI system development procedure by querying the adequacy and/or sufficiency of its methods and methodology. This school of thought holds the view that, it is perhaps possible to reduce human intelligence (totally or partially) into a mechanical process. However, modern tools and techniques are inappropriate for the task. The second school of thought questions the fundamental principles upon which AI rests. This school posits that the question of "developing a machine for something we cannot describe is ill-conceived". indeed, John Searle proposed a counter-experiment to the Alan Turing test for Intelligence. This is discussed in the following Subsection.

1.5.3 John Searle's Chinese Room Problem

The Chinese-Room Problem proposed by John Searle (Searle, 1980) has been one of the portent argument used against the validity of the Turing test. The Chinese-Room Problem is as stated in Table 1.3.

The human subject Ag_{IN} simulates or imitates a Chinese language speaker using text. However, the competence for language is beyond the ability to arrange letters or text according to a prescribed rule. To understand a human language, such as Chinese, requires a competence and performance that transcend symbols or text manipulation. The competency in a language includes familiarity with the vocabulary, grammar, syntax and semantics as well as the knowledge and ability to use them within the Chinese world-view. This competence makes it possible for an individual to creatively and spontaneously articulate an infinite instances of sentence.

In the case of the Chinese-Room Problems presented in Table 1.3, the subject's respond is pre-

Table 1.3: John Searle Chinese-Room Problem

Given that a human subject Ag_{IN}, competent in the English language, is locked up in a room. In that room, there are books $B1$ and $B2$ for translating text from English to Chinese languages, and *vice versa.* Ag_{IN} also has access to a procedure and cross-reference rules for constructing Chinese text for the answers to English questions. The question now is, if the human Ag_{IN} is interrogated in the English language and Ag_{IN} is able to respond with Chinese language text, can we then say that Ag_{IN} is competent in the Chinese language?

determined and pre-planned. “Previous” and “Next” are inherent in pre-planed process. The competence in the use of language is a happening. There is neither “Previous” nor “Next” in the emergence of a happening. The John Searle’s argument posits that neither “Intelligence” nor “Consciousness” can be ascribed to a material agency.

Figure 1.3: The Chinese Room Problem

1.5.4 Limitation of “Turing Test” and its conventional challenges

The “Turing Test of intelligence” and its “Chinese Room” counter-argument are grounded in the “Language Theory of Intelligence”. In that theory, it is assumed “intelligence” is situated in human language expression. Hence, language expression is posited as the domain through which the human capacity for intelligence can be admitted or refuted. Indeed, “Turing Test” and the counter-arguments for its used in the assessment of intelligence are situated in pseudo-language and/or proxy medium.

The expressions of an instrument of language is authentic to the extent that it is articulated directly by its human agency. Expression communicated through "Written text" and "Proxy medium", such as teletype-writer, is NOT a total account of its human agency's language performance.

Fundamentally, "intelligence" is neither situated nor inherent in the language performance of an *Àbẹ̀mí* ("Organism"). *Ọgbọ́n* ("Intelligence") is ascribed to *Àbẹ̀mí* (an "Organism") on the grounds of the creativity manifesting his/her/its performances. The expression of *Ọgbọ́n* ("Intelligence") through *Ìṣe* ("Action") and/or *Àìṣe* ("Inaction") does NOT conform to a prescribed or standard set of criterial and/or formulation. Indeed, *dídákẹ́* ("being quite") is a mark of *Ọgbọ́n* ("Intelligence") in some context.

Moreover, *Ìmòye* ("the Understanding") on which *Ọgbọ́n* ("Intelligence") is grounded is outside the ambit of instruments of language. *Ọgbọ́n* ("Intelligence") is, therefore, is inexpressible through an instrument of language. Indeed, treating *Ọgbọ́n* ("Intelligence") as the mutually inclusive and complementary binary opposite aspect of *Agọ̀* ("Stupidity" or "Un-intelligence") is the extent that context-sensitive language can render.

This implies that, an instrument of context-sensitive language is necessary for the expression of intelligence. However, intelligence transcends the context-sensitive universe of discourse. Therefore, the instances rendered through instruments of regular and/or context-neutral language are too inadequate to warrant their use for the expression and/or assessment of *Ọgbọ́n* ("Intelligence").

Every biological agency is original, complete and unique. An inexpressible instance is inexplicable. The completeness of an organism can neither be duplicated, outsourced and/or seconded to another organism and/or mechanism. Grounding the assessment of the activity of a biological agency on his/her/its ability to imitate another is a symptom of *Ìdàndan Àtinúndá* ("Abstraction illusion of completeness").

The computational mechanisms used to imitate "Intelligence" are created and realised through Pseudo language of symbols. Treating an organism as though it were its Pseudo language of symbols imitation is a symptom of *Ìdàndan Àfirán* ("Illusion of Pseudo completeness").

1.6 Subfields of Conventional AI

1.7 Task 3.1

Using available textbooks and other sources, such as the internet, discuss your understanding of the following Subfields of AI.

1.7.1 Games

1.7.2 Robotics

1.7.3 Decision support system

1.7.4 Automate story generator

1.7.5 Human Language Processing

1.7.6 Recommender system

1.7.7 Automatic grammar correction system

1.7.8 Speech recognition systems

1.7.9 Speech synthesis system

1.7.10 Computer Assisted Learning Systems (CALS)

CHAPTER 2

ỌGBỌ́N ("INTELLIGENCE") IN *ÀRÒKÚN*

Based on evidences documented in the modern and conventional literature of "Artificial Intelligence", the word "Artificial" is *Ohùn Àfihun* ("Pseudo-term"). Treating a Pseudo-term as the binary opposite of an instance in which completeness is inherent , that is "Nature or Natural" is treated as a symptom of *Ìdàndan Àfihun* ("Illusion of pseudo-completeness) in *Àròkún*.

Fundamentally, *Ìṣẹ̀dá ko ni ààrọ̀* ("The phenomenon of nature no substitute"). For example, there is neither a substitute nor replica for instances in the "phenomenon of nature". The identity ascribed to an instance that has NO Substitute is outside the ambit of Binary opposite formulation.

Indeed, there is no *Yorùbá* term with equivalent information content as the English words "Nature" or "Natural", either as "Verb" or "Noun". Therefore, the identity of a "Non-Artificial" instance is difficult to define in the context of that which is "Artificial". The "Artificial" ascribed to a material instance is situated in *Ìdàndan* ("Illusion"). The material manifestation of the phenomenon of nature is language-neutral. Hence, the word "Artificial" is replaced with *Tìkararẹ̀* ("Automated") in the subject-matter of this course.

2.1 *Àròkún* perspective of *Ọgbọ́n* ("Intelligence"): An overview

The creation, manipulation and refinement of *Ìṣẹ̀rọ* ("A mechanism") is grounded in the instrument of language rendering of *ìṣe-inún* ("mental activities") of *Ìṣẹ̀mí* ("Organisms"). Therefore, rather that "Artificial Intelligence", the title "**Automate Intelligence**" is adopted in this course.

NOTE THAT:

(A.) *Ìṣẹ̀rọ* ("A mechanism") is *Tìkararẹ̀* ("Automated") to the extent that human intervention is NOT required to Start, Maintain and/or Terminate its operation. **The binary opposite of *Tìkararẹ̀* ("Automated") is *Àfaraṣe* ("Manual").**

(B.) The computational rendering of an Automated Mechanism is prescribed and expressed through a Pseudo Language of symbols.

(C.) A Pseudo language of symbols is used to construct the string for cueing the representation of the terms and expression of an instrument of language. An agency of Pseudo language of symbols is rendered into a mechanism through a prescribed process and to the extent that *Ìpèrọ* ("Technology") permits.

Fundamentally:

(i.) Intelligence is NOT situated in the magnitude of the material manifestation of an organism. (*Gíga kò kan Ọgbọ́n.*)

(ii.) Intelligence is NOT situated in the speed or efficient of operation or computation. Indeed, intelligence can neither be counted nor computed nor calculated nor measured. (*Ohun Iyebíye ni ọgbọ́n. Iyebíye kò ní òṣùwọ̀n.*)

(iii.) Intelligence is NOT situated in memory, information and/or data. Finiteness is inherent in the accumulation of memory, information and/or data. (*"Moti gbọ́n tán" ni ìbẹ̀rẹ̀ agọ̀.*)

(iv.) Intelligence is NOT situated in Language. Intelligence is situated in the ability to use language to express creativity. (*Èdè kọ́ l'Ọgbọ́n. Ṣùgbọ́n, èdè burúkú ló ń pa ọgbọ́n mọ́ olówó ẹ̀ nínún.*)

(v.) The intelligence ascribed to an agency or group of biological agencies is tentative. (*Ọlọ́gbọ́n ọdún yìí, òmùngọ̀ ẹ̀mîi.*)

(vi.) The Intelligence ascribed to an agency or group of biological agency is contextual. *"Èmi ni mo gbọ́njù ní oko wa", ìyà ló fi ń jẹni l'Ọ̀yọ́.*

Ọgbọ́n ("Intelligence") defers from *Ìmọ́* ("Knowledge"). Whereas knowledge is accumulative, intelligence is situated. Whereas the mental state of knowledge is interpretable to its human agency and explicable to the extent that instrument of context-neutral language permits, Intelligence can neither be interpreted nor explained. *Ìmọ́* ("Knowledge") is rendered with the metaphor of *Inán* ("Fire") while *Ọgbọ́n* ("Intelligence") is rendered with the metaphor of *Omi* ("Water").

2.2 *Àròkún* definition of "Automated Intelligence"

Ọgbọ́n ("Intelligence") is ascribed to human activity on the grounds of *Ìmòye* (the "Understanding") manifesting in the observation of its performance. *Ọgbọ́n* ("Intelligence") is to *Ìmòye* ("Understanding") what *Ìtumọ̀* ("Meaning") is to *Ìmọ̀* ("Knowledge"). Also, *Ìtumọ̀* ("Meaning") is to *Ìmọ̀* ("Knowledge") what *Àṣamọ̀* ("Information") and *Àlá* ("Imagination") are to *Ìrántí* ("Memory"). *Àìgbọ́n* ("Unintelligent") is the binary opposite of *Ọgbọ́n* ("Unintelligent"). *Àìmọ̀* ("Unknown") is the binary opposite of *Ìmọ̀* ("Knowledge"). *Àṣamọ̀* ("Information") is the binary opposite of *Àlá* ("Imagination"). The term *Agọ̀* ("Stupid") is used to reckon an observation that does not conform to any of the above listed performances.

The term *Ọgbọ́n* ("intelligence") is also used to reckon the creativity observed in the problem-solving skills of *Ẹranko* ("Animals"), *Ẹyẹ* ("Birds"), *Kòkòrò* ("Insects") and *Ẹja* ("Fish"). Indeed, there are several evidences in the narratives of *Yorùbá* folk and intellectual discourses in which *Ọgbọ́n* ("intelligence") is ascribed to the cleverness and ingenuity demonstrated by animals (such as *Ọbọ* ("Monkey") and *Ìjàpá* ("Tortoise") and *Kòkòrò* (insects (such as *Ikán* ("Termites") and *Oyin* ("Bees")). The term intelligence is therefore, used based on the context and circumstance in which cleverness and ingenuity is being demonstrated by Organism (or "Biological agencies").

Within *Àròkún*, the term *Ọgbọ́n* ("Intelligence") is NEVER ascribed to Mechanism. However, *Ọgbọ́n* ("Intelligence") can be ascribed to performance of the biological agency that created the Mechanism.

The above explanation is the grounds on which there is no logical process to which the "competence" inherent in the capacity for *Ọgbọ́n* ("intelligence") subscribes and by which it can be prescribed and/or described using a context-neutral language. The *Àròkún* conception of *Ìdàrọ-Ọgbọ́n* ("Automated Intelligence") is limited to the observable and observed aspects of the performance of biological agencies within the ambit of an instrument of regular language. Based on the above discussions, *Ìdàrọ Ọgbọ́n* ("Automated Intelligence") is defined from the *Àròkún* perspective, as presented in Illustration 2.0:

2.0 **Automated Intelligence (AI)** is the computational study of the process prescribed for the creativity ascribed to the observed performance of an organism with the aim to formulate and implement a mechanism that imitates the process to the extent that technology permits.

The term "process" in the above definition is the instances of "State and Transition" or "Location and Displacement" ascribed to the performance of the biological agency ("Organism") of interest. However, there is no criteria to which the happening of a performance subscribes and/or conforms. Therefore, the observation ascribed to a performance is admissible to the extent that the instruments of language used by its human agency permits. Indeed, an observation is such by virtue of a prescribed process. The expression of a computational study is admissible to the extent of the Language of symbols used to formulate its process. The term "creation", as used in the definition in Illustration 2.0, includes "Specification", "Formulation", "Designing", "Modelling", "Simulation", "Implementation" and "Evaluation" of a mechanism.

Based on the narrative in Illustration 2.0, "Intelligence" is ascribed to the aspect of the observed performance of one or more biological agency through an instrument of language. Therefore, *Ìdàrọ Ọgbọ́n* ("Automated Intelligence") is rendered and formulated to the extent expressible through an instrument of language.

Base on the above definition, therefore, the manifestation of *Ọgbọ́n* ("Intelligence") is outside the ambit of instruments language. Within *Àròkún*, ascribing *Ọgbọ́n* ("Intelligence") to an agency through an instrument of context-sensitive language is a symptom of *Ìdàndan Àtorídá* ("Conceptual illusion of completeness"). Ascribing intelligence to an agency through an instrument of context-neutral language is a symptom of *Ìdàndan Àtinúndá* ("Abstraction illusion of completeness"). Ascribing intelligence to an agency through an instrument of regular language is a symptom of *Ìdàndan Àtọwọ́dá* or *Ìdàndan Ohun* ("Objective illusion of completeness"). Ascribing intelligence to an agency through a Pseudo language of symbols is a symptom of *Ìdàndan Àfihun* ("Illusion of pseudo-completeness").

Fundamentally, within *Àròkún*:

(i.) It is NOT admissible to ascribe *Ọgbọ́n* ("Intelligence") to *Ẹ̀rọ* (a "machine" or "mechanism"). Intelligence is cannot be ascribed to an entity or agency in which "Faculty of language" and "Sensory organs" are not inherent.

(ii.) It is NOT admissible to ascribe *Ọgbọ́n* ("Intelligence") to an agency through an instrument of regular language. Binary opposite is inherent in the rendering and formulation of *Ọgbọ́n* ("Intelligence"). The universe of regular language is binary opposite neutral. A regular process is constant and deterministic.

(iii.) *Ọgbọ́n* (The "Intelligence") ascribed to an agency through an instrument of context-neutral language is monotonic and non-deterministic. A monotonic process is finite. Therefore, a monotonic and finite process will eventually become stale and insipid (*tẹ́*).

(iv.) *Ọgbọ́n* (The "Intelligence") ascribed to an agency through an instrument of context-sensitive language non-deterministic and non-monotonic. Intelligence is deeper than meaning. Therefore, Intelligence is transcends an agency of context-sensitive language. The intelligence ascribed to biological agency through an instrument of context-sensitive language is falsifiable on the grounds of incoherence

(v.) The sensory accounts of every instance in the manifestation of *Ọgbọ́n* ("Intelligence") is situated in *Ìṣẹ̀lẹ̀* ("Happening"). *Lọ́wọ́* ("Now")) is the only accessible instance in the beingness of a happening. The sensory accounts of every instance of "Now" is *Ọ̀tun* ("New").

(vi.) *Àìpé* ("Incompleteness") is inherent in the human sensory accounts of of *Ìṣẹ̀lẹ̀* (the "Happening") of a performance to which intelligence is ascribed. Therefore, there NO unfalsifiable observation or prescription or formulation or criteria to which the *Ìṣẹ̀lẹ̀* ("Happening") of intelligence conforms.

2.2.1 What "Automated Intelligence" is NOT

The following are NOT in the purview of "Automated Intelligence":

(i.) Automated Intelligence (AI) is **NOT Engineering**. The aim in Engineering is to design and implement the solution appropriate to a problem in the context of its usefulness in human environment.

(ii.) Automated Intelligence (AI) is **NOT Science**. The aim in Science is to explain the human observation of the phenomenon of nature to the extent that measurement and theory permits.

(iii.) Automated Intelligence (AI) is **NOT Computation**. The aim in computation is to create and manipulate symbols in the process of constructing mechanisms for effecting the output to valid input strings.

In Automated Intelligence (AI) the aim is to create mechanisms for imitating the activities of biological agency to which intelligence is ascribed. The aim is achieved to the extent that modern Science, Engineering, Computation and Technology permits. Such mechanism is useful for the exploration of the performance/action of biological agencies. Nonetheless, the intelligence ascribed to a biological agency is grounded in the narrative expressed for the observation of its performance.

Therefore, AI mechanism are created to imitate or replicate the performance documented in a narrative. For example, the used of body organs, such as legs, to effective movement by human and animal, is situated in happening. An instrument of regular language is used to ascribe stepwise movement to the motion observed. Of course, the most efficient means to effect movement is through cyclic rotation. An attempt to effect efficient movement by imitating humans or animal such as Cheetah or Rabbit will NOT be as efficient as that of a Car or Train.

Of course some vehicles have four wheels like the legs of dogs of horses. But the movement of dogs and horses with their legs differs. Of course, in aeronautics, aeroplane have wings, but their wing are not flapping during flight like those of birds. An aeroplane is "a flying" mechanism while a bird is a flying organism. The creation of aeroplanes is an Engineering activity. If aeroplanes were to be an AI agency, then it will be created by imitate the flying of birds.

2.3 Kinds of the manifestation of *Ọgbọ́n* ("Intelligence")

There are Three (3) kinds of circumstances during which *Ọgbọ́n* ("Intelligence") is ascribed to the activity of organism within *Àròkún*. The activities include:

1. *Àdáṣe* ("Individual"),
2. *Ìdíje* ("Competitive"), and
3. *Àjọṣe* ("Cooperative")

These are briefly discussed in the following Subsections.

2.3.1 *Àdáṣe* ("Individual") Intelligence

Ọgbọ́n ("Intelligence") is ascribed to *Àdáṣe* (the "Individual") activity of a biological agency on the grounds of the creativity and/or ingenuity manifesting in his/her/its performance. This kind of intelligence manifest during the performance of a context-sensitive process. The performance of a Context-sensitive process cannot be planned and/or preplanned. This is because it comprises infinite instances of mutually inclusive and complementary states and locations. However, aspect of context-sensitive performance, that has been rendered and simplified through a context-neutral language, can be planned, expressed and revised.

Hence, *Ọgbọ́n* (the "Intelligence") ascribed to such a simplification comprises finite instances of mutually exclusive and contradictory states and location. Therefore, *Àdáṣe* (the "Individual") activity expressed through an instrument of context-neutral language will eventual become monotonic and stale after several repetition. A monotonic and stale process has lost its creativity.

The context-neutral rendering of a process is expressed through an instrument of regular language. A process formulated with an instrument of regular language renders constant instances of state or isolated instances of location. Therefore, a process rendered with a regular language is inherently deterministic. Creativity is outside the ambit of a deterministic process.

2.3.2 *Ìdíje* ("Competitive") Intelligence

Ọgbọ́n ("Intelligence") is ascribed to a biological agency during *Ìdíje* (a "Competitive") activity on the grounds of his/her/its ability to demonstrate better creativity and/or ingenuity than another biological agency of equal status.

A Game environment is contrive on the grounds that the admissible engagement and interaction between the competing agencies have been prescribed. Also, the prescription cannot be altered during the game. Therefore, a game is a monotonic environment through which creativity and ingenuity can be expressed to the extent of a player's familiarity and experience.

Aspect of the creativity deployed during a competitive process, formulated with a context-neutral language, is expressible through a regular language. Strategic games, such as *Ayò*, Go and Chess, are examples of *Ìdíje* (the "Competitive") activities during which *Ọgbọ́n* ("Intelligence") had been ascribe to individuals.

Competitive Equilibrium is the foundation of the ultimate state of consistency achievable during competitive interaction. At the state of Competitive equilibrium, new interaction scheme cannot emerge. At the state of competitive equilibrium, the interaction between the agencies becomes monotonic. A monotonic and stale process has lost its creativity. Hence, all the strategy in the universe of discourse of the interaction have been exhausted and no new strategy can emerge. When this occurs, the interacting agency's losses interest in the interaction.

2.3.3 *Àjọṣe* ("Cooperative") Intelligence

Ọgbọ́n ("Intelligence") is ascribed to *agbo* ("a group") of biological agency during *Àjọṣe* ("Cooperative") activity on the grounds of the creativity and/or ingenuity manifesting in their collective performance. *Ọgbọ́n* (The "Intelligence") ascribed to *Àjọṣe* ("Cooperative") activity is to the extent of coordination and coherence manifesting in the performances of the community of biological agencies. Of course, coherence is grounded in context-sensitive formulation. For example, *Ọgbọ́n* ("Intelligence") can be ascribed to the cooperative activity manifesting in the construction of *Ọ̀gán Ikán* ("Anthill") and *Afárá Oyin* ("Beehives"). *Ọgbọ́n* (The "Intelligence") ascribed to these biological agencies is grounded in human mental conception. The faculty of language with which to ascribe purpose to Anthill is inherent in Ants. The faculty of language with which to ascribe purpose to Beehives is inherent in Bees.

Fundamentally, the performance manifesting in *Àtorídá* (the "Creative" and/or "Ingenious") activity of a community of biological agency, does not conform to a standard set of criteria, prescription and/or formulation. The intelligence ascribed to a collective activity is contextual and its performance manifest during *Ìṣẹ̀lẹ̀* ("Happening").

Ìṣẹ̀lẹ̀ ("Happening") is *Lọ́gán* ("Spontaneous"), *Àìṣeétúnṣe* ("Unrepeatable") and *Àbálá* ("Irreversible" or "Irrevocable"). Indeed, *Àtorídá* ("Creativity" and/or "Ingenuity") is *Àìṣeésọ* ("'Inexpressible") on the grounds of the infinity inherent in its instance.

Cooperative Equilibrium is the foundation of the ultimate state of coherence achievable during cooperative interaction. At the state of cooperative equilibrium, new interaction scheme emerges. Therefore, the interaction between cooperating agencies is non-monotonic. Hence, cooperative agencies cannot run out of interacting strategy. The creation of new strategy motivates more interest in the interaction between the agencies.

2.3.4 Features of Organism imitated in "Computation Mechanisms"

Some of the features of organism imitated in Conventional Computation Mechanism or Machines are as presented in Table 2.1. However, computation machines are fundamentally different human, or other biological, agency. An Computation Mechanism, from its **Process** ("Hardware") down to its **Agency** ("Software"), is realised through various interconnected Switches, Relays, Components and Parts. In modern computation machines, these Switches, Components and Parts are manipulated through modulate electrical signals. The signals are manipulated to the extent that the technology used to realise its mechanism permits.

Various technologies have been deployed in the realisation of computation mechanism. These include: (i) Mechanical, (ii) Electromechanical, (iii) Electromagnetic, (iv) Transistor and Capacitor, (v) Integrated circuit electronic, (vi) Microelectronic and (vii) Nanotechnology. More recently, the idea of "quantum technology" is being touted. Note that, these technologies do NOT influence the Structure, Logic or Polarity of the mental activity of computing. For example, the logic of the Arithmetic operations of Addition $(+)$ and Multiplication $(\times)$ is neither influenced nor enhanced by the technology used to realise its computation. However, an advanced technology makes it possible to realise the Addition and Multiplication operations more efficiently (faster and with fewer or no errors). Indeed, a computation mechanism cannot ascribe Arithmetic and/or Logic to an operation.

Fundamentally, and based on the above explanation, the activity of "an organism" is neither equal nor equivalent to the "Mechanism" used to imitate it.

2.3.5 Similarities between AI and Conventional Computing

(i.) They are both prescribed and expressed through the Language of symbols, howbeit with different formulations.

(ii.) They are both realised and implemented as mechanism.

(iii.) Their realisation is to the extent that available technology permits.

(iv.) They are both used to implement prescribed processes.

(v.) They are both grounded in computational study.

Table 2.1: Human agency and material agency (computer functions) imitated

Ser. No.	Things human do	What humans use to do it	Imitation in computers
1.	*Pè* (Evoke) the state of the environment.	**Sensory organs**: Eyes, Ears, Nose, Skin, Tongue, *Òyì* (Balance).	**Input devices**: keyboard, mouse, scanner, camera, joystick and sensors.
2.	Register and store the instances of reckoned through sensory organs	**Brain and neurons:** (a) Current Information (b) Previous or past information (c)Intuition	**Material Memory and registers** (a) RAM (Random Access Memory) (b) Mass memory (c) ROM
3.	Manipulate instances in register and/or memory.	**Mental activity** {Computing, Thinking and Reasoning}	(a) Running program (Processor) (b) Running process
4.	Communication instrument.	Human language	Programming language.
5.	Things to affect/effect the environment	**Action**: Walk, speak, dance, etc.	Output devices and actuator
6.	Reckoning of desirable/undesirable state (i.e. correct/wrong thing)	**Conscience/** Awareness/ Reference/ Reflection/ Identity/ Sickness/ Pain	**Error**: program crashes, hangs
7.	Trigger for process commencement	Incoherence/ Inconsistency/ Imprecision / imminent danger, etc.	**Initial state**: An input symbol or string, power on, input device action.
8.	Trigger for process termination	**Reward**: pleasure, pay-off, self-satisfaction	**Final state**: printed output, Halt or Stop state
9.	Coordination of Interaction with others	Education/ Consensus/ Convention/Agreement	Protocol, Format, Coding scheme, and Standards
10.	Grounds of operation continuity	**Heart and Self-consciousness**	**Battery and timer**

(vi.) They both have a set of criteria for evaluation. Efficient is the criterial for evaluation in conventional computing. Extent to which its imitates an organism ("Mean Opinion score") is the criterial for evaluating AI systems.

2.3.6 Distinction between AI and Conventional Computing

The distinction between conventional "Computing" and "AI" is summarised in Table 2.2.

Table 2.2: Distinction between Conventional Computing and AI

Ser. No.	Feature	Conventional Computing	Automated Intelligence
1.	Input	Precisely defined and missing data not admissible	Ill-defined some level of missing or fuzzy data are tolerated.
2.	Output	Exactly one output for every input instance irrespective of the number of its applications to the computing process	Output may vary for an instance of input at different applications to the AI process. Output is tolerated within a range.
3.	Process	Deterministic and monotonic	Non-deterministic and non-monotonic
4.	Technique	Iterative and Algorithmic	Heuristics
5.	Evaluation	Efficiency, extent of computing resources such as memory, register, time used to accomplish a process	Extent to which the machine imitate the intelligence ascribed to an organism.
6.	Language of symbols	Regular or context-neutral	Context-sensitive.

2.3.7 *Àròkún* approach to creating agency of "Automated Intelligence"

As discussed in Subsection 2.2, the substance of the performance of an activity to which *Ọgbọ́n* ("Intelligence") is ascribed is innate to its biological agency. Therefore, *Ọgbọ́n* ("Intelligence") cannot be seconded and/or outsourced and/or replicated in another agency. Therefore, *Ọgbọ́n* ("Intelligence") *can neither be learnt nor taught*. The idea of "learning or teaching a machine", or even humans, to become "intelligent" is, therefore, NOT admissible within *Àròkún*. Indeed, there is neither a "curriculum for teaching or learning" nor "a finite set of criteria for the assessment" of intelligence. The competency with which to ascribe intelligence or creativity to a performances is situated in human *Àfiyèsí* ("Self-awareness"). The universe of human *Àfiyèsí* ("Self-awareness") is spontaneous, unrepeatable and irreversible. Therefore, every instance in the sensory accounts of the performance of intelligence is *Ọ̀tun* ("New").

From the *Àròkún* perspective, therefore, the meaningful approach to the creation of a computational mechanism for imitating the intelligence ascribed to the activities of a biological agency comprises the following steps:

(i) Determine the distinction between an Organism (Biological agency) and a Mechanism (Material agency).

(ii) Carefully observe a biological agency (Ag_I) in $(i.)$ to which Intelligence had been ascribed on the grounds on the conformity of its performance/actions with a set of criteria (I_C). Also, identify another biological agency (Ag_U), of the kind Ag_I, that is certified to be unintelligent (or stupid) on the grounds of the non-conformity of its performance/actions with the criteria C_I;

(iii) Reduce the observation and criteria in (ii) into a succinct narrative using an appropriate instrument of language.

(iv) Using the narrative in (iii), create a procedure (P_I) by which the unintelligent agent Ag_U can be transformed into an intelligent agent.

(v) Represent and refine the procedure in (iv) into a process (P_s) using a Pseudo language of symbols.

(vi) Formulate, design and implement an Automated Mechanism for the process in (v) to the extent that technology permits. Assess and refine the Automated Mechanism based on the extent to which it imitates the agency described in (iii).

2.4 Social issues in AI

Debates on the merits and demerits of AI in the context of its applications in human society still rages within and outside the AI community. Of focus is the social issues that will arise in the consequent that AI systems are able to perform intelligent tasks at human-level of competence. The use of AI machine in the stead of humans is very scary for many individual. The fears are justified on the grounds of humans experience of the application of previous advances in technology such as Mechanical, Nuclear, Genetics and, indeed computing.

Whereas some social perspective are positive and supportive of AI activities others are negative and oppose AI activities. These are listed as the following Sections.

2.4.1 Arguments supporting AI activities

The supporting or positive view of AI activities are:

(a.) Increase productivity through better decision (e.g. Decision Support System);

(b.) AI system have been demonstrated to be useful in dangerous operations that are risky for human to perform, e.g. exploring volcanoes, earthquake, bomb detonation, etc;

(c.) Development of AI system will provide useful tool for better understanding "human intelligence";

(d.) Recreation, e.g Playing strategic games with AI;

(e.) Useful in Computer Assisted learning, e.g Computer Assisted Language Learning (CALL);

(f.) Useful in the support for the physically challenged (E.g. speech synthesis and speech recognition applications).

2.4.2 Arguments opposing AI activities

The opposing or negative view of AI product are:

(a.) Reduction in the quality of human interaction;

(b.) Possibility of abuse or misuse;

(c.) Effect on the value of human labour, e.g. reduction in the need for human labour when machine with human-level competency is available. This can culminate in the loss of jobs for many humans;

(d.) Danger of faulty systems; the guiding principle is that "*if it can fail, it will eventually*".

(e.) Fear of machine superiority; This motivate **Isaac Asimov** to set the following rules for Robots:

  (i.) A robot must not, by action or inaction, cause a human being to be harmed.

  (ii.) A robot must obey all the command given by humans except when it violates rule (i.) above

  (iii.) A robot must defend itself except when it violates rule (i.) and rule (ii.).

> "But remember, please, the Law by which we live, We are not built to comprehend a lie, We can neither love nor pity nor forgive, If you make a slip in handling us you die! We are greater than the People or the King, Be humble, as you crawl beneath our rods! Our touch can alter all created things' We are everything on earth- except the Gods! Rudyard Kipling (2020)

(f.) Social consequence of alien intelligence in human community (e.g. rule of engagement and interactions);

(g.) Moral issue; who should and how to determine the morality of intelligent machines;

(h.) Religious concerns; the idea that "God created humans and has not asked humans to re-create themselves or another kind of humans"

CHAPTER 3

ÈDÈ ÀHUNKALẸ̀ (“FORMAL LANGUAGE”) IN AI

The grounds on which Automated Intelligence is ascribed to a mechanism are discussed in Sections 2.1, 2.2 and 2.3.7. “Identity” and “Binary opposite” are the foundation of the formulation and prescription of the process effected by an “Agency”. The process effected by an “Automated computing agency” is prescribed and formulated using a Pseudo language of symbols. A Pseudo language is used to imitate an instrument of language.

Èdè Àhunkalẹ̀ (“A formal language”) is used to prescribe and represent the process effected by *Ìṣẹ̀dọ* (“Mechanisms”). *Àhunkalẹ̀* (“A formalism”) is a standard and/or conventional prescription to which the rendering and expression a process must conform. The closest word in information content to the *Yorùbá* expression *Àfihun Èdè* is “Language Grammar”. A Grammar is used to prescribe and/or formulate the arrangement of Symbols in a String or the admissible sequence of string in a Sentence. A String is used to represent a term while a Sentence is used to represent an Expression.

3.1 “Formal language” for AI Process

Àfihun Èdè (“The Grammar of language”) is used to formulate *Èdè-Àfihun* (“A pseudo-language”). The grammar for prescribing *Èdè Àhunkalẹ̀* (“A formal language”) comprises Four (4) aspects, namely:

(i.) Inventory of “Symbols” constituting its Alphabet. Symbols are language neutral. The are assigned to, and used to construct, **Primitive-term**.

(ii.) Inventory of “Strings” constructed using the symbols drawn from the Alphabet in (i). Each valid string is in the dictionary of the Pseudo language of symbols. A nominal-term is a string constructed through the concatenation symbols (Letter). A nominal-term is used to ascribe identity to an instance, agency, location entity, and so forth. **A nominal-term is a Primitive-term.** A String (or Primitive-term) is used to ascribe nominal identity to an instance. A string that has been used to ascribe identity to an instance is an **Auxiliary-term** of the Pseudo-language.

(iii.) Inventory Variable-term into which the instance of Strings in the dictionary are categories. This corresponds to Part-of-speech, Such as “Noun”, “Verb”, “Adverb”, “Adjective” , “Sentence”, an so forth, in conventional grammar. The Variables constitute the **Auxiliary-axiom** of the Pseudo-language. A distinguish Auxiliary-axiom is used to Reckon and abduct an instance of Prime-axiom into universe of discourse. This corresponds to the “Sentence” or “Pattern” in conventional grammar.

(iv.) The entirety of the terms and expressions that can be constructed with the Grammar is represented with **Prime-axiom** of the Pseudo-language.

A set of regular prescription for the construction of strings from the Alphabet and A set of context-neutral prescription for constructing expression from the Dictionary. Base on the above explanation, a regular language is used to render the connection between Items (i) and (ii). A Context-neutral language is used to render the connection between Items (ii) and (iii). The Context-sensitive language is used to render the connection between Items (iii) and (v). Through this formulation, the totally countable instances of Primitive-terms in an alphabet is used to generate the dictionary comprising finite sets of “String” or “Motif”. The finite sets of “String” or “Motif”, in the vocabulary, are, in turn, used to generate infinite instances of “Sentences” or “Pattern”. Each valid “Sentence” or “Pattern” reckons One (1) of the possible infinite instance represented by the Prime-axiom of the Context-sensitive pseudo-language defined by the grammar.

In *Àtọlẹ̀dọ́lẹ̀* (“the Organic”) metaphor based formulation, Regular grammar is subsumed in Context-neutral grammar which is, in turn, subsumed in Context-sensitive Grammar. Based on the above explanation, the regular and context-neutral aspects of a context-sensitive pseudo-language can be independently prescribed, tested and then combined into one Grammar.

Individual “String” or “Motif” is constructed using the regular aspect of the formal language while individual “Sentence” or “Pattern” is constructed with the context-neutral aspect of the language. The Prime-axiom of the context-sensitive universe is used to impose constraint on admissible regular terms (String) and/or context-neutral expression (Sentence).

Grammar is to “Formal language” what “Polynomial is to Algebra”. Indeed, A grammar can be prescribed to express an Algebraic process.

3.1.1 Definition of formal language

Èdè Àhunkalẹ̀ (“A formal language”) is defined with a Four-Tuple grammar. The grammar for Temporal and Spatial aspect of universe of discourse are expressed in Equation 3.1 and 3.2, respectively.

$$FL ::= \langle \mathbf{T}, \mathbf{A}, \mathbf{R}, \mathbf{G} \rangle \tag{3.1}$$

$$FL ::= \langle \mathbf{T}, \mathbf{A}, \mathbf{C}, \mathbf{G} \rangle \tag{3.2}$$

where

(i.) **T** are *Ohùn* (the "Terms") ("Nominal-term" or "Numerical-term" or "Hybrid-term") used to cue the instances in a universe of discourse. *Ohùn* ("Terms") are instance-neutral. A term is immutable and unique to its universe of discourse.

(ii.) **A** are the Assertions used to ascribe "constant state" or "isolated location" to the individual instances of term in a universe of discourse. An Assertion is used to ascribe identity to an instance. Assertion is formulated with *Unary operator* of the form $f(x)$. This implies that the identity $f()$ is ascribed to the instance x. The identity ascribed to an instance is neutral to the binary opposite rendering and manipulation in the universe of discourse. Therefore, Assertion is the grounds of which *Nǹkan* ("Sameness" or "Thing") is ascribed to an instance in a universe of discourse.

(iii.) **R/C** are the mutually exclusive and contradictory binary opposite "Relations" or "Connectors" for rendering the identities ascribed to instances a universe of discourse. A context-neutral **Relation (R)** or **Connector (C)** manipulated Two (2) instances of identities. Context-neutral relation is rendered with temporal axioms. Temporal axiom is formulated with the language tools logic and polarity. Relational **Binary operators** are used to manipulate the state of constant instances. Context-neutral connection is rendered with spatial axioms. Spatial axiom is prescribed with the language tools Structure and Polarity. Connection **Binary operators** are used to manipulate the location of isolated instances. The formulation $F(x, y)$ depicts a context-neutral relation between a pair of constant states or isolated locations. Context-neutral instances are finite. **Relation (R)** or **Connection (C)** used to render and simplify a Context-sensitive instance comprises Three (3) instances variables (context-neutral terms). The formulation $F(x, y, z)$, depicts a context-sensitive relation or connection between three instances of variable state or locations. Context-sensitive instances are infinite.

(iv.) **G** is the set of Quantifiers used to "Constrain" or "Contextualise" the admissible treatments of "Assertion", "Relation" and "Connection" in a meaningful expression. A Quantifier is the template for "the reckoned" and "yet to be reckoned" instances. Therefore, the instance reckoned with a Quantifier is potentially infinite. Indeed, Quantifier is used to simplify the infinity inherent in *Gbogbo-ohun* ("Everything") into finite instances of *Àwọnohun* ("Something").

The **Operators** ("Assertion" (Nominal or Numeral), "Relation" (Temporal) or "Connection" (Spatial)) for creating and manipulating instances in the universe of discourse of a Formal Language are listed and explained in Table 3.1.

The Operators for relating Operand (Terms) in Formal Language are listed and explained in Table 3.2.

Fundamentally, *Asán* ("Nothing" or "No thing") is NOT admissible in *Èdè Àhunkalẹ̀* ("A formal language"). However, the sensory accounts of *Òfo* ("Empty") cued with *Asán* ("Nothing" or "No thing") is used to indicate *Àlàfo* ("Gap") between a pair of spatial location. *Àlàfo* ("Gap") is used to render *Ìṣí* (the "Interval") between a pair of states. "Blank" is used to reckon the spatial accounts of *Àlàfo* (a "Gap"). *Òdo* (Zero of (0)) is used to render the temporal accounts of *Ìṣí* (an "Interval"). "Blanks" are used to create the placeholder into which the terms in an expression are located. Every

Table 3.1: Formal language Operators for creating and manipulating Operands (*Ohùn*/Terms)

Ser. No.	Operator	*Yorùbá*	English	Operand	Used to
1.	$!\forall$	*Ohun*	Term	Instance neutral	The String used to cue an instance in the universe of discourse. This is the name of the instance being cued.
2.	$\bigtriangledown$	*Nǹkan*	One	Exactly one instance	Assertion used to ascribe identity to an instance of "State" or "Location". An assertion is used to abduct label or string into universe of discourse.
3.	$\bigtriangleup$	*Àwọn*	Some	Two or more, but finite, instances	Relate states or Connect Locations. A pair of constant state or spatial locations are reckoned together.
4.	$\forall$	*Gbogbo*	Every	All instances. Infinity is inherent.	Interrelate and Interconnect States and Locations in the universe of discourse. Three variables are reckoned simultaneously.

term is used to cue an instance in the universe of discourse. *Òdo* (Zero of (0)) is used to cue a logic neutral instance.

Assertion is used to ascribed identity to a term. Indeed, *Ohùn* ("A term") is instance-neutral. This implies that *Ohùn* ("A term") neither influences nor obstructs nor enhances nor obscures the instance ascribed to it. *Orúkọ* ("A Nominal-term") is used to cue the location of an instance. *Oǹkà* ("A Numerical-term") is used to cue the state of an instance.

Negation or Inversion, that is NOT ($\neg$ or !) is used to render the binary opposite of the identity ascribed to an instance. The temporal binary opposite of *Akọ* ("Emitter" or "Positive") is *Abo* ("Receptor" or "Negative") and *vice versa*. The spatial binary opposite of *Ojú* ("Obverse") is *Òdì* ("Reverse") and *vice versa*. The Disjunction (OR ($\vee$)) operator is used to Unrelate or Disconnect the Assertion ascribed to instances in a universe of discourse. The Disjunction (OR ($\vee$)) operator is used effect Separation, Subtraction, Differentiation during *Ìmọ̀tú* ("An analysis") process.

The Conjunction (AND ($\wedge$)) operator is used to Relate or Reconnect the Assertion ascribed to instances in a universe of discourse. The Conjunction (AND ($\wedge$)) operator is used effect Accumulation, Addition, Integration during *Ìmọ̀jọ* ("A synthesis") process.

As depicted in Figure 3.1, in *Àtọlẹ̀dọ́lẹ̀* ("the Organic") rendering of *Èdè Àhunkalẹ̀* ("Formal language"), content-sensitive Quantifiers (**G**) subsumes context-neutral Relation (**R**) or Connection

Table 3.2: Formal language Operators for Relating or Connecting instances

Ser. No.	Symbol	*Ohùn*	Term	Operator	Used to
1.	$\wedge$ / ,	*ÀTI/PẸLÚ*	AND	Conjunction	Effect Accumulation.
2.	$\vee$	*TÀBÍ*	OR	Disjunction	Effect Differentiation.
3.	$\neg$ / !	*KỌ̀/KỌ́*	NOT	Negator	Effect Reversal or Negation.
4.	$\Rightarrow$	*JÁSÍ*	IMPLY	Consequence	Indicate the outcome of Assertion, Relation or Connection.
5.	:	*JẸ́PÉ*	SUCH THAT	Precedence	Express constraint.
6.	$\in$	*WÀNÍ*	INSTANCE OF	Ascription	Indicate that an instance is a member of a group or set. For example $x \in X$ indicates that x is a constant instance in the Variable X.

(**C**) which, in turn, subsumes regular Assertion (**A**). Assertions are expressed through instances of terms (**T**). Figure 3.2 is *Àtọlẹ̀dọ́lẹ̀* ("the Organic") formulation for Figure 3.1.

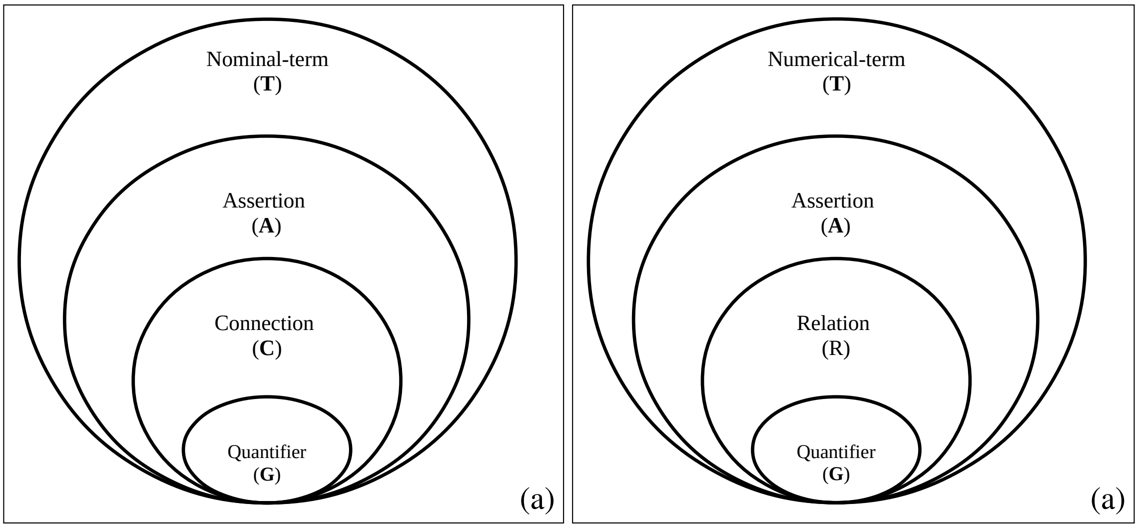


Figure 3.1: Organic rendering of Quantifier, Relation/Connection, Assertion and Term

3.1.2 The treatment of Major and Minor operands

The first operand in the expression of a formulation is its "Major operand". Other operand in the formulation are "Minor operand". This implies that, a "Minor" operand is such on the grounds of the "Major operand".

An Assertion is rendered and expressed through an instrument of regular language. An Assertion takes only one operand. Therefore, the operand of an Assertion is its Major operand. In the

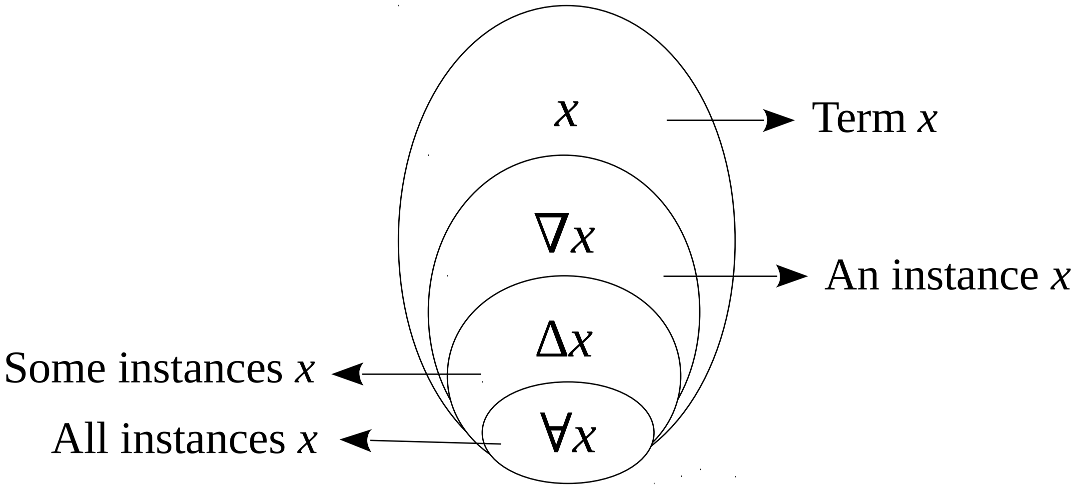


Figure 3.2: Organic formulation of Quantifier, Relation/Connection, Assertion and Term

formulation $f(b)$, for example, $f()$ is the Assertion or Function and b is the major operand.

A Relation or Connection is rendered and expressed through an instrument of context-neutral language. A Relation or Connection takes two(2) operands. Therefore, the first operand of A Relation or Connection is its Major operand. In the formulation $f(b, a)$, for example, $f()$ is the A Relation or Connection, b is the major operand and a is the Minor operand. The arrangement of "Major operand" and "Minor operand" in A Relation or Connection conforms to mutually exclusive and contradictory treatment of binary opposite. Therefore, when the arrangement of operand is reverse, a binary opposite state or location is expressed. For example, the expression "b is to the left of a" is formulated as $left(b, a)$. The formulation $left(a, b)$ is a contradictory expression that "a is to the left of b".

A Quantification is rendered and expressed through an instrument of context-sensitive language. A Quantification takes Three(3) operands. Therefore, the first operand in A Quantification is its Major operand. The other operands are its minor operator. In the formulation $f(b, a, c)$, for example, $f()$ is the A Quantification, b is the major operand while a and c are the Minor operands. The arrangement of "Minor operands" in A Quantification formulation conforms to mutually inclusive and complementary treatment of binary opposite. For example, if b, a and c are ascribed to instances in the spatial locations "Middle", "Left" and "Right", respectively, then the formulation $Middle(b, a, c)$, states that, "b is in the middle of a and c". A complementary formulation $Middle(b, c, a)$, which is, "b is in the middle of c and a" is also admissible.

The use of $\exists$ ("There exits") is NOT admissible in *Àròkún* formulation of *Èdè Àhunkalẹ̀* ("Formal language"). *Ìwà* ("Existence") and *Àìwà* ("Non-existence") are the mutually inclusive and complementary aspects of an instance. Mutually inclusive and complementary binary opposite is formulated using an instrument of context-sensitive language. *Àìlópin* ("Infinity") is inherent in a context-sensitive instances. An infinite instance is inexpressible. An inexpressible instance CANNOT be formalised.

CHAPTER 4

ÀDÁṢE (INDIVIDUAL INTELLIGENCE): KINSHIP NARRATIVE

Regular, Context-neutral and Context-sensitive temporal relation is best demonstrated through the human *Ìbátan* ("Kinship") universe of discourse. Assertions, Relations and Quantification can be used to render the terms used to cue individual human in the universe of discourse. The Assertion ascribed to individual terms can then be formulate and express Relation between humans. The relation can the be rendered into informative explanation through Quantification. This way, Regular, Context-neutral as well as aspects of Context-sensitive rendering of the intelligence ascribed to *Àdáṣe* ("Individual") human performance can be illustrated in the context of the capacity to use language.

Though *Ìdíje* ("Competition") and *Àjọṣe* ("Cooperative") can also be demonstrated through a dialogue system, *Àdáṣe* ("Individual") is the focus of the exercise in this Chapter.

The English and *Yorùbá* kinship universes of discourse are used to demonstrate the Formal Language rendering of "Automated Intelligence".

4.1 The English Kinship universe of discourse

The expression constructed for instances in a universe of discourse is to the extent of its habitual instrument of language. By virtue of the information contents in message communicated with the Instrument of English language, by its habitual human user, its Prime-axiom is situated in the context-neutral universe of discourse.

Therefore, while consistency is a desirable attribute, Primitive-terms are treated as though they confer logic and/or polarity on English language expression. However, the Depth of an expression grounded in Primitive-terms does NOT extent beyond the axioms of its regular language. The axioms of an Instrument of regular are situated the Auxiliary-term of the human universe of discourse.

4.1.1 Primitive-terms

The primitive in formal language rendering of the English kinship universe of discourse are the letters in written English language alphabet.

4.1.2 Construction of Nominal-terms

The nominal-terms of the English kinship universe of discourse are cued with the string constructed with the English alphabet.

Given that "Ade", "Akin", and "Kemi" are the strings used to cue the terms in the English Kinship universe of discourse. Strings are a concatenation of symbols. Since each symbols is a primitive, then strings constructed with them are also primitive. Every instance of primitive is a primitive.

Therefore the strings, "Ade", "Akin", and "Kemi" are used to cue terms in the English kinship universe of discourse.

Other standard are used to cue the symbols ascribed to Assertion, Relation and Quantification. The following convention is adopted in the rendering of these instances:

(i.) A symbol assigned to a primitive is enclosed in single quote, that is ' ', for example 'A' and 'b';

(ii.) "A string of symbols" used to cue a term is enclosed in double quote, that is " ", for example "Ade";

(iii.) The symbol used to cue a term is represented with lowercase a letter, for example x, y. For example, x = "Ade", y = "Kemi"

(iv.) A variable-term is represented with uppercase letters, for example X, Y. A variable-term comprises finite instances of term. For example, $X = \{x_1, x_2, \ldots x_n\}$ and $Y = \{y_1, y_2, \ldots y_m\}$

(v.) A term is Assertion neutral. Assertion is ascribed to a term through the axiom of an instrument of regular language. This is the Auxiliary-terms of the universe of discourse. Assertion is binary opposite neutral. Regular language axiom is represent with lowercase function, for example f().

(vi.) Binary opposite is ascribed to Assertions through the axiom of Context-neutral language. This is the Auxiliary-axioms of the universe of discourse. Context-neutral language axiom is represent with uppercase function, for example F(). The simplification of Context-sensitive axiom, through quantifiers, is represent with bold uppercase function, for example **F**(),

Specifically:

1. f("Kemi") and f("Ayo") are regular Assertion on through the string used to reckon an isolated instance.

2. F("Kemi", "Ayo") is a context-neutral formulation over a pair of regular Assertions.

3. If $x =$ "Kemi" and $y =$ "Ayo", then x and y are regular terms. Therefore, $F(x, y)$ is a context-neutral formulation on regular terms.

4. If $x \in X$ and $y \in Y$, then **F(X, Y)** is a context-neutral quantification or simplification of a sensitive instance.

4.2 Assertions (Auxiliary-term)

An Assertion is a unary operator. Therefore, Assertion admits only one. The only operand in an Assertion is therefore a "Major operand".

4.2.1 Single instance Assertion

For example, $boy()$, $girl()$, $man()$, $woman()$, $human()$, $mortal()$ are unary operator for ascribing Assertion to terms. For example, the sentence "Ade is a boy" will be formally rendered as $boy(``Ade'')$. Here "Ade" is the Major operand and $boy()$ is its Assertion. Note that $\neg boy(``Ade'')$ ("Ade is not a boy") does not imply that "Ade is a girl", except this is formally prescribed through a Quantifier.

4.2.2 Multiple instance Assertion

Two or more Assertions can be ascribed to a term using Conjunction and/or Disjunction. For example, the expression "Ade is a boy and Ade is human" is formally expressed as $\boxed{\text{boy (``Ade'')} \wedge \text{human (``Ade'')}}$. This can also be recursively expressed as $\boxed{\text{boy (human (``Ade''))}}$.

The expression "Ade is a tall, brilliant boy" is formally expressed as $\boxed{\text{boy (``Ade'')} \wedge \text{brilliant (``Ade'')} \wedge \text{tall (``Ade'')}}$.

This can also be recursively expressed as $\boxed{\text{boy (brilliant (tall(``Ade'')))}}$. Logic OR cannot be rendered recursively. For example "Ade is young but/or tall" cannot be expressed recursively.

4.3 Relations (Auxiliary-axioms)

A context-neutral relation admits exactly Two (2) terms (operands) with which Assertion had been ascribed. The first of the two terms (operands) is the Major while the other is a Minor operand. $Son()$, $Daughter()$, $Mother()$, $Father()$, $Brother()$, $Sister()$, $Previous()$, $Next()$, $Left()$, $Right()$, $Front()$, $Back()$ are examples of binary relation.

For example, the sentence "Akin is the father of Ade" will be formally rendered as $\boxed{Father(``Akin'', ``Ade'')}$. In this formulation Akin is the Major operand while Ade is the minor operand. Also the sentence "Ade is the son of Akin" will be formally rendered as $Son(``Ade'', ``Akin'')$. In this formulation Ade is the Major operand while Akin is the minor operand.

4.3.1 Using relation to ascribe Assertion on term

When a Relation is used to ascribe Assertion on a term, a variable instance results. A variable instance comprises finite instances of term. The term to which the Relation is used to ascribe Assertion on is one of the admissible instances of a variable instance.

For example, the statement "Kemi is a mother", formulated as $\boxed{\text{Mother (``Kemi'')}}$, is a Relation ascribed to a term. The term "Kemi" is used to cue the identity of one of several instances of $\boxed{\text{Mother()}}$. Therefore, the formulation $\boxed{\text{Mother (``Kemi'')}}$ is an abstraction and NOT an expression.

For the relation $Mother()$ to become an expression, its Two (2) instances of terms must be specified. Also, the statement “Ade is a son” is formulated as $Son(``Ade'')$ is an abstraction. An abstraction requires further simplification to become an expression.

However, formulation $Mother(``Kemi, Ade'')$ is the expression “Kemi is the mother of Ade”.

4.3.2 Using relation to Simplify context-sensitive instance

A context-sensitive relation admits exactly Three (3) terms (operands) with which Relation or Assertion had been ascribed. Therefore, a context-sensitive instance is rendered with a ternary formulation. The first item is the Major operand while the other two are Minor operands.

Examples of context-sensitive relation include,

1. **Current** (),
2. **Middle** (),
3. **Between** ().

For example, the statement “Akin is between Ade and Kemi” will be formally rendered as **Between** (“Akin”, “Ade”, “kemi”). In this formulation “Akin” is the Major operand while “Ade” and “Kemi” are the Minor operands. The formulation of **Between ()** is not affected by swapping the location of the minor operands “Ade” and “Kemi”.

A context-sensitive relation can be simplified into a pair of consistent context-neutral formulations.

For example,

Between (“Akin”, “Ade”, “Kemi”)

is simplified into instance of context-neutral formulation using the binary opposite relation $Left()$ when the Right constrain is imposed. The is formulated as:

Between (“Akin”, “Ade”, “Kemi”) $\rightarrow$ Left (“Ade”, “Akin”) $\wedge$ Left(“Akin”, “Kemi”). The expresses that “Akin is between Ade and Kemi when Ade is to the left of Akin and Akin is to the left of Kemi”.

When the Left constrain is imposed, then

Between (“Akin”, “Ade”, “Kemi”) $\rightarrow$ Right (“Kemi”, “Akin”) $\wedge$ Right (“Akin”, “Ade”)

(Kemi is right of Ade and Ade is to the right of Akin) is formulated.

Attempt to use binary opposite Relations to formulate Context-sensitive relation will culminate in inconsistency.

For example, if the binary opposite $Left()$ and $Right()$ are combined in the simplification of $\mathrm{Between}()$, inconsistent formulation will result.

Terms are the only admissible operand in the definition of Assertions and Relations.

4.4 Quantification (Prime-axiom)

Quantifiers used to impose constraint on the Relation rendered through Variable operand. “Unary Assertion” and “Binary relation” are constrained through quantifiers. For example, the expression “A man has a mother” is formulated as

$\boxed{\text{man(X)} \rightarrow \triangledown \text{ Y. Mother(Y, X)}}$. "Every man has a mother" is formally expressed as $\boxed{\forall\text{X. man(X)} \rightarrow \triangledown\text{y. Mother(Y, X)}}$.

"Some men are fathers" is formally expressed as $\boxed{\triangle \text{ X. man(X)} \rightarrow \text{Father(X)}}$. "All fathers are male" $\boxed{\forall\text{X.Father(X)} \rightarrow \text{male (X)}}$. The expression "A child can either be a male or a female" is expressed as $\boxed{\triangledown\text{X.child (X)} \rightarrow \text{male(X)} \vee \text{female(X)}}$

The Minor and Major operands in a context-sensitive relation are mutually inclusive and complementary. When a context-sensitive relation is constrained, the Minor operands are location neutral. Therefore, the position of the minor operand can be swapped without affecting the message expressed. For example, $\boxed{\textbf{Married}\text{ ()}}$ is a context-sensitive relation between two (2) instances through a process called Wedding. The expression "X and Y are married through the Wedding process W" is formally expressed as $\boxed{\textbf{Married}\text{ (W, X, Y)}}$. If the wedding process is constrained to that between Two (2) individuals only, then the expression "X is married to Y" is equivalent to "Y is married to X" $\boxed{\text{Married (X, Y)} \rightarrow \text{Married(Y, X)}}$. This swapping of terms is NOT admissible in a context-neutral formulation. Indeed, swapping of terms will culminate in inconsistent expression in a context-neutral formulation.

Also, "X and Y are sibling" is equivalent to "Y and X are sibling"

$\boxed{\text{Sibling(X, Y)} \rightarrow \text{Sibling(Y, X)}}$

when Parent is used to constrain the relation. Constraint can also be used to further restrict the relation. For example, if the Assertion $Male()$ is used to restrict $Sibling()$, then "x and y are brothers" is equivalent to "y and x are brothers"

$\boxed{\text{Brother(X, Y)} \rightarrow \text{Brother(Y, X)}}$.

In this case the assertions $Male(X)$ and $Male(Y)$ is implied

4.4.1 Transitive relation

Transitive relation is used to express the interrelation between more that Two (2) individual.

For example, the statement "The father of a father is a grandfather" interrelates Three (3) individuals X, Y and Z. In that interrelation, if X is the father of Y and Y is the father of mother of Z then, Y is the grandfather of X

$\boxed{\text{Grandfather (X, Z)} \rightarrow \text{Father (X, Y)} \wedge \text{(Mother (Y, Z)} \vee \text{Father (X, Y))}}$

This can be generalised as

$\boxed{\forall\, x \in X, y \in Y, z \in Z, \text{Grandfather (x, z)} \rightarrow \text{Father (x, y)}}$

$\boxed{\wedge \text{ (Mother (y, z)} \vee \text{Father (y, z))}}$

Also, if the gender of the parents is ignore the expression "X is the grandparent of Y if Y is the parent of Z and Y is the parent X". This culminates in statement "The parent of a parent is a grandparent.

$\boxed{\text{Grandparent (X, Z)} \rightarrow \text{Parent (Y, Z)} \wedge \text{Parent (X, Y)}}$

Two (2) individuals X and Y are Siblings if they are of a Parent P.

$\boxed{\text{Sibling (X, Y)} \rightarrow \text{Parent (P, X)} \wedge \text{Parent (P, X)}}$. Expresses that "X and Y are sibling P is the parent of Y and P is the parent of X"

Parent () is further formulated as a Mother () or Father ().

$\boxed{\text{P(X, Y)} \rightarrow \text{Father (X, Y)} \vee \text{Mother (X, Y)}}$. Expresses that "X is the parent of Y if X is the father of Y or X is the mother of Y"

Brother() is mutually inclusive when male() is imposed on its terms,

$\boxed{\forall\ \text{(male(X)} \wedge \text{male(Y)): Bother (X, Y)} \rightarrow \text{Bother (Y, X)}}$

$\boxed{\text{Brother (X, Y)} \rightarrow \text{male (X)} \wedge \text{Parent (P, X)} \wedge \text{Parent (P, Y)}}$

or

$\boxed{\text{Brother (X, Y)} \rightarrow \text{male (X)} \wedge \text{Sibling (X, Y)}}$

4.5 *Yorùbá* kinship universe of discourse

Yorùbá language expressions, including that of the universe of kinship discourse, are formulated through *Kóró-Igi* (the "Seed-Tree") aspect of *Àtọlẹ̀dọ́lẹ̀* ("the Organic") metaphor. An expression is rendered from *Ojú-odù* ("the Prime-axiom") towards its *Ìtẹ-odù* ("Primitive-terms") or *Ohùn* ("Terms"). *Ohùn* (The "Terms") in *Yorùbá* kinship universe of discourse are used to cue *Ìdámọ̀* ("the identity") of individual, group as well as the ancestry or progenitor of *Ènìyàn* ("Human agency").

In the *Yorùbá* kinship universe of discourse, *Ìwọ̀n* ("Quantifiers") are used to render and restrict potentially infinite relationship. *Ẹbí ()* is the Prime-axiom of the *Yorùbá* kinship universe of discourse. Every other formulation is grounded in *Ẹbí ()*.

Kindred is the English language word closest in information content to the *Yorùbá* term *Ẹbí*. *Ẹbí* is, however, formulated through a relationship seeded and rooted in *Alálẹ̀* ("Ancestry"). *Iyèkan* is used to contextualise the formulation of kinship relation on *Ìyá* (the "Mother"). *Ọbàkan* is used to contextualise the formulation of kinship relation on *Bàbá* (the "Father"). Of course, *Iyèkan* ("Mother's root") and *Ọbàkan* ("Father's root") are the mutually inclusive and complementary binary opposite aspects of an instance of *Ènìyán* ("Human agency").

4.5.1 Relations (Auxiliary-Axioms)

Relations in *Yorùbá* kinship discourse are defined over variable Nominal-Term, NOT isolate Nominal-term. A variable Nominal-term is a set comprising instances of Isolated Nominal-term.

The *Ọmọ()* relation

(i.) X is the Ọmọ of P, if X and P share ancestry and P is old enough to be the *Òbí* ("Parent") of P.

$\boxed{\text{Ọmọ (X, P)} \rightarrow \text{Ẹbí (E, X, P)} \wedge \text{Òbí (P, X)}}$

(ii.) The formulation of an individual *Ọmọ* ("Child") $x \in X$ is rendered as:

$\boxed{\text{Ọmọ (x,P)} \rightarrow \text{Ẹbí (E, x, P)} \wedge \text{Òbí (P, x)}}$ This formulation expresses that, "every instance $x \in X$ in the Ẹbí (E) that is young enough to be the *Ọmọ* of P is his/her *Ọmọ*.

(iii.) The formulation of a set of *Ọmọ* ("Child") belonging to an individual *Òbí* ("Parent") $p \in P$ is rendered as: $\boxed{\text{Ọmọ (X, p)} \rightarrow \text{Ẹbí (E, X, p)} \wedge \text{Òbí (p, X)}}$

This formulation expresses that, "every instance $p \in P$ in the Ẹbí (E) that is old enough to be the *Òbí* of X is his/her *Òbí*.

(iv.) The formulation of an individual *Ọmọ* ("Child") $p \in X$ belonging to an individual *Òbí* ("Parent") $p \in P$ is rendered as:

$\boxed{\text{Ọmọ (x, p)} \rightarrow \text{Ẹbí (E, x, p)} \wedge \text{Òbí (p, x)}}$

(v.) The formulation of *ỌmọỌmọ* ("Child of Child" or "Grandchild") rendered with generalisation through Three (3) variables X, Y and Z:

$\boxed{\text{Ọmọọmọ (X, Z)} \rightarrow \text{Ọmọ (X , Y)} \wedge \text{Ọmọ (Y , Z)}}$

This formulation is generalised as

$\boxed{\forall\, x \in X\text{: Ọmọọmọ (x, Z)} \rightarrow \text{Ọmọ (x , Y)} \wedge \text{Ọmọ (Y , Z)}}$

The *Ìyàwó ()* relation

Obìnrin ("A woman") is the *Ìyàwó* in the *Ẹbí* of her *Ọkọ* on the grounds of agreement between the Two (2) *Ẹbí*. The coming together of the woman's *Ẹbí* and that of her *Ọkọ* is established on an irrefutable and irrebuttable obligation established through *Ànọn* (M).

(i.) If E_H is the *Ẹbí* of the *Ọkọ* identified as X and E_W is the *Ẹbí* of the *Ìyàwó* identified as Y, then the formulation is rendered as follows:

$\boxed{\text{Ìyàwó (Y, X)} \rightarrow \text{Ànọn (M, } E_W, E_H) \wedge \text{Ẹbí (} E_W\text{, Y)} \wedge \text{Ẹbí (} E_H\text{, X)}}$

(ii.) If $x \in X$ is an instance of *Ọkọ* in the *Ẹbí* E_H and E_W is the *Ẹbí* of the *Ìyàwó* identified as Y, then the formulation is rendered as follows:

$\boxed{\text{Ìyàwó (Y, x)} \rightarrow \text{Ànọn (M, } E_W, E_H) \wedge \text{Ẹbí (} E_W\text{, Y)} \wedge \text{Ẹbí (} E_H\text{, x)}}$

4.5.2 Auxiliary-Terms (Assertion)

As the *Yorùbá* kinship is grounded in the mutually inclusive and complementary treatment of binary opposite aspects of an instance, formulation such as Gender or Sex, Age, Status, and so forth, are NOT admissible.

For example, *Akọ* and *Abo* are grounded in the Syllable *Kọ* ("Emitter") and *Bo* ("Receptor"), respectively. *Akọ* and *Abo* are ascribed to *ìrírí* ("the sensory accounts") of "That which emits" and "That which Receives", respectively. The binary opposite of "Emitting" and "Receiving" are the mutually inclusive and complementary aspects of an instance.

The *Yorùbá* single syllable term *rin* is used to cue the centre or core of an instance. Of course the term *Àrin* ("Middle") is formulated based on *rin*.

The term *Ọkùnrin* and *Obìnrin* are formulated through the use of *rin* to render *Akọ* and *Abo*.

The term *Ọkùnrin*, therefore, transcribes into English as: "That whose core agency is situated in Emitting". The term *Obìnrin* transcribes into English as: "That whose core agency is situated in Receiving".

Also, binary opposite formulation, such as *Ẹgbọ́n*, *Àgbà* ("Older") and *Àbúrò*, *Èwe* ("Younger"), are treated as mutually inclusive and complementary. The formulation of the terms *Ẹgbọ́n* ("Older") and *Àbúrò* (“Younger”) are quantified by imposing context such as *Dídé* (“Arrival”), *Ipò* (“Status”) or *Àiwáyé Ìbí* (“Occurrence of birth”) on them.

Serious information deficit will be incurred if English words such as “Grandparent”, “Parent”, “Mother”, “Father”, “Child”, “Daughter”, “Son”, “Boy”, “Girl”, “Man” and “Woman”, are used to render *Yorùbá* kinship discourse.

4.5.3 *Orúkọ* (“Nominal-Term” as “Auxiliary-term”)

In addition to the *Ẹbí* into which he/she is born, *Orúkọ* (The “Nominal-Term”) used to cue the identity of an individual is grounded in:

(i.) Character of birth (Twins, After twins, Bridge, Leg first,)

(ii.) Circumstance of birth (During, festival, journey, while father has traveled, when agent parent died, an do forth)

(iii.) Circumstance of parent (During calamity, At the moment of wealth, When hope lost, and so forth)

Therefore, *Orúkọ* (The “Nominal-Term”) used to cue the identity of an individual is NOT a Primitive-term. Indeed, Assertion is inherent in Nominal-term used to identify individual in the *Yorùbá* universe of discourse. Since Assertion is inherent in *Yorùbá* names, they are “Auxiliary-terms” rather than Primitive-terms.

Within *Àròkún*, term are not confused for the beingness of the phenomenon of nature. However, the formulation of terms is situated in the metaphor grounded in the human sensory accounts of the phenomenon of nature. Therefore, *Yorùbá* language *Ohùn* (“Terms”) are used to ascribe instance or agency to human sensory accounts of material manifestation of the phenomenon of nature.

The Auxiliary-term in the formal language rendering of the *Yorùbá* kinship universe of discourse. Instance in written *Yorùbá* language alphabet are signs NOT letters. The written *Yorùbá* language comprises Two (2) alphabets: (i) Alphabet of Tone and (ii) Alphabet of Phone.

Motifs are a combination of signs. Each syllable (sign) is used to cue a partial-term. Therefore, combination of syllable are also partial-term. Every instance of partial-term is a partial-term. Hence, “Adé”, “Akin”, and “Kẹmi” are used to cue partial-terms in the *Yorùbá* kinship universe of discourse.

4.6 Interrelation

The Ẹ̀gbọ́n () relation

(i.) E is the Ẹ̀gbọ́n of A, if E precede and A on the grounds of the context J rendered with the relation *Ju* (“Surpasses”). The relation Ju () can be contextualised and formulated based on “Arrival at a location”, “Status” or “Occurrence of birth”, as discussed above. The variable-term formulation for Ẹ̀gbọ́n is expressed as:

$\boxed{\text{Ẹ̀gbọ́n (E, A)} \rightarrow \text{Ju (E, A)}}$

(ii.) The formulation of an individual $e \in E$ as the *Ẹ̀gbọ́n* of a set of individuals A is rendered as:

$\boxed{\text{Ẹ̀gbọ́n (e, A)} \rightarrow \text{Ju (e, A)}}$ This formulation expresses that, “the individual $e \in E$ is *Ẹ̀gbọ́n* of every individual in the set A on the grounds of C.

(iii.) The formulation of the set of E who are *Ẹ̀gbọ́n* to an individual $a \in A$ is rendered as:

$\boxed{\text{Ẹ̀gbọ́n (E, a)} \rightarrow \text{Ju (E, a)}}$ This formulation expresses that, “every instance $e \in E$ in the Ẹgbọ́n of an individual $a \in A$ on the grounds of C.

(iv.) The formulation of an individual $e \in E$ who is the Ẹ̀gbọ́n to another individual $a \in A$, on the ground C, is rendered as:

$\boxed{\text{Ẹ̀gbọ́n (e, a)} \rightarrow \text{Ju (e, a)}}$

(v.) Ẹ̀gbọ́n(), is transitive. If X is the Ẹ̀gbọ́n of Y and Y is the Ẹ̀gbọ́n of Z, then X is the Ẹ̀gbọ́n of Z. This is formulated as:

$\boxed{\text{Ẹ̀gbọ́n (X, Z)} \rightarrow \text{Ju (X, Y)} \wedge \text{Ju (Y, Z)}}$

(vi.) The term rendering of the above transitive formulation is:

$\boxed{\forall\, x \in X, y \in Y, x \in Z\text{: Ẹ̀gbọ́n (x, z)} \rightarrow \text{Ju (x, y)} \wedge \text{Ju (y, z)}}$

4.6.1 Primitive-terms

Primitive-term is NOT admissible in the instrument of *Yorùbá* language. Ohùn (“The Terms”) of the *Yorùbá* instrument of language are used to cue the temporal and spatial aspects of instances in its human agency universe of discourse.

4.6.2 Laboratory Exercise II: Formalising Expression: * Òrófó* the Chatbot

Ability to use an instrument of language to express and ascribe message to an expression is situated in individual human. *Ọgbọ́n* (“Intelligence”) is ascribed to *Àdáṣe* (the “Individual”) activity of human agency on the grounds of the creativity and/or ingenuity manifesting in performance the ability to use an instrument of language. This includes

1. The ability to generate an expression that effectively conveys an informed message.
2. The ability to ascribed message to a valid expression.

Example with sentences

1. You can deceive some human all of the time, but you cannot deceive all humans all of the time
2. All human are equal but some are more equal that others
3. Some children are resemble (are alike or similar to) their parents
4. An expression comprises instances of term

5. An explanation comprises finite instances of expression
6. An Expression comprises infinite instances of explanation.

CHAPTER 5

GAMES: *ÌDÍJE* ("COMPETITIVE INTELLIGENCE")

Ọgbọ́n (The "Intelligence") ascribed to an organism during *Ìdíje* (a "Competitive") activity is grounded in his/her/its ability to demonstrate better creativity and/or ingenuity than another biological agency of equal status.

Ìdíje (A "Competitive") activity differs from other kinds of *Ọgbọ́n* ("Intelligence") in that:

(i.) Its aim is to rank agency according to a predefined hierarchy. Hierarchy is situated in human mental abstraction.

(ii.) Its universe comprises finite instances of state, location, transition or displacement. These are connected and/or related through the rules of engagement.

(iii.) The activity expressed is to the extent that consistent interaction permits.

(iv.) Its participants express their creativity to the extent that (i), (ii) and (iii) permits.

(v.) Its environment or world is closed. Instances are predefined before the game commences and cannot be modified.

(vi.) The outcome of the interaction is mutually exclusive and contradictory for its participants: A participant cannot "Win" and "Lose" simultaneously. Drawn is reckoned as No contest.

Aspect of a competitive process, formulated with a context-neutral language, is expressible through a regular language. Strategic games, such as *Ayò*, Go and Chess, are examples of *Ìdíje* (the "Competitive") activities that have been used to ascribed *Ọgbọ́n* ("Intelligence") to an agency.

5.1 Overview of strategic games

Strategic games are perhaps the most fertile activity through which human creative skill is expressed in a competitive environment. A game world usually comprises a set of tokens and their admissible manipulation. The process prescribed for strategic games is easy to explain and follow. Despite its simplicity, strategic game environment exhibit attributes and features commonly encountered in creative problem-solving. The interaction within a game world can, therefore, provide insight into the creativity expressed through human performance.

Ability to play and win strategic games at a high level of competence, is often considered a distinctive evidence of the manifestation of human intelligence in conventional AI. Indeed, the extent to which game playing skill is replicated or imitates in computing machine had been used as a measure of progress in conventional AI. In 1996, “Deep Blue”, developed at the International Business Machine (IBM) Inc., defeated Gary Kasparov, the then world champion in Chess. The defeat was regarded as a major fit of success in conventional AI.

Popular strategic games include:

(i.) Chess,

(ii.) Go,

(iii.) *Ayò*

Indeed, the ability to play strategic game such as *Ayò* requires insight, creativity, innovation and estimation. These abilities are similar to those encountered in social interaction and conflict resolution. However, game playing is situated in predefined process whose performance had finite instances of outcome.

The “game world” is situated n the context-neutral universe of human mental abstraction. Therefore, game activity is useful in the following situation:

(a.) Exploring the possibilities of “what can be done” in the process of determining “what should be done”;

(b.) Generate possible cases and horizon in informed problem-solving process.

(c.) Recreation and entertainment.

5.1.1 Principles of strategic games

In this laboratory exercise, we will explore the game of *Ayò*. *Ayò* is a popular indigenous board game in West Africa, particularly in the *Yorùbá* nation. The following are the character of a Game:

(i.) There is a **World** (domain, field, board or arena) where it is admissible for the game to take place.

(ii.) There is a finite set of **Tokens** (object) and Two (2) or more **Agency** in the game world.

(iii.) There are Two (2) contending parties or individuals, called **Players**.

(iv.) Other game-neutral agency, referee or moderator, and/or game-free agencies, audience or spectator, in the game world.

(v.) The playing parties (players) manipulate the Tokens according to a set of **Game rules**.

(vi.) Each Token occupies predetermined location in the configuration of the **Initial-state** of the game

(vii.) A **Move** is used to effect a **Transition** of Game-state. A move culminates in the removal or inclusion of Tokens into the game world. The removal and/or inclusion is determined by the game rule during the game playing process.

(viii.) The **Payoffs** or **Rewards** (scores or points) accumulated during the game playing process determines the **Winner** or **Loser** of a game.

The aim is for a player to demonstrate that he/she has a superior winning strategy by virtue of the ability to acquire higher points or scores. The superior player at the end of the game is declared the winner while the other player is the loser. The binary opposite of *Ọ̀ta* ("Winner") and *Ọ̀pè* ("Loser") is mutually exclusive and contradictory. This implies that a player cannot be "a winner" and "a loser" at the end of a game. However, *Ọ̀mìn* ("a draw") can be declared, when the players score equal points. However, whereas a Player cannot declare himself/herself as a winner, a Player may concede to losing during the game. A game may have several rounds. Usually, the winner is determined according to the rules for combining the points scored in the rounds.

5.1.2 Features of strategic games

The following are the features distinguishes a **Strategic game** from a **Game of Chance**.

(i.) Outcome of the game is determined after the game playing is concluded. Therefore, activities before and after the game are not reckoned in the determination of the outcome. Activities of players during the game are admissible on the grounds that they conform to the rule of the game.

(ii.) Every move (play and response) must conform to the rule of the game.

(iii.) In strictly turn-taking game, a player cannot skip when it is his/her turn to make a move. Also, a player cannot prevent the move (play or response) of his opponent.

(iv.) The influence of luck and/or chance is not reckoned in the playing of the game.

(v.) A player's move is atomic and irreversible. This implies that a player cannot un-move after initiating and making a move.

(vi.) Ability to compute (e.g. count, calculate) by reckoning instances of game object is not sufficient for winning. Player's ability to plan, project, estimate and predict his/her opponent's next moves is crucial. Competence not performance is key; skill not speed is important.

(vii.) Mistakes cannot be ruled out during the game even with the best of caution and utmost care. The player (or party) that makes the mistake is responsible for the effect on his/her play. Hence, mistakes are not reckoned in the determination of the outcome of the game.

(viii.) A player might withdraw before the end of a round or game. Such decision may be the consequence of the player coming to the conclusion that the opponent has demonstrated superior skill and/or capability.

(ix.) Withdrawal is taken as an admission of the lost of a round or that of the game. This depending on the game rule.

(x.) A player CANNOT take recourse in aggression and/or violence in a **pure strategic game**.

5.1.3 *Àkóyawọ́* ("Fair play") in strategic game

A strategic game is fair when it is grounded in the following principles.

(i.) The prescription for playing the game is mutually agreed before the commencement of the game.

(ii.) No player or playing party or moderator or referee can modify and/or alter the prescription of the game and/or instances of game token or arena during the play.

(iii.) The moderator or referee is game-neutral. This implies that the referee's or moderator's participation does influence the outcome of the game.

(iv.) No player is assisted or hindered from deploying his/her competency in the game to the best of his/her performance ability.

(v.) A player does NOT deliberately play to loose.

(vi.) A player does NOT reject the outcome of the game whether or not he/she wins or looses.

Figure 5.1: *Àkóyawọ́* in *Ayò* game playing

5.2 *Ayò* game process

Using terms in the conventional language of game, *Ayó* is a *Two-player*, *Turn-taking*, *Zero-sum*, *Partial information* game. These terms are explained as follows:

Two-player: Is a situation where there are Two (2) contending "Individuals" or "Teams" (Group of individuals). **Each Individual or Team is a Player in the Game**. The Payers are contending for rewards (or points or scores). Game seeds (or Tokens) are the reward in *Ayó* game. The process from the beginning to the end of the Game is called Playing. The admissible interaction between the Players of the game is defined and guided by the prescription of the Game.

Turn taking: *Ayò* is a strictly turn-taking game. The two (2) players alternate moves. A player must make a move when it is his/her turn. Skipping move is NOT admissible (or permitted). However, a player cannot make a move such that it will be impossible for the opponent to make a countermove.

Zero-sum: This implies that the *total reward* of the Two (2) players, at the end of the game, is Zero (0). This is determine as follows: A winner is awarded $+1$ and a loser is awarded -1. In the case of a draw, each player is awarded 0. Since we have two players, the sum of these awards will always be $+1 - 1 = 0$ or $0 + 0 = 0$ (See the Table below). A game abandoned by both Players is considered "a no-contest" and treated as *Ọ̀mìn* ("a draw").

Player	Award	Award	Vertical Sum
A (wins)	+1	-1	0
B (wins)	-1	+1	0
Draw	0	0	0
Horizontal Sum	0	0	0

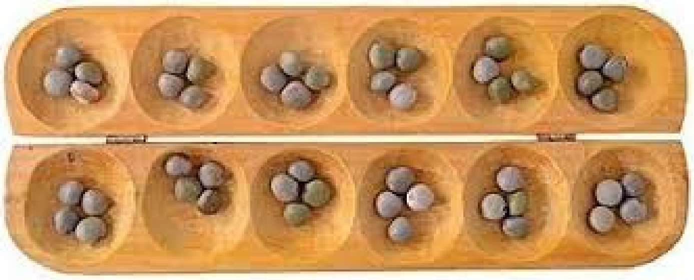
Partial information: In a Total Information game, all the data required to make informed move are available to both players (as well as the spectators). No data is hidden from the players in respect of the game status or progress. In the case of *Ayò*, however, a player cannot *kó* ("scoop") and/or *ka* ("count") the seeds in the hole of his/her opponent. Scoping is necessary for counting the seeds in a hole with many seeds. Therefore, access to the total information on the board is reduced as the game progresses and seeds are accumulated in the holes. Therefore, a player needs to take recourse in *àfojúwọ̀n* ("estimation") to determined the status of seeds on his/her opponent's side of the game-board. *Àfojúwọ̀n* ("Estimation") is *kù* ("a partial") accounts of *Ìwọ̀n* ("Quantification"). The need for *Àfojúwọ̀n* ("Estimation") rendered *Ayò a Partial information* game.

Àkóyawó ("Fair play") in *Ayò* requires a Player not to manipulate seed counts and/or add seed into the board.

5.2.1 Features of *Ayó* game

Ayò differs from conventional boardgames such as Chess and Go. The games of Chess and Go are situated in the arrangement and rearrangement of predefined patterns during interaction between Two (2) competing agencies. *Ayò* is situated in the temporal rendering of the interaction and interrelation between Two (2) competing agencies. Therefore, in addition to the capacity to reckon and manipulate patterns, a player of *Ayò* also require the capacity to count, estimate and relate different states in an appropriate context.

Ayó is played on a wooden board called *Ọpọ́n*. Figure 5.2a depicts the initial setting of *Ayò* board without capture hole. Figure 5.2b depicts the initial setting of *Ayò* with a capture hole at either sides of the board. The cup-shaped holes on the board is called *Ilé* ("house"). Each hold, except the capture two holes, is usually between $6cm$ and $9cm$ in diameter and about $3cm$ deep. Two (2) rows of Six (6) holes are carved into the board. Each player is responsible for a row (set of six holes). The tokens of the game are called *Ọmọ* ("Child/Seed"). They are usually shrub (*caesalpina crista*) or small pebbles or palm-nuts. There is usually a protruding hole, about twice the size of the play-hole, on either ends of the board. This is where the players keep captured seeds.



(a) Image of initial setting of *Ayò* board **without capture hole**

(b) Image of initial setting of *Ayò* board **with capture hole**

Players take turns in making moves according to the established prescription of *Ayò* game. There are Two (2) popular versions of the game in *Yorùbá* community. These are (i) *Jẹ̀rin-Jẹ̀rin* ("Non strategic") and (ii) *Ìjodù* ("Strategic" or "Coded") versions. The *Ìjodù* strategic version is used here.

5.2.2 Setting and prescription of *Ayò* game playing

Ayò game prescription is set as follows:

Initial set up: Before a bout begins, Four (4) seeds are placed in each of the Twelve (12) holes on the *Ayò* board, as shown in Figure 5.3 (see also Figure 5.8 a). Each player's side of the board comprises Six (6) consecutive holes on a row. Since each of these holes contains 4 seeds, then there are 24 seeds per player. The total number of seeds on the board at the start of a bout is then $24 + 24 = 48$.

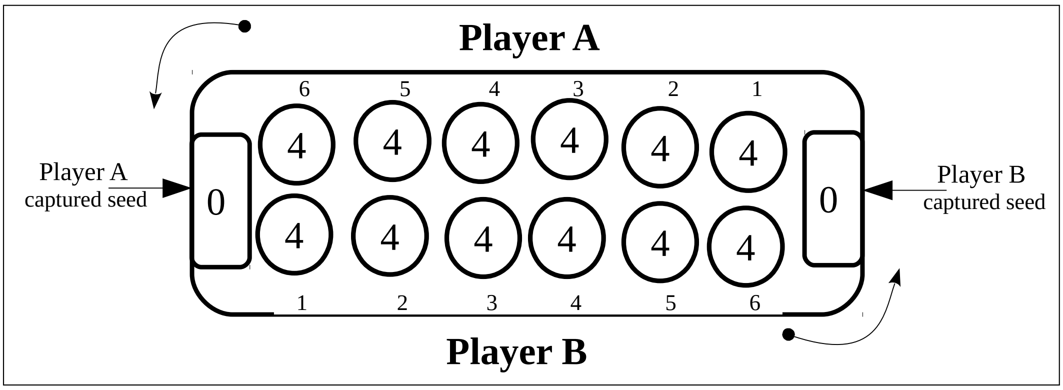


Figure 5.3: Drawing of initial *Ayò* Game board setting with seed capture holes

Turn taking: *Ayò* is a strictly turn-taking game. The Two (2) players moves alternately. A player must make a move when it is his/her turn to play. A player CANNOT make a move such that it will be impossible for his/her opponent to make a countermove while there are still more seeds on the board. If this occurs, **the player that makes it impossible for his/her opponent to make a countermove has lost the bout.**

Aim and Objectives: The aim of the game is for a Player to *Pa* ("Win") his/her opponent. This is achieved by winning Three (3) consecutive bouts or rounds. The objective of a round of contest is to capture a higher number of seeds. A player that has captured 25 (that is one plus his/her own 24) seeds has won the round. A player that wins *Three (3) consecutive* rounds has won contest.

Play moves: A move is made when a player chooses a non-empty hole from his or her side of the board. This is called the *play-hole* (*Ojúta*). S/he empties the seeds in the play-hole into his/her hand. The player then *ta* ("moves") by visiting and *dropping one seed per hole* into the next and consecutive holes in a *counter-clockwise* direction. The move continues until the scooped seeds is exhausted. If the number of the scooped seeds is large such that the play-hole is visited again, as is the case in an *Odù*, the seeds dropping proceeds except that a seed cannot be dropped into the play-hole. This implies that at the end of a move, the play-hole must be empty.

Capture: A player cannot capture seeds on his/her own side of the board. During a move, if the last hole visited by a player is on the opponent's side of the board and it contains *Two* (2) or *Three* (3) seeds, then the seeds in that hole are captured. Also captured, are seeds in the consecutive clockwise holes, preceding the captured hole, with Two (2) or Three (3) seeds. Captured seeds are removed from the board into the capture hole belonging to the player that captured the seeds. No captured seeds can be returned onto the playing hole during a bout. It is NOT permitted to capture all the seeds in an opponent's side. Indeed, this will make it impossible for the opponent to make a countermove. If a player captures all the seeds in an opponent's side, he/she either losses all the captured seeds or be declared the loser of the bout. Back-tracking is NOT permitted after a move had been committed.

End of Game: At each turn, a player must make a permissible move. A permissible move is that which does not violate any of the prescription *Ayò* game stated above. If it is not possible for a player to move in such a way to give his/her opponent a permissible countermove, then the end game is reached. If the seeds on the board is so few that a player cannot capture a seed even when he/she makes a permissible move, then the game is suspended. Dialogue between the players then ensue for sharing the remaining seeds on the game board. In summary, a round of *Ayò* game ends when one of the following conditions occurs:

(i.) There are so few seeds remaining on the board that it is not possible to capture any more seed. Attempt to continue playing will culminate in *Àìlópin* ("Infinite play move" or "Game Singularity"). At this point, the game is suspended. The players then agree to share the remaining seeds. Usually, the player who thinks he/she is winning initiates the dialogue for terminating the bout and sharing the remaining seeds.

(ii.) A player has no hole containing enough seeds to reach his opponent's side and his opponent has no hole with seeds ("deadlock configuration"). NOTE: A player must not play to deliberately prevent his/her opponent from making a counter-move.

(iii.) The players decide by mutual agreement to stop playing and share the remaining seeds according to their analysis of the situation. This is the most common way of ending games between good players if situation in Item (i). above occurs.

(iv.) The players agree to abandon the game. An abandoned game is reckoned as *Ọ̀mìn* ("a draw").

Win of Game: A player has won a round or bout when the following Two (2) criteria are met: (i) He/she has captured more seeds than his/her opponent. The capture of 25 seeds indicates a win. (ii) The player has not played in such a manner to prevent his/her opponent from making countermove (*Jẹ ayò pa*). A player that plays in such a way that his/her opponent is unable to make a countermove has lost the round or bout even if such a player has captured more than 25 seeds. When a player wins Three (3) consecutive rounds or bouts then he/she is declared *Ọ̀ta* ("the winner") of the game.

5.3 Computational rendering of *Ayò* Game playing

Skilful *Ayò* game playing requires the ability to remember or accurately estimate the number of seeds in each of the Six (6) holes the player's sides as well as an estimate of the number of seeds in each of the Six (6) holes in the opponents side of the board. A player seeks to capture opponent's seeds while at the same time protecting his/her own seeds from being captured. The earlier and latter stages of a round is easier to analyse as there are fewer seeds on the board, hence fewer possible alternative moves.

The playing of *Ayò* is a reflective process. A reflection is a mirror image. In a mirror image, the left of pattern is located at the right of the mirror and the right side of the image is located at the Left. Since the Two (2) players are sitting opposite to each other, the left of one player is the right location of his/her opponent and *vice versa*. Therefore, the clockwise transition in the playing process is maintained. The reflective as expressed in *Ayò* is depicted in Figure 5.4.

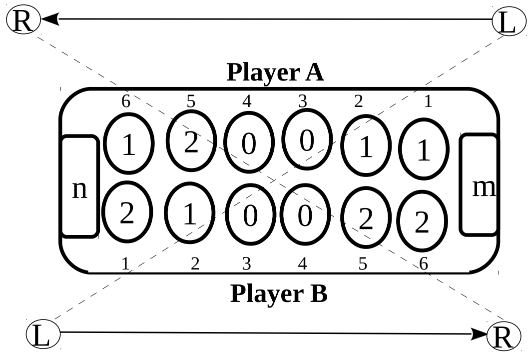


Figure 5.4: Reflection in Game board configuration

In the Figure 5.4, the Arrow for the direction of Player A play move is $\longleftarrow$ and that for Player B's play move is $\longrightarrow$.

5.3.1 Computation of *Ayò* move

When it is his/her turn during the game, a Player is required to make *ta* ("a move"). A move is made when a Player **Selects** and **Play** a hole on his/her own side of the board. A player's goal is to use the move to to gain a **Positive Pay-off**. The following are legitimate activities during a move.

(i.) Scooping (*Kíkó*) and Counting (*Kìkà*) a hole In a Scooping step, all the seeds inside an identified hole is removed into the hand. The holes are numbered anticlockwise from 1 to 6 as depicted in Figure 5.3. The *Count()* function returns the number of seeds in a *Scooped()* hole. The *Count()* function returns Zero (No Seed to count) if there is no seed in a Scooped hole.

(i) y = Count(Scoop(A, 4))

(ii) y = Count (Scoop(B, 1))

The instruction in (i) above states that "The Player A scoops and counts the seeds in hole number 4" on his/her own side of the board. The instruction in (ii) above states that "The Player B scoops and counts the seeds in hole number 1" on his/her own side of the board.

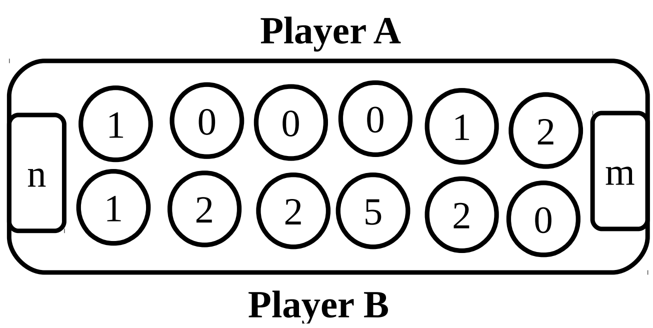


Figure 5.5: Game board configuration

For example, in the board configuration in Figure 5.5 the instruction in (i) will return y = 5 while that in (ii) it will return y = 2.

After Scooping and Counting a hole, its may be Played. If a hole is Not played, its seeds are returned into it. The "Current Board State" in Figure 5.5 is depicted in "Current Game status data table" in Table 5.1.

Hole	Players	
	A	B
1	2	1
2	1	2
3	0	2
4	0	5
5	0	2
6	1	0
Captured	n	m
Total 48 =16+n+m	4+n	12+m

Table 5.1: Current Game status data table

(ii) Playing (*Títa*) a hole A Scooped hole is played by Visiting its Next and subsequent hole traversing in an anticlockwise direction. When a hole is visited, a seed is placed into it. The traversal continues until the scooped seeds is exhausted. **A scooped hole cannot be visited during a move.**

```
WHILE (moreSeed(y) AND A[i] ≠ playHole())
{ VisitNexthole(); y = y-1; }
```

The above algorithm effects the playing of hole number i by Player A.

5.3.2 Task 4.1

(i.) Explore the above algorithm for the Game board configurations in Figures 5.7, 5.8, and 5.10. Calculate the numerical value for ? in Figure 5.6.

(ii.) Noting that the total seeds on a board during a game is $48(24 + 24)$, use Figure 5.6 to compute the content of Table 5.1.

5.3.3 Computation of seed *Jíjẹ* ("Capturing")

If the last of the seeds being played ends in a hole in the opponent side of the board, a capture has occurred when:

(a.) There are **Two (2)** or **Three (3)** seeds inside the last visited hold.

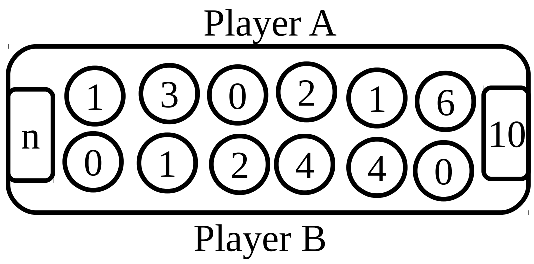


Figure 5.6: Game board configuration

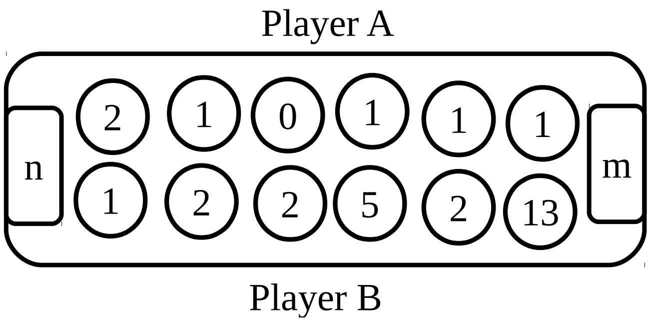


Figure 5.7: Game board configuration

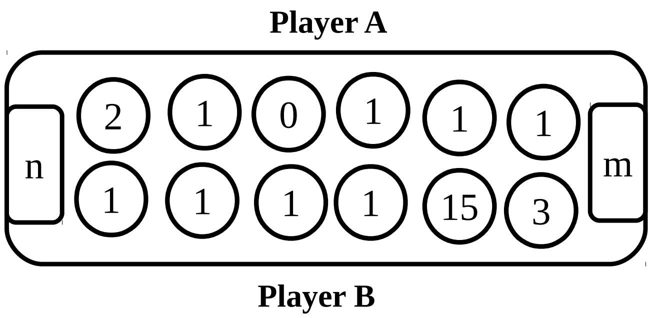


Figure 5.8: Game board configuration

(b.) The seeds in the next consecutive clockwise holes, in the opponent side of the board, with Two (2) or Three (3) seeds are also captured.

5.3.4 Inadmissible capturing (*Ìjayòpa*)

During a bout/round, it is not admissible for a player to play in such a way that his/her opponent will not be able to make a countermove. A player will not be able to make a countermove if there is no seed on his/her own side of the board. The only exception to this rule is that, there is no more seed on either side of the board after the capture of all the seeds on the opponent's side.

Given the Game board configuration in Figure 5.9. If it is the turn of Player B to play and he/she plays hole 6, containing 2 seeds. He/she will capture holes 1 and 2, containing 1 seed each on the side of Player A. Then Player B has lost the round/bout despite having 26 seeds already. This is because after Player B has played hole 6, Player A will not have seeds on his/her side of the board to make a countermove. However, there is still one seed on the board.

If Player B Plays hole 6 in the Game board configuration in Figure 5.10 he/she has won the bout/round. This is because there will be no more seeds on the Board after Player B has capture holes 1 and 2 on the side of Player A.

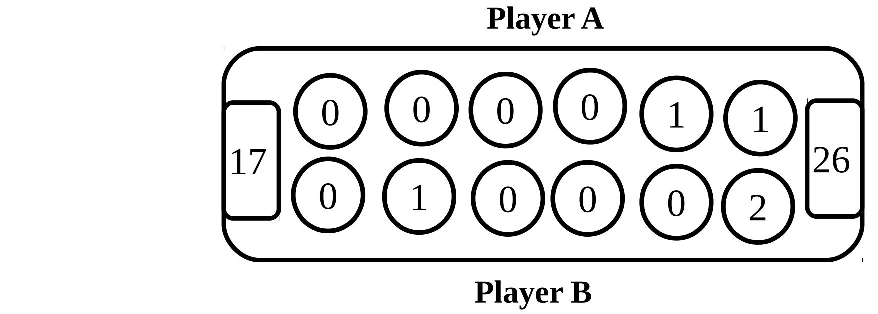


Figure 5.9: Game board configuration

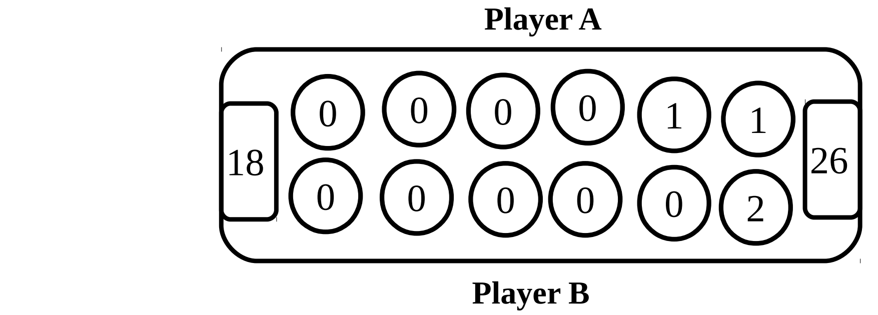


Figure 5.10: Game board configurations

5.3.5 Task 4.2

Discuss the capturing of seeds on the Game board configurations in Figures 5.7, 5.8, and 5.10 for both Players.

5.4 *Ayò* game tree

Igi ("The tree") metaphor is popular in conventional computing. *Igi* ("The tree") metaphor is used in data prescription ("Data structure?"), File configuration, Memory formulation and configuration as well as in Game process representation, among others.

In a strategic game, a player attempts to compute his/her move in the context of the next best response (countermove) by his/her opponent, against his/her own next move and the next best response of his/her opponent, and so forth. If a Player decides to compute four or more moves and countermoves in advance, he/she will be taking a daunting, and indeed, an impossible task upon himself/herself. Of course, such computation is NOT necessary because, the opponent's next move cannot be determined exactly. Just like the player, his/her opponent's decision is based on many confounding factors which include:

(i.) Accuracy of seed counting in his/her own and opponent side of the board;

(ii.) Judgement of the strength of the opponent's computing and estimation skills;

(iii.) Intuition;

(iv.) Possibility of mistake by the opponent.

These are further influenced by player's skill. Hence, a player can only estimate what the opponent is likely to (or can) do. He/she cannot determine what the opponent will do. What he/she and his/her opponent does is situated in their moves and countermoves.

As discussed in Section 5.2, *Ayò* is a **Partial information** game. This is because a player CANNOT scoop the seeds in the holes inside the opponent's side of the board. Therefore, a player can only take recourse in estimation whenever the number of seeds in a hole on the opponent side of the board is substantial. To implement the *Ayò* game playing process on a computing machine, therefore, it has to be simplified and treated as though it were a **total information** game. Base on this simplification, each player has access to the number of seeds inside all the holes on the game board. Therefore, *Àkóyawọ́* ("Fair play") CANNOT be implemented for the entire *Ayò* game playing process through conventional computing techniques and tools.

However, the simplification *Ayò* game playing process makes it amenable to analysis, representation and implementation through the tools used in conventional Computing and Mathematics. Such tools includes Conventional of the abstraction of Tree, Matrix and Table.

Here, recourse is taken to the metaphor of tree (*igi*) as a tool for formulation and modelling in the computational rendering of the *Ayò* game process. The Tree metaphor is popular in conventional AI particularly in Game playing process formulation and implementation. Game tree is most effectively deployed in **Total information** games, such as Chess, where estimation is not required to determine the state of the game board. Therefore, *Àkóyawọ́* ("Fair play") can be implemented for Chess through conventional computing.

5.4.1 *Ayò* Game tree

A game tree is:

(a.) An explicit representation of possible moves at the **Current state** of the game.

(b.) The **Root** of the tree corresponds to the Current state of the game.

(c.) The **Branches** emerging from the root represents the game moves (transition). The nodes at a level of the tree represents a players move. The nodes at the next level is the counter-move by his/her player.

(d.) A level in the tree corresponds to all possible moves available to a player.

(e.) The edge (line) between two (2) nodes is an admissible move. The number of seeds that will be captured if the move is taken is indicated on the line.

(f.) The leaves of a game tree represents the terms in the current expression of the game. The Payoff of the moves and countermoves from the Root to a leave is indicated with the number inside the leave.

A new tree is constructed by a player at every turn in the game. The Game tree in Figure 5.16 is Player **A**'s analysis of the Game board configuration in Figure 5.11. The board game move configuration in Figure 5.12 indicates that, Player A plays Hole 6 and captures 3 seeds.

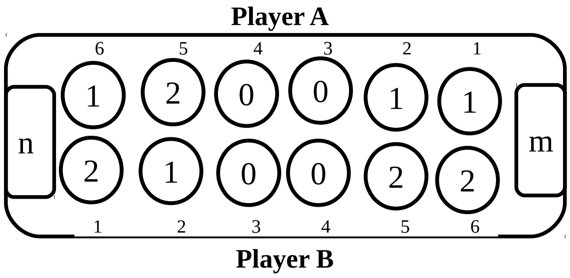


Figure 5.11: Game board configuration

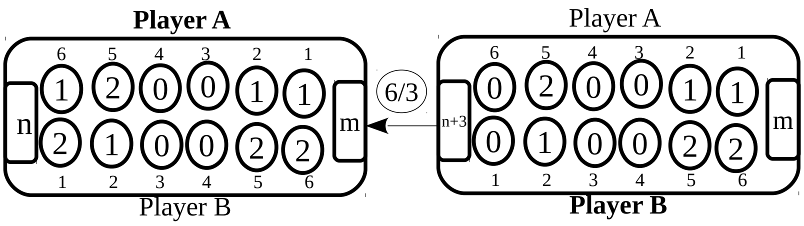


Figure 5.12: Game board configuration

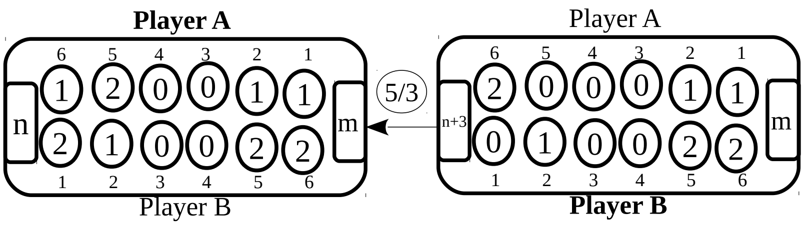


Figure 5.13: Game board configuration

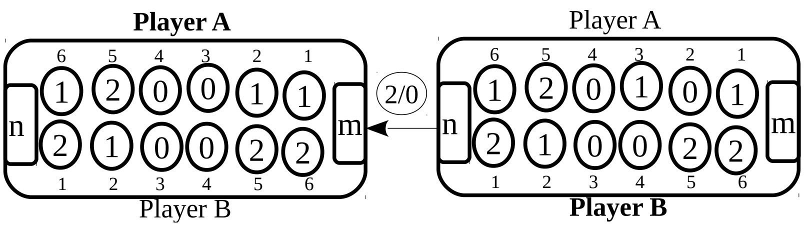


Figure 5.14: Game board configuration

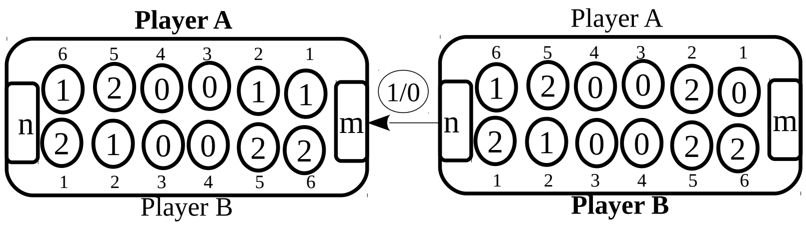


Figure 5.15: Game board configuration

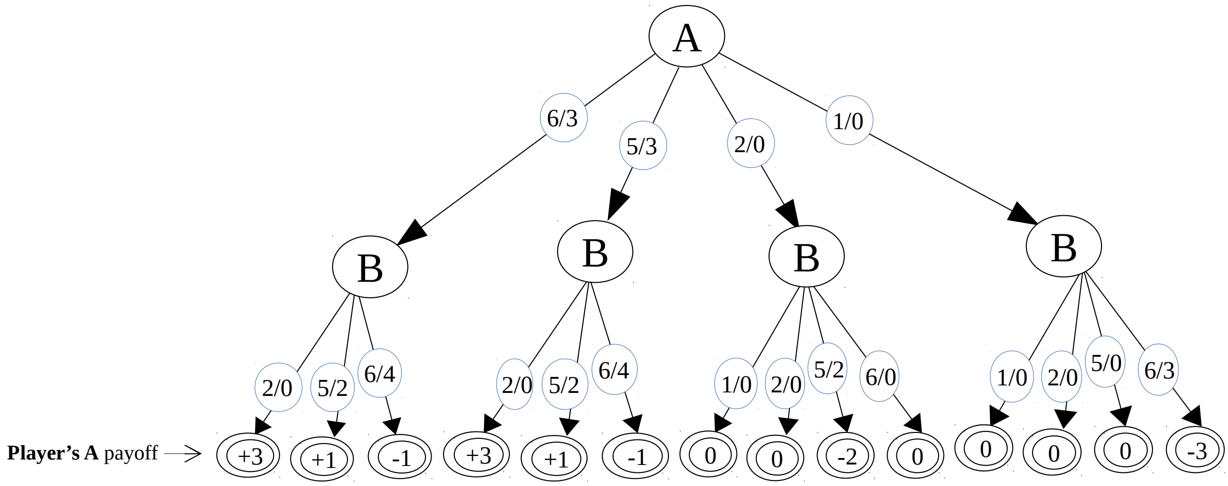


Figure 5.16: Game Tree for Player A

5.4.2 *Ayò* Game Board Matrix

The board configuration in Figure 5.11 is represented as the matrix in Equation 5.1. In the Two-dimensional matrix inside Equation 5.1 the content (number of seeds) in the wholes of Player **A** and **B** are indicated in the index (i, j). Where i is the number of seeds in the hole of Player **A** while j is the number of seeds in the corresponding hole of Player **B**. For example, the $(1, 2)$ in location $Board[1, 1]$ is the numbers of seeds in the corresponding holes, hole 1 of Players **A** and **B**, respectively.

Note that the directions of traversal of the holes on the Player's side of the board is binary opposite, mutually exclusive and contradictory. This way, Hole 6 on Player A's side of the board is Hole 1 on Player B's side of the Board, and *vice versa*. Hole 5 on Player A's side of the board is Hole 2 on Player B's side of the Board, and *vice versa*, and so forth.

Note that the Current state of the board is expressed in the diagonal of the matrix.

$$Board[i,j] = \left(\begin{array}{c||cccccc} & \multicolumn{6}{c}{PlayerB} \\ \hline PlayerA & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & \mathbf{(1,2)} & (1,1) & (0,0) & (0,0) & (2,2) & (1,2) \\ 2 & (1,2) & \mathbf{(1,1)} & (1,0) & (1,0) & (1,2) & (1,2) \\ 3 & (0,2) & (0,1) & \mathbf{(0,0)} & (0,0) & (0,2) & (0,2) \\ 4 & (0,2) & (0,1) & (0,0) & \mathbf{(2,0)} & (0,2) & (0,2) \\ 5 & (2,2) & (2,1) & (2,0) & (2,0) & \mathbf{(2,2)} & (2,2) \\ 6 & (1,2) & (1,1) & (1,0) & (1,0) & (1,2) & \mathbf{(1,2)} \\ \hline\hline \end{array} \right) \quad (5.1)$$

As depicted in Figure 5.4, the Game Board-matrix in Equation 5.1 is self-reflexive. This implies that element pairs at the upper diagonal are the reflection of those at the lower diagonal. The diagonal elements of the Game Board-matrix in Equation 5.1 is the status of the Game board configuration in Figure 5.11 and depicted in Current Game Board Data Table in Table 5.2.

Hole	Player	
	A	B
1	1	2
2	1	1
3	0	0
4	0	0
5	2	2
6	1	2
Captured	n	m
Total (48)	5+n	7+m

Table 5.2: Current Game Status Data Table for board configuration in Figure 5.11

5.4.3 Computing players move with Game Board Matrix

A game board matrix depicts the status of an on going *Ayò* game. In the game board matrix $Board[i, j]$ in Equation 5.1:

(a.) The pair of numbers in the diagonal columns, that is $Board[i, j]$, $j = i$, *j,i = 1,2, . . . 6* is the Current state of the game.

(b.) The first element in the pair (i, j) belong to Player A and the Second element belong to Player B. For example, in $Board[1, 1]$, containing the pair $(1, 2)$, the 1 is the number of seeds belonging to Player A while 2 belongs to Player B.

(c.) To play a hole, the player empties its. Visits the holes on his/her side of the board until reaching hole no 6. Then he/she will start visiting holes in the opponent side of the board until reaching hole 6. The process is repeated until the seeds scooped and played is exhausted.

For example, if Player A plays Hole 5, the Game Board matrix in Equation 5.2 will be the result.

$$Board[i,j] = \left(\begin{array}{c||cccccc} & \multicolumn{6}{c}{PlayerB} \\ PlayerA & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & \mathbf{(1,3)} & (1,3) & (1,3) & (1,3) & (1,3) & (1,3) \\ 2 & (1,2) & \mathbf{(1,1)} & (1,0) & (1,0) & (1,2) & (1,2) \\ 3 & (0,2) & (0,1) & \mathbf{(0,0)} & (0,0) & (0,2) & (0,2) \\ 4 & (0,2) & (0,1) & (0,0) & \mathbf{(2,0)} & (0,2) & (0,2) \\ \mathbf{5} & (0,2) & (0,1) & (0,0) & (0,0) & \mathbf{(0,2)} & (0,2) \\ 6 & (2,2) & (2,1) & (2,0) & (2,0) & (2,2) & \mathbf{(1,2)} \end{array}\right) \quad (5.2)$$

In the Game Board Matrix in Equation 5.2

1. Board[5, 1] which was $(2, 1)$ becomes $(0, 2)$ because the content of hole 5 of Player A had been scooped. **ScoopSeeds= 2**

2. Board[6, 1] which was $(1, 2)$ becomes $(2, 2)$ because the content of hole 6 of Player A had been increased by 1 seed. **ScoopSeeds= ScoopSeeds- 1=1**

3. Board[1, 1] which was $(1, 2)$ becomes $(1, 3)$ because the content of hole 1 of Player B had been increased by 1 seed. **ScoopSeeds= ScoopSeeds- 1=0**

Note that

1. The manipulation of the content of one hole ripples through the Board matrix $Board[i, j]$.

2. The transition of the play moves of Player A goes from Up to Down ($i = i + 1$ to 6) and Left to Right (that is, $j = j - 1$ to 1) until the scoped seeds is exhausted.

3. The transition of the play moves of Player B goes from Right to Left (that is, $j = j - 1$ to 1) and Up to Down ($i = i + 1$ to 6) until the scooped seeds is exhausted.

4. If the three seeds captured by Player A in the Hole 1 of Player B are removed from the board Game matrix in Equation 5.3 is the outcome.

$$Board[i,j] = \left(\begin{array}{c||cccccc} & \multicolumn{6}{c}{PlayerB} \\ \hline PlayerA & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & \mathbf{(1,0)} & (1,0) & (1,0) & (1,0) & (1,0) & (1,0) \\ 2 & (1,2) & \mathbf{(1,1)} & (1,0) & (1,0) & (1,2) & (1,2) \\ 3 & (0,2) & (0,1) & \mathbf{(0,0)} & (0,0) & (0,2) & (0,2) \\ 4 & (0,2) & (0,1) & (0,0) & \mathbf{(2,0)} & (0,2) & (0,2) \\ \mathbf{5} & (0,2) & (0,1) & (0,0) & (0,0) & \mathbf{(0,2)} & (0,2) \\ 6 & (2,2) & (2,1) & (2,0) & (2,0) & (2,2) & \mathbf{(1,2)} \end{array}\right) \tag{5.3}$$

5.5 *Ayò* Game Payoff Matrix

A Game Payoff Matrix represents the leafs in a Game tree. The Player A Game Payoff matrix in Equation 5.4 is computed for the Game tree in Figure 5.16. The Row (Horizontal) aspect of the matrix indicates the move of Player a while the Column (Vertical) indicates the counter-move of Player B.

Base on Equation 5.4, for example:

(a.) The item in Row 1, Column 1 is zero (0). This is because, as indicated in the Game tree, if the Player A Plays hole 1 and Player B responds with hole 1, the Payoff will be Zero (0).

(b.) The item in Row 1, Column 6 is -3. This is because, as indicated in the Game tree, if the Player A Plays hole 1 and Player B responds with hole 6, the Payoff for Player A will be -3 (A loss of 3 seeds).

(c.) Note that the Columns and Rows indicate as * cannot be played.

$$\mathbf{G_{pom}}(\mathbf{A}) = \left(\begin{array}{c||cccccc} & \multicolumn{6}{c}{Player B} \\ \cline{2-7} PlayerA & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 0 & * & * & 0 & -3 \\ 2 & 0 & 0 & * & * & -2 & 0 \\ 3 & * & * & * & * & * & * \\ 4 & * & * & * & * & * & * \\ 5 & 0 & 0 & * & * & * & * \\ 6 & 0 & 0 & * & * & * & * \\ \hline\hline \end{array}\right) \tag{5.4}$$

5.5.1 Task 4.3

(i) Explore the game board status shown in Figures 5.8, 5.10 and 5.6 and discuss the game situation for the Two (2) players.

(ii) Construct the game trees for the game board status shown in Figures 5.8 and use it to determine the best next move for a player of your choice (you may select either of player A or B).

(iii) Compute the remaining data in the Game Matrix in Equation 5.4.

(iv) Discuss any Three (3) game status that can result in *Jẹ ayò pa* ("Inadmissible move").

(v) Discuss any Three (3) game status that can result in *Àìlópin* ("Infinite play move" or "Game Singularity").

5.5.2 Task 4.4

Using the *Python* programming language carry out the following tasks.

(i.) Write a program to display the game board (use array) as shown in Figure 5.3.

(ii.) Update your program by writing function to count the number of seeds in a whole.

(iii.) Update your program to select and play a hole by Player A.

(iv.) Update your program to compute the next move for Player A based on $(ii.)$.

(v.) Update your program to display the status of the game board after the *move* computed in $(iii.)$ is made.

5.6 *Ète* ("Strategy") in *Ayò* game playing

Ète ("A strategy") is an action (in this case "game move") whose aim extend beyond the Current state of affairs (in this case "Game board status"). The aim in a non-strategic game does not extent beyond the Current game status. The next move in *Ayò* is non-deterministic but monotonic. A non-deterministic process has more than one possible and admissible alternative paths. In a monotonic process, new instance cannot be added to, and/or removed from, the finite instances of possible transitions from the Current state or status of affairs. Also, in a monotonic process, a Next state depends on the Current state alone. Indeed, Previous and Next moves ("Actions") are treated as mutually exclusive and contradictory instance at the Current State. Therefore, causal necessity ("Cause-Effect") is admissible in a monotonic process.

Causal necessity ("Cause-Effect") is NOT admissible in a Non-monotonic process. This is because, in a Non-monotonic process, Previous and Next Actions are the mutually inclusive and complementary aspects of the Current state.

5.6.1 Popular Playing strategies in *Ayò* game

There are Two (2) popular playing strategies in the *Ayò* game : (i) *Ẹyọ* ("Constant" or "Term") and (ii) *Odù* ("Code" or "Axiom"). In the *Ẹyọ* strategy, the player adopts a *hit-and-run* approach, capturing opponent's seeds as soon as they become vulnerable to attack. In the *Odù* strategy, the player collects seeds in a hole, usually the hole nearest to the opponents side, and lunches an attack when seeds in a sequence of opponent's holes becomes vulnerable. *Odù* strategy, can become *Ìkàrè* ("Stale" or "Spoilt") when the opponent makes it impossible for a player to use it to capture seeds.

There is *no chance factor* in the game, as obtained in non-strategic games such as Ludo. A player's success is dependent on ability to *plan*, *project*, *judge*, *do arithmetic* and *count*. *Ọ̀ta* ("A skilful player") is also able to estimate and use a combination of strategies to match and overcome the strength of his/her opponent. The ability to learn from experience makes *Ọ̀ta* ("a skilful player of *Ayò*").

The move of a skilful player is NOT only about capturing seeds but also about preventing an opponent from capturing (or at least capturing more seeds) in countermoves. Ineptitude at capturing *Ààlè* ("Bait seeds"), can expose a player to losing a large number of seeds when the opponent lunches a countermove.

5.6.2 Example of the limitation in the use of *Ẹyọ* ("Constant") strategy

Given the board configuration in Figure 5.17, if Player A basis his/her Next move on the Current status of the board alone, he/she will play Hole 6. By playing Hole 6, the Payoff of Player A will be 2, irrespective of the countermove made by Player B.

The board status after Player A has played Hole 6 is depicted in Figure 5.18.

If Player B responded by Playing Hole 5, he/she will not capture any seed. Hence, the total Payoff of Player A remains higher than that of Player B. The board status after Player B has played Hole 5 is depicted in Figure 5.19.

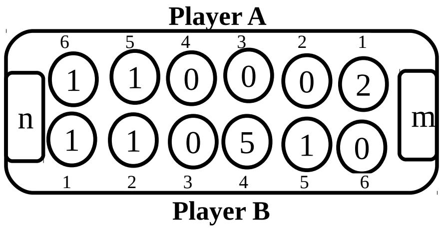


Figure 5.17: Strategy in *Ayò* Game playing

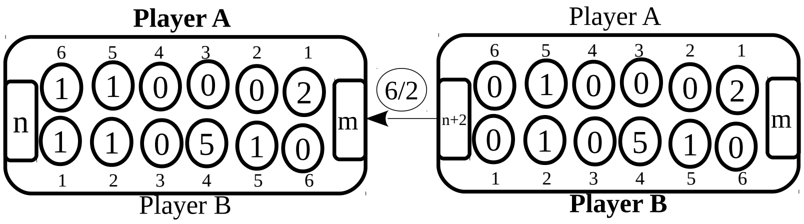


Figure 5.18: Strategy in *Ayò* Game playing

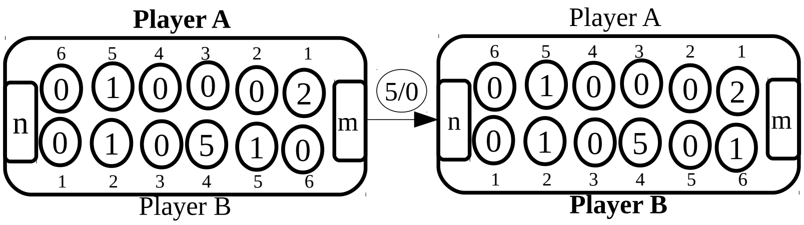


Figure 5.19: Strategy in *Ayò* Game playing

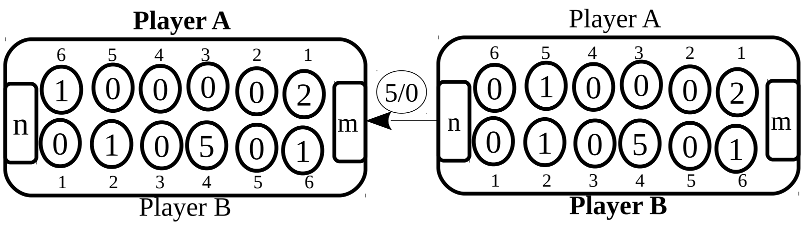


Figure 5.20: Strategy in *Ayò* Game playing

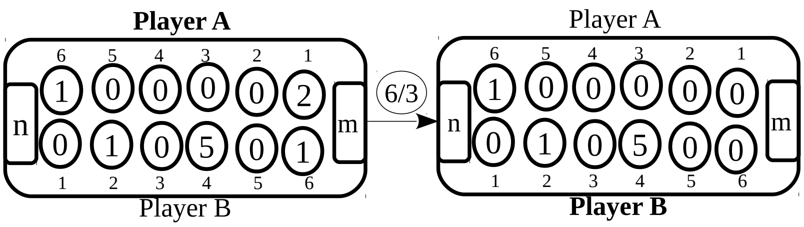


Figure 5.21: Strategy in *Ayò* Game playing

Given the board configuration after Player **B** has played in Figure 5.19, Player **A** will not capture any seed irrespective of his/her countermove. For example, if Player **A** plays Hole 1 (being his/her best countermove based on the Current board status) in his/her countermove, no seed will be capture. However, Player **B** will capture 2 seeds if he/she responds with Hole 6. The Game Play transcript, showing the Payoff of Player **A** is depicted in Table 5.3.

Table 5.3: Game play transcript for board configuration in Figure 5.17

Ser. No.	Player move		Player **A** accumulated payoff
	A	**B**	
1 2 3	(6/2) (1/0) (5/0)	(5/0) (4/2) (6/3)	+2 0 -3
Captured	n	m	
Total (48)	5+n	7+m	

5.7 Approaches to the computation of strategy in game moves

A game can be engaged from Two (2) types of perspective: (i) *Gbèsè* ("Loss") or (ii) *Èrè* ("Gain"). There are Four (4) game move strategies in each of the Two (2) perspectives. The adopted perspective will inform the strategy deployed by a player during a bout. The strategy adopted influences the treatment of *Àjùlé* (the "Maximisation") and *Àdínpin* ("Minimisation") of loss and gain. These perspectives and strategies are discussed in the following Subsections.

5.7.1 *Gbèsè* ("Loss") approach to game strategy computation

In *Gbèsè* (the "Loss") perspective to strategy formulation, a player seeks to minimise his/her own loss while maximising the loss of his/her opponent. The aim is to ensure that the final outcome of losses will culminate in gain for him/her. Each of the Four (4) strategies in Loss Perspective is indicated as X_1, X_2, X_3 and X_4 in Table 5.4.

Table 5.4: The Four (4) strategies in Loss Perspective

Self	Opponent	
	Maximise	Minimise
Maximise	X_1	X_2
Minimise	X_3	X_4

In the column indicated as X_1, a Player (The self) is trying to maximise his/her loss while also maximising the loss of his/her opponent. The is called *Awẹ́nibákú* ("Suicidal") move. This is a situation whereby a player has nothing to loose, as probably he/she has lost the game already and just using tactics to delay the eventuality that is certain to occur.

In the column indicated as X_4, a Player (The self) is trying to minimise his/her loss while also minimising the loss of his/her opponent. This is called *Awẹ́nibágbé* ("Toleration") move. This is a situation whereby a player has 26 or more seeds. He/her is only playing to keep the opponent productively engaged.

In the column indicated as X_2, a Player (The self) is trying to maximise his/her loss while also minimising the loss of his/her opponent. This is the case when a player is playing to lose. This strategy of play is used by a naïve or a novice player.

In the column indicated as X_3, a Player (The self) is trying to minimise his/her loss while also maximising the loss of his/her opponent. This is the case when a player is playing to win.

During the playing of a game, a player endeavours to have a higher payoff than his/her opponent. Hence, a player that is always seeking to **minimise his cumulative losses while maximising the cumulative loss of his/her opponent** is playing to win. **This is the MINI-MAX strategy** in conventional game theory.

5.7.2 *Èrè* ("Gain") approach to game strategy computation

In *Èrè* (the "Gain") perspective to strategy formulation of game, a player seeks to maximise his/her own gain while maximising the loss of his/her opponent. The aim is to ensure that the "final outcome" or "culminated payoff or gains" is in his/her favour. Each of the Four (4) strategies in Gain Perspective to game is indicated as Y_1, Y_2, Y_3 and Y_4 in Table 5.5.

Table 5.5: The Four (4) strategies in Gain Perspective

Self	Opponent	
	Maximise	Minimise
Maximise	Y_1	Y_2
Minimise	Y_3	Y_4

In the column indicated as Y_1, a Player (The self) is trying to maximise his/her gain while also maximising the gain of his/her opponent. This strategy is used in *Àjọ̀ẹ* ("a Cooperative") situation. This is not admissible or rarely occurs in *Ìdíje* ("a Competitive") game.

In the column indicated as Y_4, a Player (The self) is trying to minimise his/her gain while also minimising the gain of his/her opponent. The is *Jẹkínjẹ* ("Tit-for-tat") strategy. This strategy is usually deployed when a player has already captured 25 seeds and requires just one more seed to win the bout. He/her is only playing to capture one more seed, which he/she is sure to capture.

In the column indicated as Y_2, a Player (The self) is trying to maximise his/her gain while also minimising the gain of his/her opponent. This is the case when a player is playing to win. This strategy of play is used in *Ìdíje* ("Competitive") game mode.

In the column indicated as Y_3, a Player (The self) is trying to minimise his/her gain while also maximising the gain of his/her opponent. This is the case when a player is playing to lose. This strategy of play is used *Òpè* (a naïve or novice) player.

In a competitive game, a player seeks to have a higher payoff or gain than his/her opponent. Hence, a player seeking to **maximise his/her cumulative gain while minimising the accumulative gain of his/her opponent** is playing to win. **This is the MAXI-MIN strategy** in conventional game theory.

CHAPTER 6

PLANNING AND SPATIAL RENDERING OF AI DISCOURSE

Contact, connection and the interconnection in the Regular, Context-neutral and Context-sensitive spatial aspect of universe of discourse is best demonstrated through the conventional Robotics. The Planning of the Location and Displacement of Individual Robot demonstrates *Àdáṣe* ("Individual") intelligence. The planning of the coordination of the activities of Two (2) or more robots in the achievement of a Task demonstrates *Àjọṣe* ("Cooperative") intelligence.

6.1 Overview

A problem is an undesirable location of instances on the spatial aspect of universe of discourse. The instances on the spatial aspect of a universe of discourse is undesirable when they do not conform to an admissible arrangement. The aim of problem-solving in a spatial process is to manipulate the location of instances until a desirable arrangement emerges. A plan is a sequence of instrument that describes the instances of displacement that will transform the undesirable or problem to its solution or desirable. The locations of instances in Problem expression and that in its Solution are mutually exclusive and contradictory. A problem (undesirable) and solution (desirable) cannot be expressed simultaneously on the spatial aspect of universe of discourse. Therefore, a planning process is used to destroy the expression of a problem. A problem is destroyed when is solution emerges. An agency of spatial process uses a plan to effect the instances of displacement from a problem to a solution expression.

Planning is	:	The act of exploring "what can be done" in the process of determining "what should be done".
A plan is	:	A sequence of steps comprising the prescription of the actions for effecting displacements from an undesirable to a desirable spatial location.

***Ètò* (A plan) is the outcome of *Ìpètò* (a planning) process.**

6.1.1 Distinction between "Plan" and "Game"

Game is a competitive interaction between Two(2) independent agency. The instances of states in a game is expressed by its agencies during playing of the game. Therefore, an agency determines his/her next transition during a game process. Even when an agency of a strategic game has a plan, it is tentative. He/she modifies the plan to respond to the game situation. The game situation is determined by the skill of the opposing player.

The process for a Plan is totally constructed before its agency gives expression to its instances of state and transition.

6.1.2 Kinds of Plan

On the grounds of the formulation of process there are three(3) kinds of *Ìlànọ̀n* (Plan):

1. A **Regular plan** is formulated and effected through an iterative and inductive process. The location and displacement in a regular process is isolated. A process regular , in the absence of fault or availability of energy, it will always effect displacement from one location to another. A regular process is binary opposite neutral. The formulation of a regular plan is expressed through a stepwise and iterative process. Therefore, the instances of displacement in a regular plan is deterministic.

2. A **Context-neutral plan** is prescribed and effected through an Algorithmic and Deductive process. A content-neutral plan is consistent, when in the absence of mistake or availability energy, its displacement from one location to another does not change. The displacement between a pair of variable location is consistent when they are treated as mutually exclusive and contradictory binary opposite. For example, the binary opposite terms "Up and Down" or "Left and Right" cannot simultaneously be ascribed to the location of an instance. Indeed, "Forward and Backward" spatial-temporal instances cannot be simultaneously evoked in a consistent context-neutral planning process. Only one of binary opposite location is admissible is the prescription of a consistent context-neutral planning process. The instances displacement from Problem to Solution expression in a content-neutral plan is finite. Therefore, the instances of displacement in a context-neutral plan is non-deterministic and monotonic.

3. A **context-sensitive plan** is prescribed and effected through a Heuristic and Default inherent process. A content-sensitive plan is coherent when, in the absence of mistake or availability of energy, its binary opposite locations are treated mutually inclusive and complementary spatial aspects. For example, binary opposite "Up and Down", "Left and Right" as well as "Front and Back" must be simultaneously reckoned at the location to which Middle is ascribed. Also, "Forward and Backward" displacement are treated as mutually inclusive and complementary binary opposite spatial-temporal instances in a coherent planning process. Infinite displacement is inherent in content-sensitive planning process. Therefore, the instances of displacement in a context-sensitive plan is non-deterministic and non-monotonic.

6.1.3 *Àtọlẹ̀dọ́lẹ̀* ("Organic") prescription of plan

As depicted in Figure 6.1, in *Àtọlẹ̀dọ́lẹ̀* ("Organic") prescription, a regular plan is subsumed in context-neutral plan which is, in turn, subsumed in context-sensitive plan. Partial aspect of a context-sensitive plan is simplified and express through an instrument of context-neutral language. Isolated instances of the aspect of a context-neutral plan is simplified and express through an instrument of regular language. The terms used to cue the location on the spatial instance in a regular plan are used to simplify and express its instances of displacement.

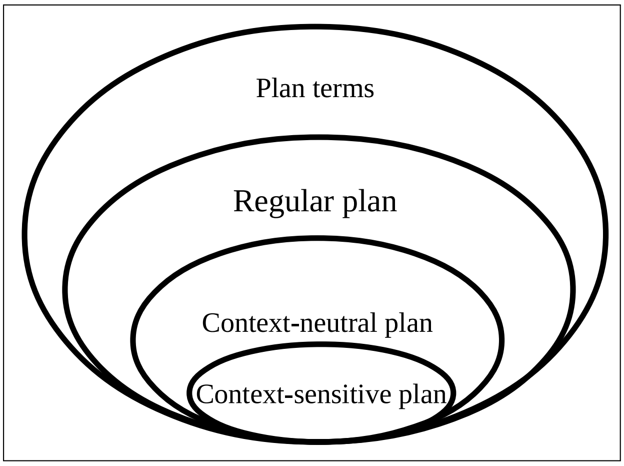


Figure 6.1: Kinds of plan

Context-sensitive plan is the subject-matter in "Automate Intelligence". However, a Context-sensitive instance is inexpressible on the grounds of its infinite instances of displacement. An instrument of context-neutral language is used to reduce a context-sensitive plan into finite instances of location and displacement. An instrument of regular language is further used to reduce a context-neutral plan into instances of isolated location and displacement. By way of these reduction and simplification, partial aspect of Context-sensitive plan is expressed through Context-neutral and Regular instruments of language.

6.2 Illustration of aspects of a plan prescription process

Given Three (3) instances identified with the terms a, b and y occupying locations on the spatial aspect of universe of discourse. The rendering of a description of the universe of discourse based on instruments of regular, context-neutral and context-sensitive language are discussed in the following Subsections.

6.2.1 Regular rendering of spatial instance

Figure 6.2 is the regular rendering of instances whose identities are cued with the symbols a, y and b. The terms a, y and b are ascribed to these instances through regular language. a and y are distinguished as differentiable instances on the grounds of *Àlàfo* ("The Gap") between them. Also, y and b are distinguished into differentiable instances on the grounds of the gap between them.

An instrument of regular language allows these instances to be treated as isolated and unconnected instance based on the gaps between them.

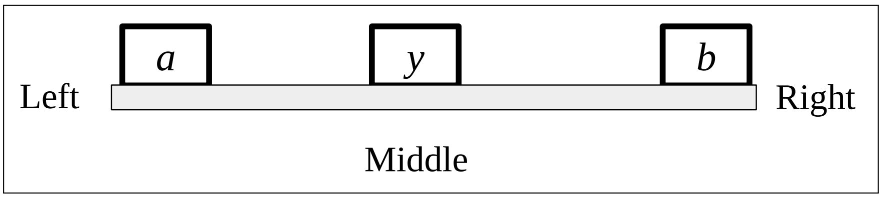


Figure 6.2: Regular rendering of location of instances

The terms "Left", "Middle" and "Right" can be ascribed to a, y and b through the Prime-axiom of the instrument of regular language. However, an agency of regular language can neither relate nor connect these terms. The terms are used to ascribed Assertion to instances in a universe of discourse through the Prime-axiom of an instrument of regular language. One-dimensional structure is used to render the location and/or instances ascribed a, y and b. Attempt by an agency of regular language to rotate the will culminate in Self-identity. However, an agency of regular language can articulate the terms ascribed to each instance in the universe of discourse though the identity ascribed to them.

6.2.2 Context-neutral rendering of spatial instance

An instrument of context-neutral language is used to treat the binary opposite spatial relations of *Left()* and *Right()* ascribed to a and b as mutually exclusive and contradictory instance at the Middle location ascribed to y. By the way of the context-neutral rendering the agency of the universe of discourse can extricate himself/herself from the instances described. Self-reflection is reduced into reflection in the process of generating a description. As depicted in Figure 6.3 a rotation of the universe culminates in the rotation of its instances. The rotation excludes the agency describing the universe.

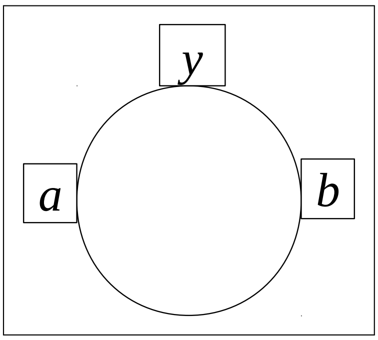


Figure 6.3: Context-neutral rendering of location of instances

Figure 6.4 depicts the tree for the formulation of context-neutral language rendering of Figure 6.3. As depicted in Figure 6.4(a), an agency located at Left, will describe y and b as the Right of a. As depicted in Figure 6.4(b), an agency located at Right, will describe y and a are the Left of b.

Figure 6.4(a) is formulated as: $\boxed{\text{Middle}(y, a, b) = \text{Left}(a,y) \land \text{Left}(y, b)}$

Figure 6.4(b) is formulated as: $\boxed{\text{Middle}(y, a, b) = \text{Right}(a,y) \land \text{Right}(y, b)}$

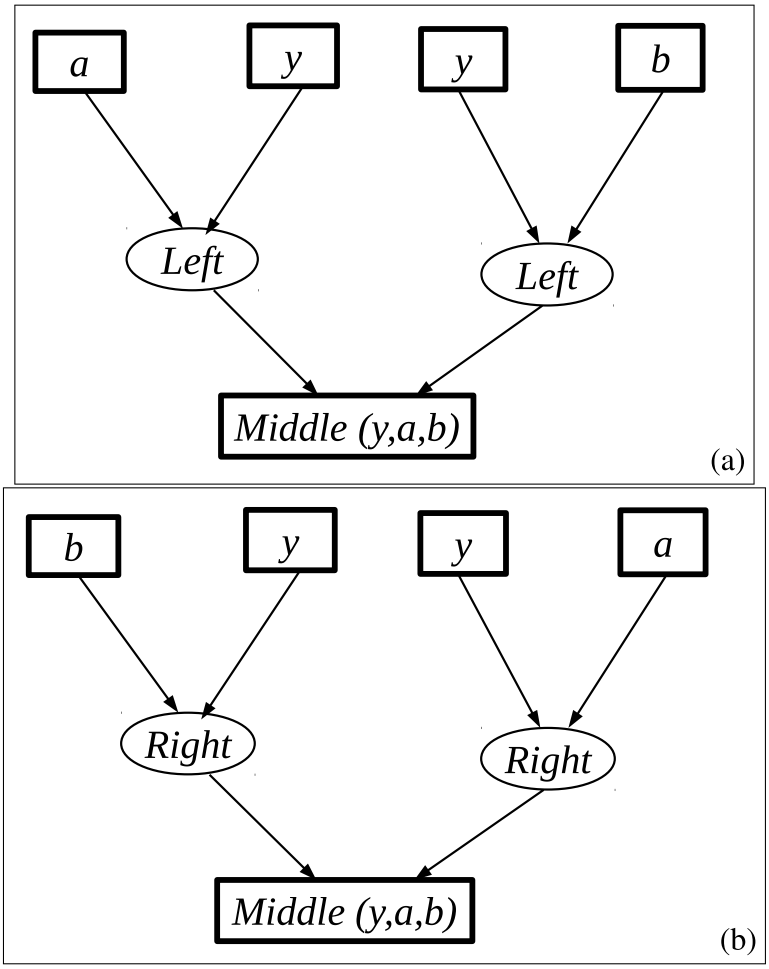


Figure 6.4: Context-neutral rendering of location of instances

Therefore, the context-sensitive Prime-axiom Middle() is reduced into instances of mutually exclusive and contradictory context-neutral expressions through the used of the Auxiliary-axioms $Left()$ or $Right()$. Attempt to use $Left()$ and $Right()$ simultaneously in a formulation will culminate in contradiction for an agency of context-neutral language. Each of the formulations depicted in Figure 6.4(a) and 6.4(b) is consistent. However, they contradict each other.

The formulation rendered as is a simplification comprising two (2) instances of context-neutral recursions.

$\boxed{\text{Middle(y, a, b) = (Left(a,y)} \wedge \text{Left(y, b))} \vee \text{(Right(a,y)} \wedge \text{Right(y, b))}}$

6.2.3 Context-sensitive rendering of spatial instance

An instrument of context-sensitive language is used to treat the binary opposite locations of *Left()* and *Right()*, ascribed to the spatial locations of a and b, as the mutually inclusive and complementary aspects the *Middle()* location ascribed to y.

By the way of the context-sensitive rendering, the agency of the universe of discourse is subsumed in the instances described. Self-reference is reduced into reference in the process of generating a description for instances in the universe. As depicted in Figure 6.5 a rotation of the universe culminates in the rotation of the agency and its instances. The rotation, therefore, includes the agency describing the universe.

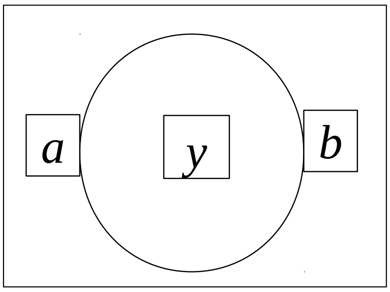


Figure 6.5: Context-sensitive rendering of location of instances

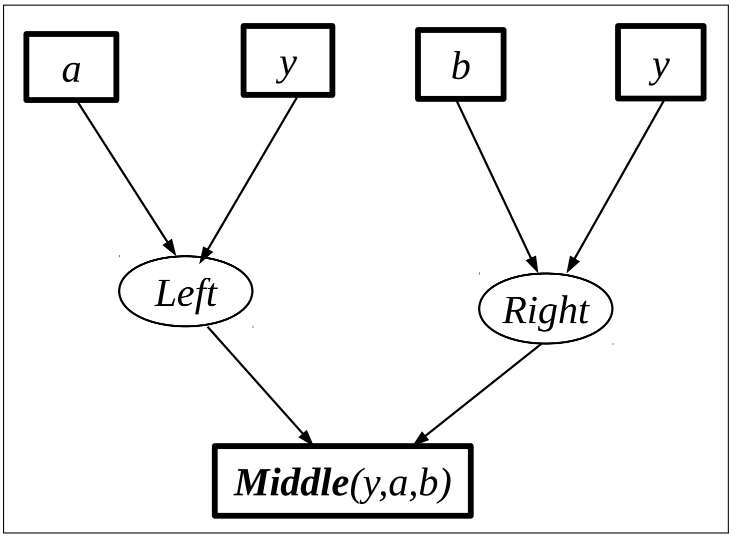


Figure 6.6: Context-sensitive rendering of location of instances

Figure 6.6 is the tree for the context-sensitive formulation for 6.5. An agency located at the Middle location y will always reckon a and b on the grounds of his/her own Left() and Right(). The context-sensitive rendering and formulation is: **Middle**(y, a, b) = Left(a,y) ∧ Right(b,y)

This way, an agency of context-sensitive language is able to treat a and b as the mutually inclusive and complementary aspect of the location occupied by y. Whereas each of the prescription in Figure 6.4 is consistent, the prescription in Figure 6.6 is coherent.

6.3 *Apálará* the robot

"Robot" is derived from the Czech word 'robota". The word "Robota" transcribes into English as "Forced labour". The aim of creating a robot is to simulate the movement of biological agency including human, animal, fish and insect. An essential component of a robot is the mechanism through which it effects the manipulation material instances such as a Stone, Paper or Box through Picking, Holding, Carrying and Putting. Industrial robots are fixed and their entire operation had be programmed into memory. Therefore, Industrial robots are treated under conventional computing in collaboration with Mechanical and Electronic engineering.

Intelligent robot on the other hand, are created to imitates how biological agency, individually or collectively, carry out a prescribed task. Sensors are used to read signals directly from the environment. Cameras are often used record movement in the environment. The signals recorded are manipulate in accordance to a plan. The outcome of the manipulation is used to make a choice of action, inaction or reaction in response to the environment. The outcome of the manipulation is used to effect the actuation of action or movement.

The tree depicted in Figure 6.7 depicts the World of *Apálará* the robot. The world comprises *Ẹni* ("Agency"), that is *Apálará*, and *Ohun* ("Props" or "Things") in its *àyíká* ("environment" or "surrounding"). There are Two (2) categories of Props: (i) Static and (ii) Dynamic. The Static props are neutral to the activity of the agency. They are the environment or surrounding for the agency to manifest and operate. Static props are neutral to the activity of the agency. The Dynamic props are sensitive to the activity of the Agency. The agency uses them to transform its world from one state to another.

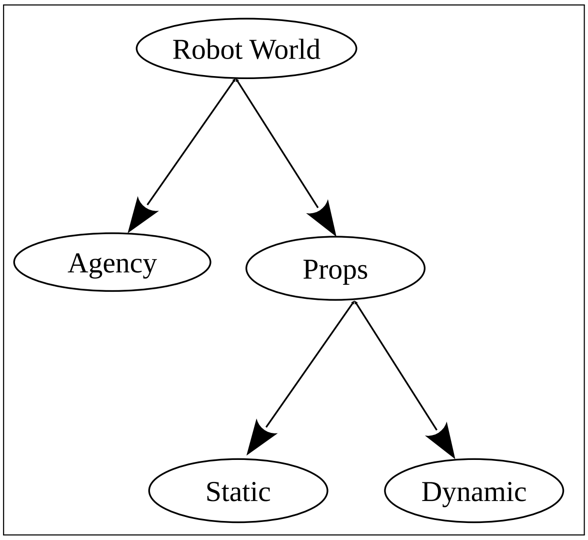


Figure 6.7: Robot world tree

The problem of a planning is the prescription of a process by which an agency can act as follows: Given a World (W) comprising static and dynamic props: If the world (W) is in an undesirable (initial) state S_0, generate and effect the sequence of actions, which when executed, a desirable (final) state S_f will emerge.

The following are assumed in the above statement:

(i.) The language with which to ascribe identity to instances in world W is within the ambit of the agency.

(ii.) Individual agency and props in the world has a unique identity.

(iii.) The agency is the only member of the world that can effect the actions to transform the state of the world.

(iv.) The agency transforms the state of the world to the extent of the capacity of its language.

(v.) The instances of dynamic props in the World are operation-neutral. This implies that the dynamic props neither influences nor obstruct nor enhance the operation effected on them by the agency.

(vi.) The operation effected on a dynamic prop is to the extent of the capacity of the agency.

(vii.) The location and connection as well as position and relation ascribed to instance of props determines the state of the robot World.

The world of *Apálará* the robot is as depicted in Figure 6.8.

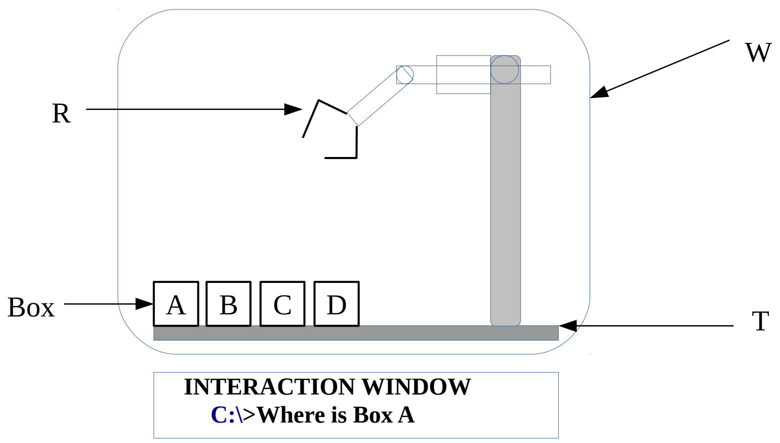


Figure 6.8: The world of *Apálará* the robot

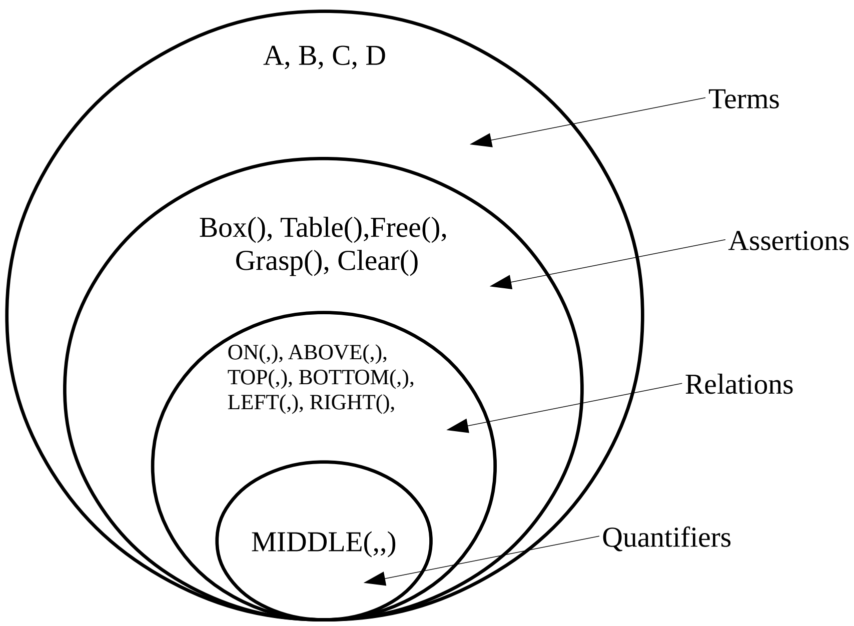


Figure 6.9: Formal language for robot world

Given the world in Figure 6.8, the environment comprises the following:

1 **W** : The world domain

2 **R** : The Agency (Robot or manipulator)

3 **T** : Static Prop (The supporting table)

4 **B** : Dynamic Prop (A set of four (4) boxes with equal sizes, dimensions and weights).

The Agency, Ag, can arrange the boxes in its world, for example, by stacking or un-stacking the boxes. To achieve such action the robot needs to execute the steps in a plan steps. This environment can be used to simulate and demonstrate planning, which is an important task in intelligent system.

The *state* and transition in the robot world are represented and manipulated through an instrument of language. The language is used to:

(i.) **Term or label** String used to cue the nominal identity ascribed to instances in the world, that is *Props* (Boxes) and Agency (Robot arm);

(ii.) **Assertion**: Formulation used to *Ascribe identity* or status of a prop or Agency. It is also used to ascribe relation between props.

(iii.) **Connection** Prescription for the connection between props through the terms used to reckon their identity;

(iv.) **Actuation** Give effect to an action by manipulating a prop ;

(v.) **Caveat** prescribes the admissible instances of location and displacement of an agency and/or the admissible manipulation that can be effected on a prop.

The above listed feature and attribute are expressible with instrument of language. The tools of logic, polarity and structure are inherent in instrument of language.

6.4 Language expression for a plan

The expression of an instance in a plan comprises instances of terms. There are Two (2) kinds of expression in a plan:

(a.) Plan State description (Used to express the status of the plan) and

(b.) Plan Displacement instruction (Used to effect the changes in the location of instance in a plan world)

The term in an expression are used to cue the identity of the instances of Props, Agency and displacement in the Plan world.

6.4.1 The terms in *Apálará*'s world

An expression comprises *terms* only. The terms in the language expression of a plan *closed* or *constant*. This implies a term can only be used to cue an the identity of instance or that of the relation between instances. An expression comprise one or more admissible terms. An agency use the terms in the language expression in the narrative for a plan to effect state transition in the world.

Table 6.1: Terms used in the description of instances on the spatial aspect of a process

1.	Before	*Kù*
2.	After	*Sin*
3.	Above	*Òkè*
4.	Below	*Abẹ́*
5.	Top	*Orí*
6.	Bottom	*Ìdí*
7.	Right	*Ọ̀tún*
8.	Left	*Òsì*
9.	Front	*Iwájú*
10.	Back	Ẹ̀yìn
11.	Middle	*Àrin*
12.	Contact	*Kàn*
13.	Connected	*Kanra*
14.	Similar	*Jọra*

6.4.2 Assertion in Apálará world

When a *term* is used to ascribe assertion to a prop, only one of two binary opposite states is possible: Accepted (*True*) or Rejected (*False*). A state is either Accepted (True) of Rejected (False) and there is *No State* in between these two (2) state. Every Assertion term in the expression of the narrative for a plan state must be Accepted (*True*).

The assertion on instances of prop (box) are listed and explained in Table 6.2.

Table 6.2: Prop (box) status assertions

Ser. No.	Assertion	Explanation
1	BOX(X)	Is Asserted if X is the identity ascribed to a box
2	CLEAR(X)	Is Asserted if there is no instance of Box on the Box identified as X
3	ONTABLE(X)	Is Asserted if the box identified as X is on the Table

The instantiation of assertion on an Agency (robot) are listed and explained in Table 6.3.

6.4.3 Relation in *Apálará* world

Instances of props in the world of *Apálará*, i.e. the boxes, are identified, related and manipulated using conventional formal language. If a relation is ascribed to the state of instances of prop. When the action of an agency had been carried out successfully, then *True* (Accepted) is returned. *False* (Rejected) will be returned otherwise. The relations listed in Table 6.4 and Table 6.5 are used in this Laboratory.

Table 6.3: Agency (Robot) status assertion

Ser. No.	Assertion	Explanation
1	FREE()	Returns Is Asserted if the Agency (Robot arm) is not holding a Box. If the Agency (Robot arm) is holding its releases it.
2	READY()	Is Asserted when the Agency (Robot arm) is ready. If the Agency (Robot arm) is not ready, its become ready.
3	GRASPING(X)	Is Asserted if Box identified as X is in the Robot arm's Claw. If the Agency (Robot arm) is not holding X its grasps X.

Table 6.4: Relations between Props

Ser. No.	Relation	Explanation
1	ON(X,Y)	Confirms that Box X is directly on Top of Y
2	ONLEFT(X,Y)	Confirms that Box X is directly on left of Y
3	ONRIGHT(X,Y)	Confirms that Box X is directly on right of Y
4	UNDER(X,Y)	Confirms that Box X is directly below Y

Table 6.5: Movement Assertion (Robot Actuation) prescription

Ser. No.	Assertions	Explanation
1	MOVETO(X)	Effect the action for the Robot arm to move to the location occupied by Box X
2	PLACEONTABLE(X)	Effect the action for the Robot place the Box in its claw on the Table
3	GRASP()	Effect the action for the Robot to grasps the Box in its current location.
4	NOP()	The robot arm assumes ready state or "No action"
5	READY()	Effect the action to wake up the robot arm. Robot arm assumes ready state.

6.4.4 Expression of the connection of assertion and relation in *Apálará* world

The assertion and relation ascribed to instances of object and agency are connected to extent the narrative through logical operators. Operators to be used along with the assertions and relation defined above include the **AND**, **NOT** and **OR**. These connectives are listed and explained in Table **??**.

6.4.5 Generalisation over assertion and relation in *Apálará* world

In a monotonic world, the state and/or transition of a prop can be defined and formally specified. This is achieved through the Two (2) universal quantifiers in formal logic. The universal quantifier are also called "Predicate" in conventional logic.

(a.) For all instances ($\forall$)

(b.) There is at least one instance (Existential $\exists$).

These quantifiers are explained in Table 6.6. Usually, one of the quantifiers is sufficient for the specification of a plan.

There are Two (2) types of universal quantifier:

Table 6.6: State generalisation operators

Ser. No.	Relation	Explanation
1	$\exists$	It is possible to find at least one instance, e.g. $\exists x\ P(x)$ implies, it is possible to find at least an x for which the assertion $P(x)$ is Holds.
2	$\forall$	For all instances, e.g. $\forall x$ imply that the assertion $P(x)$ Holds for all instances of x.

These generalisation operator can be used to specify the criteria or constrain the activity in the robot world.

6.4.6 Examples of the use of Quantifier in Plan prescription

Caveat (Assertion constraint) prescription To formulate the sentence "A box is clear when there is no box on top of it".

Using the relation **ON(X, Y)**, two (2) instance of box is required to formally express this criteria. Lets say, X and Y. Then Box(X) and Box(Y) is True for instances of X and Y. The the criteria can be expressed as follows: *"An instance of Box Y is clear if it is not possible to find an instance of Box X on top of Y"*. This sentence can be formally represent as:

$$CLEAR(Y) \Rightarrow \neg\exists X, box(X) : ON(X, Y)$$

This is explained as: "The box identified as Y is clear when it is not possible to find a box identified as X such that the relation ON(X,Y) Holds."

OR

$$CLEAR(Y) \Rightarrow \forall X, Box(X) : \neg ON(X, Y)$$

This is explained as: "The box identified as Y is clear when the relation NOT(ON(X,Y)) Holds all instances of box identified as X."

Caveat (Agency operation constraint) prescrption To formulate the sentence "The robot clear grasping an object when there is no object on the object and the robot arm is free".

Using the relation **ON(X, Y)**, two (2) instance of box is required to formally express this criteria. Lets say, X and Y. Then Box(X) and Box(Y) is True for instances of X and Y. The agency's action, that is grasp, is that to which the constraint applies. The the criteria can be expressed as follows: *"The robot can grasp the object X when there is no instance of Box Y such that ON(Y,X) Holds"*. This sentence can be formally represent as:

$$GRASP(X) \Rightarrow \neg\exists Y, Box(Y) : ON(Y,X) \wedge Free()$$

This is explained as: "The robot arm can grasp a box X when: (i) there is no instance of box Y such that the relation ON(Y,X) AND (ii) the assertion that the robot arm is Free Holds."

6.5 Computational representation of Plan

There are a number tools available in computing for expressing state and transition in a world comprising instances of object and one or more agency. This includes:

(i.) Plan state diagram (also called Semantic network).

(ii.) Plan Transition tree.

(iii.) Formal language expression

(iv.) State-transition matrix.

The above tools are deployed in the design, modelling and simulation of the robot arm system in the laboratory.

6.5.1 State-Transition diagram for plan Location-Displacement

Plan state diagram (also called Semantic network) is a diagram that shows the identities of the props in a plan as well as the relationship between them. The semantic network for the Robot world in Figure 6.10 is depicted in Figure 6.11.

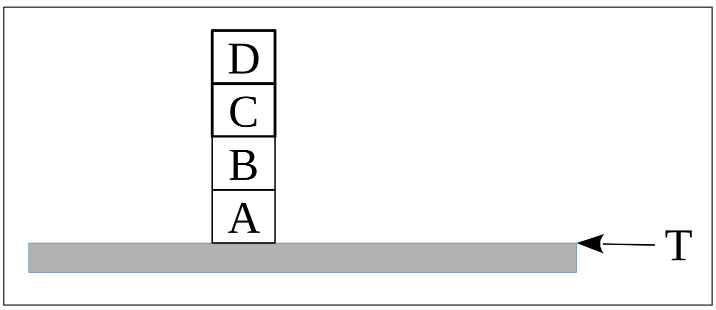


Figure 6.10: *Apálará* robot world

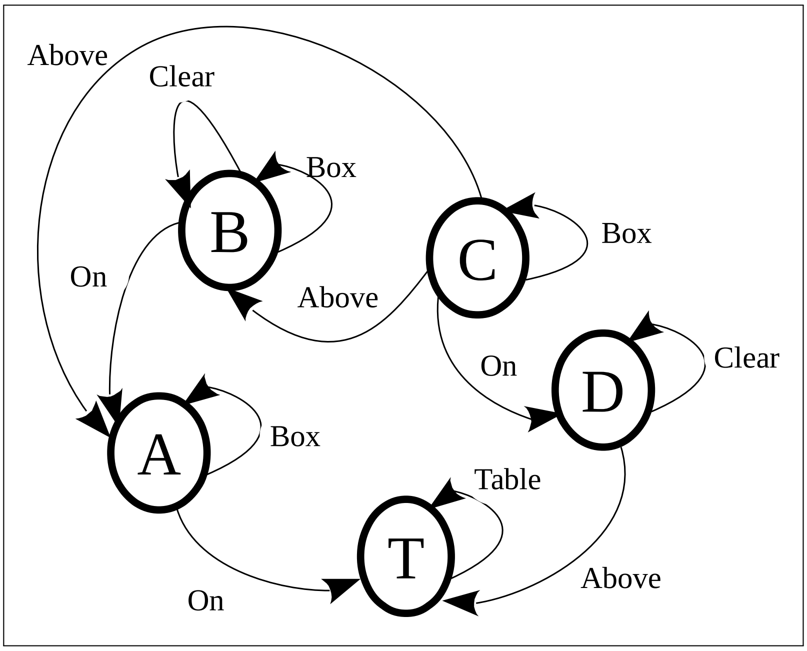


Figure 6.11: State-transition diagram for *Apálará* robot world in Figure 6.10

6.5.2 Formal language narrative for plan

The state of the world S_0 in Figure 6.12 is expressed formally as follows:

State S_0
Ready(),
OnTable(A),
On(B,A),
On(C,B),
On(D,C),
Free().

The state of the world S_4 in Figure 6.12 is expressed formally as follows:

State S_4
OnTable(C),
OnTable(D),
On(A,C),
On(B,D),
Free().

6.5.3 State-transition Matrix for plan

Avery state in a regular plan has a unique expression. The expression comprise terms that identify the props as well as the relationship between them. The symbols used to cue the identity of the

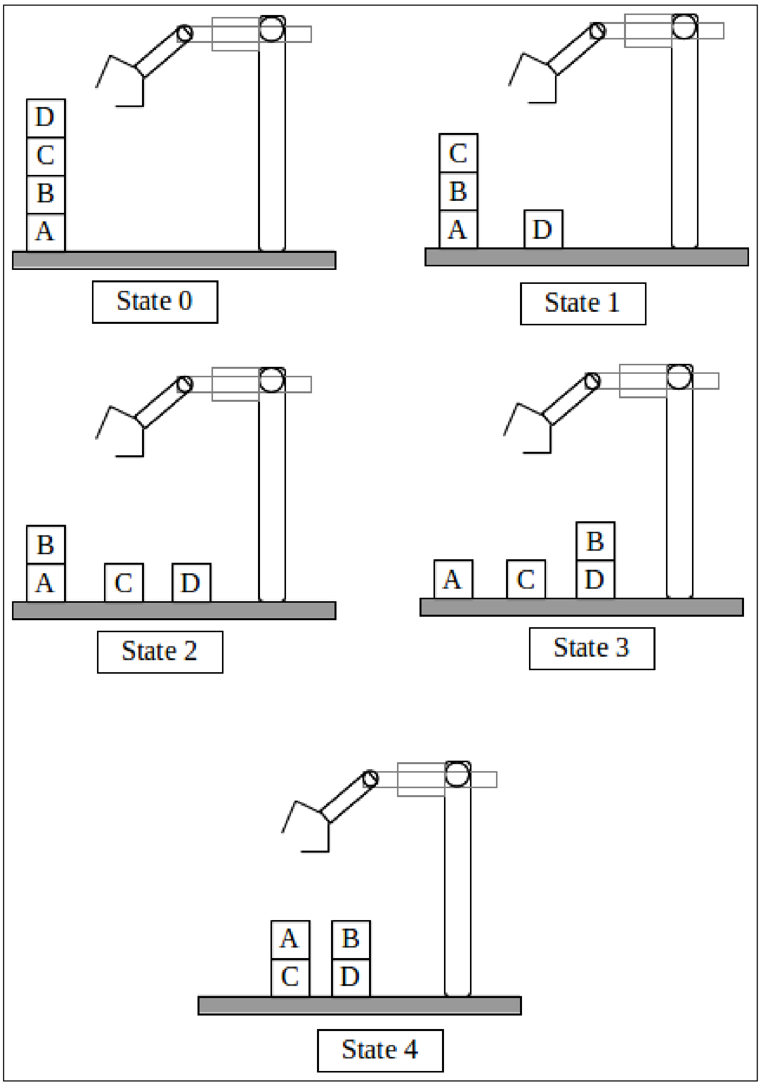


Figure 6.12: Instances of state in the robot world

props (operands) and their relation (attributes or feature) can also be formally expressed through the mathematical tool of **matrix**.

A Robot world *State matrix* is constructed as follows:

(a.) The instances of dynamic prop (The boxes A, B, C, and D) in the robot world is defined. Each prop occupies a cell in the matrix. If there are N dynamic props then the state of the Robot world is expressed using an $N \times N$ matrix.

(b.) Zero is assigned to an unoccupied location (or cell) in the Matrix of the robot world.

(c.) The location of the props is grounded in a static prop. In this case, the Table (T).

Based on the above prescription, the matrix in Equation 6.1 represents the state of the robot world in State S_0 of Figure 6.12. Equation 6.2 represents the state of the robot world in State S_4 of Figure 6.12.

$$\mathbf{S_0} = M = \begin{bmatrix} D & 0 & 0 & 0 \\ C & 0 & 0 & 0 \\ B & 0 & 0 & 0 \\ A & 0 & 0 & 0 \end{bmatrix} \tag{6.1}$$

$$\mathbf{S_4} = M = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & C & D & 0 \\ 0 & A & B & 0 \end{bmatrix} \tag{6.2}$$

Matrix representation is easier to represent, stored and manipulate using the array (two dimensional) in computer programming.

Note that, if the Matrix of the world is identified with the symbol **M**, then in State S_0 Boxes **A**, **B**, **C** and **D** are located in **M[1,4]**, **M[1,3]**, **M[1,2]** and **M[1,1]**, respectively.

Using the language and tools discoursed above, it is possible to explore *Apálará*'s world as well as use an agency to generate and give expression to the narratives for its plan.

6.6 Task 5.1

Write the robot plan to implement the following:

(i.) Swap any two boxes.

(ii.) Place a box under another box

(iii.) Place a box on top of another

(iv.) Stack all the boxes in alphabetical order with A at the bottom.

(v.) Stack all the boxes in alphabetical order with D at the bottom.

Generate the matrices for the state transitions of the above plans.

6.7 Task 5.2

(i.) Using the *Python* programming language, develop a software that implements any two (3) of the plans you wrote in Task 5.1 above.

(ii.) Test your system extensively and discuss its limitations.

(iii.) Discuss the language of your robot plan.

CHAPTER 7

ÀJỌṢE (COOPERATIVE INTELLIGENCE): ACTIVITY OF ORGANISMS

The activity of cooperating organisms differs from "Planning" or "Game playing" processes. In a cooperating activity the extent of the competency of individual participant manifest in the performance of the final outcome. Therefore, there is no final outcome to which the activity of a collective organism conform. For example, in a competitive intelligence, such as game process "Win", "Draw" or "Loss" are the only possible outcomes. In individual intelligence such as planning process, the outcome had been specified and prescribed with a plan. Whereas game playing and planning processes are non-deterministic and monotonic, a cooperative activity is non-deterministic and non-monotonic.

Ọgbọ́n ("Intelligence") is ascribed to *agbo* ("a group") of organism during *Àjọṣe* ("Cooperative") activity on the grounds of the creativity and/or ingenuity manifesting in their collective performance. *Ọgbọ́n* (The "Intelligence") ascribed to *Àjọṣe* ("Cooperative") activity is to the extent of coordination and coherence manifesting in the performances of the community of organisms.

Ọgbọ́n ("Intelligence") had been ascribed to the community of *Kòkòrò* ("Insects") such as *Ikán* ("Termite"), *Èèrà* ("Ants") and *Èèrùn* ("Soldier ants") on the grounds of the creativity and coherence manifesting in the performance of their activities.

Ikán ("Termite") moves by crawling or flying individuals or in a swarm comprising many of them.

Here, the imitation of the intelligence ascribed to *ÀDÁṢE* ("Individual") and *ÀJỌṢE* ("Cooperative") activity of *Ikán* ("Termite") are discussed and explain.

7.1 *Ikán* ("Termite")

Ikán ("Termites") have been admired for their ability to effect tasks cooperatively. Their cooperation is particularly demonstrated in *Ọ̀gán* ("Mound") construction. In the *Yorùbá* oral literature, it is narrated that the source of the water that termites use to build their mound is unknown. However, it is recorded in conventional literature that termites use their saliva to mix mud and dung in the construction of their mound. Here, intelligence is ascribed to the creativity manifesting in the cooperation among numerous Termites in the construction of their *Ọ̀gán* ("Mound"). Termite mound is depicted in Figure 7.1.

Compared to the size of an individual *Ikán* ("Termites"), the height of a full fledge *Ọ̀gán* (a "Mound") is about a Sixty (60) storey human building. It is more intriguing when it is noted that termite's mounds are usually not very close to a source of water. They do not have the skill to create cement or building rods. Despite this, *Ọ̀gán* (a "Mound") rarely collapse.

Figure 7.1: *Ọ̀gán Ikán* ("Termite Mound")

Also, termites use extensive tunnels and conduits to connect various chambers inside their mound. The conduits and chambers serve various functions such as accommodation, food storage and ventilation.

7.2 Preamble to the laboratory

This laboratory is concerned with the computational representation of the narrative for your observation of the movement and navigation behaviour of *Ikán*. Figure 7.2 depicts an image of *Ikán* ("a termite").

The agency, *Ikán*, possess the characteristics common to its kind of insects. The body organs of a typical *Ikán*, with English language annotation, is depicted in Figure 7.3. The organs for effecting movement and navigation include:

(i.) Three body regions (Head, Thorax, Abdomen): This holds the other body-organ of *Ikán* together and facilitates the coordination and control of its movements. The thorax is the middle region of the body, to which three pairs of jointed legs are connected. These Six (6), or Three (3) pairs of elbowed legs, are the primary tool for the movement of *Ikán*.

(ii.) The six jointed Legs are used by *Ikán* to move **forward** or **backward**.

Figure 7.2: Plate of *Ikán* (“Termite”)

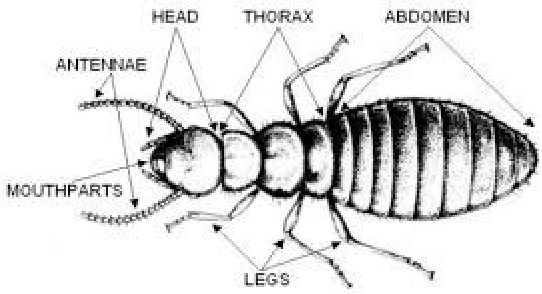


Figure 7.3: Worker Termite and its locomotion organs in conventional literature

(iii.) A pair of Antenna is used by *Ikán* for detecting obstacles on its path; and

(iv.) A pair of Mandible, sometimes called jaws, is used by *Ikán* for digging and carrying object and food as well as for biting to defend itself. It also uses the mandible when building. Its mouth organ is for chewing. *Ikán* also have “a tongue” for sucking up liquid.

Using the above body organs, *Ikán* can **smell**, **walk**, **bite**, **sense**, **dig** and **navigate** itself in its environment (its world). A group of *Ikán* achieve the task of building a mound through cooperative actions and movements.

A simplified description of the displacements in the movement effected by **one** *Ikán* is as follows.

1. *Ikán* always moves in a straight line except its antenna touches an obstacle.

2. If the **left** antenna touches an obstacle, *Ikán* will move to the **right**.

3. If the **right** antenna touches an obstacle, *Ikán* will move to the **left**.

4. If the **left** and **right** antennae simultaneously touch an obstacle, *Ikán* makes one of Two (2) moves: (i) If the obstacle is not food, the termite will move backward one step and attempt another path to its **left**. If this fails, it will attempt the path to its **right**; (ii) If the obstacle touched by *Ikán*’s antennae is food (as detected by its smell), it will carry the food and make 90^0 turn before heading home using steps 1, 2 and 3.

7.3 Hint on the laboratory

Based on the narrative above, let the symbols **l** (left), **r** (Right), **b** (Both) and **f** (Food) represent the instances of input touched by the antenna of the Agency. This instances of input is explained in Table 7.1.

Table 7.1: Input to *Ikán*

Ser. No.	Symbol	Explanation
1.	l	The Left antenna touches an obstacle
2.	r	The Right antenna touches an obstacle
3.	b	Both antenna touches an obstacle
4.	f	Both antenna touches food

Let the symbols M_L (Moving left), M_R (Moving right), M_B (Moving back), M_F (Moving Front) and M_0 (At home) represents the State/Location of the Agency of the Termite. This instances of state and its location is explained in Table 7.2.

Table 7.2: Description of the Displacements in the movement of *Ikán*

Ser. No.	Symbol	Explanation
1.	M_L	*Ikán* moves the left one step
2.	M_R	*Ikán* moves right one steps
3.	M_B	*Ikán* moves back one step
4.	M_F	*Ikán* moves forward one step
5.	M_0	Home state of *Ikán*

The transition diagram in Figure 7.4 depicts the movements effected by the agency of *Ikán*. For example, the cycle marked M_0 identifies the initial and final (home state) of the termite (*Ikán*). When the input ‘b’ (both antenna touches obstacle) is applied to *Ikán* at state M_0, the agency of *Ikán* transits to state M_B. When the input ‘r’ (right antenna touches obstacle) is applied to *Ikán* at state M_0, the agency of *Ikán* transits to state M_L. When the input ‘l’ (left antenna touches obstacle) is applied to *Ikán* at state M_0, the agency of *Ikán* transits to state M_R.

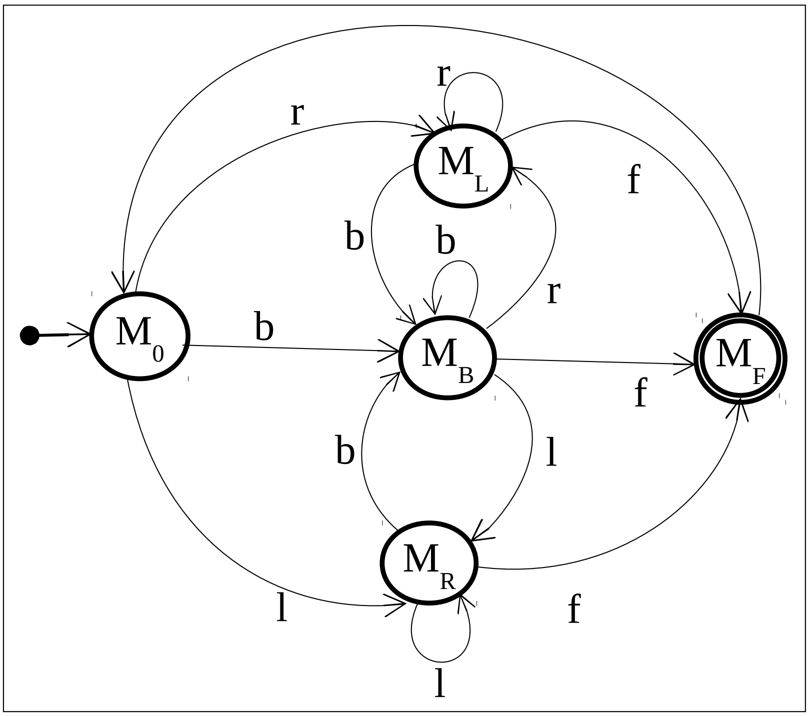


Figure 7.4: Transition diagram for the agency of *Ikán*

7.3.1 Task 1

Review and study the process used in the *Apálará* experiment in Laboratory II, Apply your experience in that laboratory to the tasks in this laboratory exercise.

1. Identify the assertions, relations and agency in *Ikán*'s world.
2. Using the transition diagram in Figure 7.4 generate the narrative for the movement of *Ikán*.
3. Discuss the process of *Ikán*'s movement and navigation behaviour.
4. Design a computational mechanism to effect the process in 3. **HINT** use the process in Section 8.4.
5. Use case Sentences or Strings to discuss the navigation and movement of *Ikán*.
6. Discuss your reflection on the differences between your system and the biological termite.

7.4 Automating the movement of *Ikán* (The Termite)

Ikán (The termite) has Three (3) pairs of legs, each in the Front, Middle and Back section. These legs are label as, namely:

1. The Front legs are identified as 1 and 2
2. The Middle legs are identified as 3 and 4

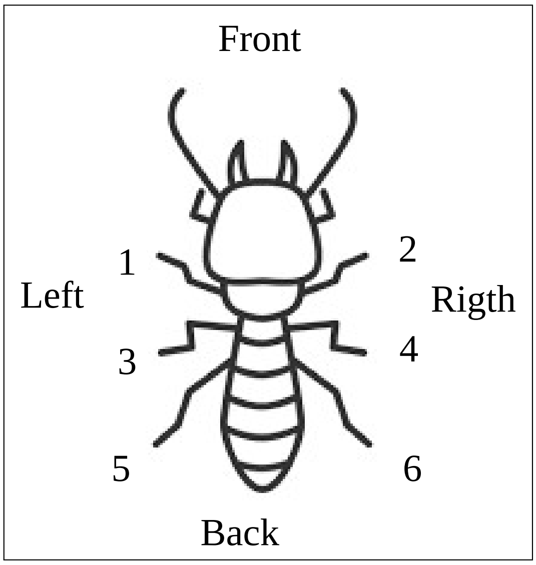


Figure 7.5: Movement of *Ikán*

3. The Back legs are identified as 5 and 6

The Three (3) pairs of leg is formed into Two (2) superimposed tripods. Figure 7.6 (a) depicts the Even leg tripod while Figure 7.6 (b) depicts the Odd leg tripod. Each tripod is used to effect balance and movement during the Termite's gait. When the even tripod is on the ground to effect balance, the termite uses the odd tripod to effect movement. Conversely, when the odd tripod is on the ground to effect balance, the even tripod is use to effect movement. The tripod on the ground is used to balance the current state of the Termite.

In the simulation of the termite's movement, the even and odd tripods are superimposed as depicted in Figure 7.7. *Ikán* ("The termite") movement is effected through the alternation of the tripod gaits. At the Current instance of the termite gait, a tripod is active while the other is inactive.

7.4.1 Forward movement simulation

If the movement is Forward

(i) If the even tripod is off the ground, rotate it anticlockwise.

(ii) If the odd tripod is off the ground, rotate it clockwise.

(iii) Repeat (i) and (ii) until desire location is reached.

7.4.2 Backward movement simulation

If the movement is Backward

(i) If the even tripod is off the ground, rotate it clockwise.

(ii) If the odd tripod is off the ground, rotate it anticlockwise.

(iii) Repeat (i) and (ii) until desire location is reached.

A step in the movement of the termite is one gait. Some assumption are required to formulate the gait for movement. For example, which tripod is to be used to effect the initial and/or final gait.

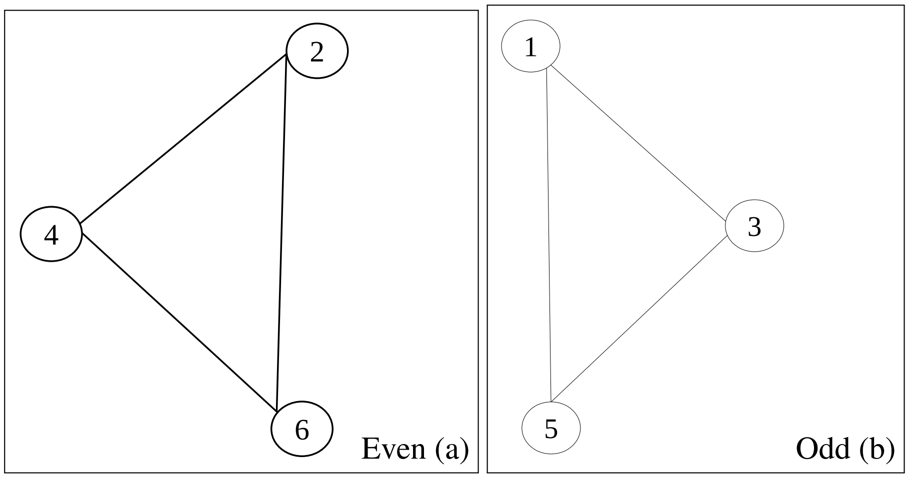


Figure 7.6: Even and Odd Tripods

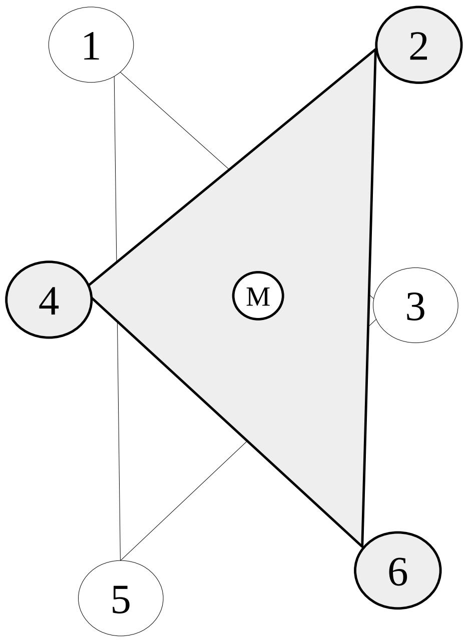


Figure 7.7: Superimposition of the Even and Odd Tripods

7.4.3 Task 2

(i.) Using a graphic-based application, simulate the movement of the Termite using the narrative in Section 7.4.

(ii.) In your Simulation, the superimposed Tripods are moves based on clicking the **radio buttons MoveFront** or **MoveBack** (iii.) Upgrade your simulation by using the narrative in Section 7.3 to effect its movement. A string of letters will then be the prompt rather than **radio buttons**.

CHAPTER 8

LABORATORY PROCEDURE AND DOCUMENTATION

8.1 Preamble

The laboratory assignments in this course are intended to, among other things, allow you to further explore the topic presented during lectures. During lectures, we have discussed that the biological agencies with which intelligence is ascribed emerges from a happening. Whereas Automated Intelligence (AI) system emerges through a prescribed process. We have discussed the distinction between an Organism (whose beingness is grounded in happening) and a Mechanism (Created through prescribed process). AI system development skills improve with practices. Such skill comes with practical experiments which the laboratory exercises here seeks to provide. To this end, the laboratory assignments in this course has Three (3) goals:

1. To provide the avenue to further explore your understanding of the principles and techniques discussed during CPE-316 lectures;

2. To improve your proficiency in computing through familiarisation with the processes involved in the development of AI systems;

3. To allow your instructor (s) to evaluate and assess your understanding of the subject-matter presented in the course.

Accordingly, you are *strongly* advised to make sure that all the work you submit for assessment arise out of your efforts. You are permitted and encouraged to seek clarification on general heuristics design with the course lecturers as well as the laboratory coordinators. You may also receive help with specific debugging problems during the practical sessions as appropriate. However, you are expected to work independently, or only within your own group (where applicable), during the laboratory assignment. Your sincere contribution to the work submitted is emphasised. Remember that the motto of this university is "*for learning and culture*".

8.2 Laboratory Report

The documentation of your experiments is very important and will be given much attention during the grading of your laboratory work. You should, therefore, ensure that your laboratory reports follow the format stated in Table 8.1. It should include information about your observations as much as possible. Also note that the items listed as $1, 2, 3, 6, 7, 8$, and 13 must be included in your reports. Items listed as $5, 9, 10, 11$ are important. While items listed as 4 and 12 should be included when used. The presentation of your laboratory report should be *clean*, *clear* and *legible*.

Table 8.1: Laboratory Document Contents

Ser. No.	Item names	Item Descriptions
1.	Name and Title of Report	NAME: AI Robotic System: Laboratory Two. TITLE: Planing System
2.	Author and date	Your full name(s), Identification numbers, and Department of major (this must be listed for all group members if it is a group work)
3.	Table of Contents	Tables of Contents List of Figures (If any) List of Tables (If any)
4.	Glossary of Notation and Terms	List all uncommon symbols and terms and specify their meanings in the context of this laboratory.
5.	Introduction	Brief discussions, stating the background of the present work.
6.	Problem statement	Statement of the problem Mathematical models of problem. Supporting theory and physical laws for solution (if and)
7.	Objectives of Experiment	General Objective(s) of the lab. work Specific Objective to be accomplished in this experiment
8.	Experiment procedure	Initial setting/material and tools. Experiment processes. Machine design Machine implementation. Final setting or configuration (if any)
9.	Discussion of results	Observations, analysis of method used, significance of result obtained.
10.	Technological interpretation and Application of results	Illustrate, with examples, the real-life interpretation and applications of the result of your experiment.
11.	Conclusion and Suggestions for further works	Summary and general observation, reflection and next course of action.
12.	Appendixes	Tables of values and results Program listings.
13.	References	List all the literature cited in your work.

8.3 Assignment Submission Dates

The dates when you will be required to submit the reports of your assignments will be announced during lectures or posted on the course webpage. You should endeavour to submit your reports on or before such dates. Late submission may attract mark deductions.

If you are having issues with any of the packages for the laboratory, please seek assistance from the laboratory coordinators.

8.4 Background to laboratory assignment

The process for AI system development is similar to that of conventional computational systems. However, the purpose of AI system development differs. During lectures, we have discussed that it is possible to design and implement a material agency to imitates the language rendering of the behaviour of biological agency with which intelligence has been ascribed. AI system development comprises the following six (6) steps:

1. Understand the Problem: Give the object of study a name. Get a clear understanding of the behaviour of the biological intelligent to be imitated. Case examples of the possible input/output characteristics of the system can be very helpful in this task. Note that, AI are unlike conventional computing agency. For example, some characteristics of biological agency are not directly observable and hence cannot be described definitively. A careful analysis of the state and transition of the biological agency, therefore, suffices. The outcome of the step in (i) The name of the IA agency, (ii) A narrative that described the biological behaviour to be imitated (iii) List and description of the attributes of state of the AI system, (iv) List and description of the attributes of transition of the AI system.

2. State Assumptions: List the assumptions you are making regarding the AI system to be developed. Assumptions are meant to simplify the AI agency by specifying the context and aspect to be reckoned in the biological agent's activity. The assumption must, first of all, recognise the extent of the instrument of human language in the description a happening. Make sure that you are able to justify your assumptions using reasonable arguments. Assumptions justified based on the environment in which the AI system will work, the resource available to you, the target users of the AI system, etc. will be appropriate.

3. Behaviour Analysis: This comprises two tasks: (i) AI system states and structure and (ii) AI system transition (dynamics) and logic. The system state will identify its component and how they are interconnect in the structure of the AI system. It will also identify and specify the character of the Input and Output of the AI system. Each and every relevant stimuli(input) to the AI system and the corresponding response (output) of the system is identify and labelled. Note that some activity with which intelligence is ascribed do not have explicit stimuli (e.g. when its response is based on reflex), while some stimuli may not generate explicitly or directly observable response. The language for an AI systems is often used to characterise its behaviour.

At the end of this step, a document containing how the AI system will be designed is produced. This is often called requirement specification.

4. System Design: This involves considering alternative formulation and design options as well as strategies, techniques and methods. After this, the most appropriate computing tools for representing the behaviour analysed in **Step 3** is selected. Possible design tools include: semantic network, heuristics, formal language, genetic algorithm, neural networks, state transition diagrams, script, graphs, plan, etc. At the end of this step, a document containing how the implementation of the AI system will be executed is produced.

5. Implementation: The system designed in **Step 4**. is reduced into a computing artefact using appropriate software tool and/or programming language. This is executed on an appropriate hardware. The programming language and software tools selected will be informed by **Step 4**. Note that the implementation of AI systems, such as Robot, require collaboration with individual from other disciplines. The implementation of a robot hardware components, for example, will require collaboration with professional from mechanical, electrical and electronic engineering fields.

6. Evaluation: This involves examining whether the AI system implemented in **Step 5** conforms to specified criteria. **The focus of evaluation is to determine how well your system has imitate the behaviour of the biological agency in the object of study**. The evaluation of an AI system is in two (2) stages: (i) Inside test and (ii) Outside test. In the inside test, the performance of the system is assessed using data (stimuli and response) from the case examples used for the system design. When the system has done well on the inside test, then the outside test is conducted. In the Outside test, the extent to which the AI system imitate that are NOT from the sample case used for the AI system design. Based on the outcomes of the evaluation, a decision to correct, modify or deploy the system is taken as appropriate. The Mean Opinion Score based on the Alan Turing Test is used in AI system evaluation.